

2023 Shutter Revision log



Effective Date	Topic	Section	Note
	Pre-drill L Frames (Inside Mount Only) or Z frames		e: add content: Please note that the prefilled hole and the plug will not be a 100% complete fit, but that would not affect the performance of the screw. Shutters can't be remade solely because of this issue.
	Mount Options For Direct Mount		i: add content: Light block and L magnet catch are available in one of the following situation. • Direct mount without top & bottom sill frame • 2-sided Hang strip without top & bottom sill frame • 3-sided Hang strip without bottom sill frame • 3-sided inverted Hang strip without top sill frame
	Custom colors	e,i,k	k: add content: For custom color orders, please provide a color sample in the form of: a 4" x 4" coated foam core, a poster board if you are requesting a custom paint color, or a raw shutter louver if you are requesting a custom stain color. You are able to send in your custom color sample in a format that does not fit the three listed forms; however, please anticipate a longer lead time as the custom color sample may be required to be processed through customs. In the event that you are sending in a custom color sample that is not in the three listed formats, providing any supplemental details about the sample will assist us in getting the right color quickly, e.g. material, the Latin scientific name of the material, detail Size, a photo of the color sample, the package method, net weight, whether a premium is included in the value, certificate of origin, plant inspection certificate etc. Due to varying materials corresponding to different colors, if a custom color sample is sent in, we will match the color on a louver and send the physical louver sample to you for confirmation. In order to streamline the process, in the event that the color variation between the physical louver swatch and the custom color sample falls outside of our acceptable range, we will verify the color digitally first with you before mailing the physical color swatch for final confirmation.
2023/01/15	Color	b,i,	b: Update colors option. i: 1. Add Panel Lock color. 2. Add 3 types of handles. 3. Update color coordination chart.

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	Color coordination chart		4. add content: ** The 017 Gray Black, 246 Matte Black, 836 Classic Black and 609 Chateau Brown Colors will have a surcharge. ** All OSMO Colors will have a surcharge.
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Features and Benefits

- Brings a touch of country elegance to your home. The textured finish of Normandy stained hardwood shutters blends the tranquility of nature with distinctive luxury.
- Normandy is treated with our exclusive Prescription Wood Conditioning™ technique according to the moisture content of the final installation location to reduce post-installation shrinking and swelling.
- A wide range of frames with inserts that keep installation screws out of sight for a clean and decorative look to match any style and preference.
- Reinforced engineered stiles -- Multiple layers of wood bonded together to reinforce durability to the core of the stile, which results in a robust support system for the shutter panel.
- Multiple stile options available in various sizes to accommodate different window configurations. Astragal (solid color shutters only) or Rabbet stiles provide a natural light block between panels.
- Employs mortise and tenon joinery that resists shear and racking forces endured by shutters and ensures prolonged stability and durability.
- Quick and easy on-site installation with the pre-installed magnets and hinges, as well as pre-mitered frames and slotted keyholes.
- Labor-intensive hand finishing process involves multiple sequences of sanding and coating with fine layers of stain or paint for a brilliant, lustrous finish.
- Personalize shutters with special options. Choose from hidden or offset tilt rods, double-hung panels, café shutters, track systems, specialty shapes and many more.
- Special panel and frame designs are finely crafted to accommodate unique window configurations. Create unity or attract attention with Normandy's exclusive decorative elements.
- Available in classic paint, radiant stain, and stimulating designer colors that harmonize into the style of any home.
- A guarantee of Norman Shutters' exemplary quality with a lifetime limited warranty on materials and workmanship.

Colors & Louver Sizes/Louver Clearance

	No.	Color ID	Color Name
Paint (24 colors)	1	001	Pure White
	2	002	Extra White
	3	003	Silk White
	4	004	Bright White
	5	006	Pearl
	6	007	Ivory Lace
	7	009	Creamy
	8	012	Crisp Linen
	9	013	Bisque
	10	017	Gray Black
	11	019	String
	12	032	Sea Mist
	13	046	Ice
	14	053	Clay
	15	063	Decorator's White
	16	066	Winchester White 2010
	17	076	Aura White
	18	080	Taupe Gray
	19	836	Classic Black
	20	090	TS White
	21	600	True White
	22	601	Chiffon
	23	603	Natural Linen
	24	609	Chateau Brown
Stain (18 colors)	1	108	Rustic Gray
	2	110	Limed White
	3	200	Natural
	4	202	Golden Oak
	5	205	Goldenrod
	6	221	Black Walnut
	7	227	Red Oak
	8	229	Rich Walnut
	9	230	Old Teak
	10	237	Wenge
	11	242	Auburn
	12	246	Matte Black
	13	248	Pretzel

*Some surcharges apply

Normandy® Shutters

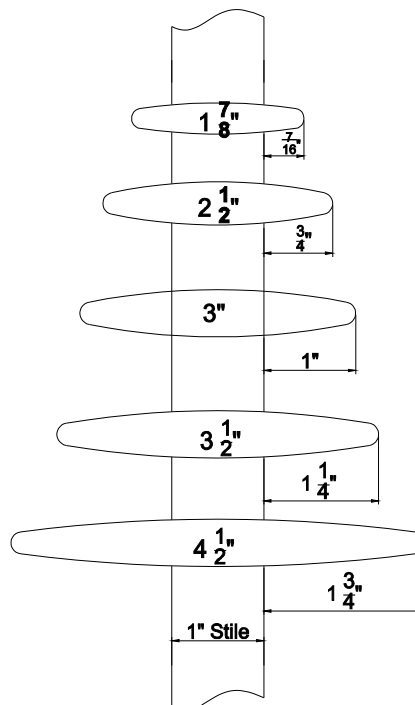
14	250	Sumatra
15	252	Toffee
16	254	Driftwood
17	255	Silver Gray
18	862	French Oak

OSMO (6 Colors)	No.	Color ID	Color Name
	1	1003	White Matte
	2	1109	Silk Gray
	3	1111	Granite Gray
	4	1203	Black
	5	1204	Ebony
	6	1502	Teak

Note: Unfinished shutters are not available.

- Please note that darker colors absorb more heat than lighter colors. As a result, shutters in darker colors that face direct sunlight for an extended period of time may absorb more heat, which may potentially affect the long-term performance of the shutter.

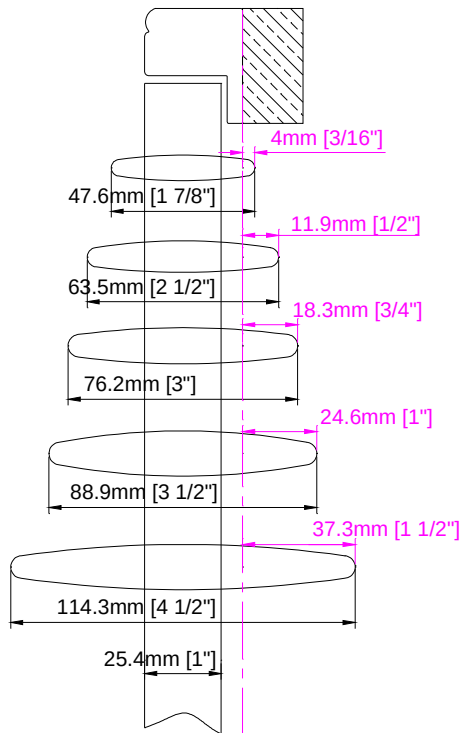
Louvers	
Elliptical	1 7/8"
	2 1/2"
	3"
	3 1/2"
	4 1/2"



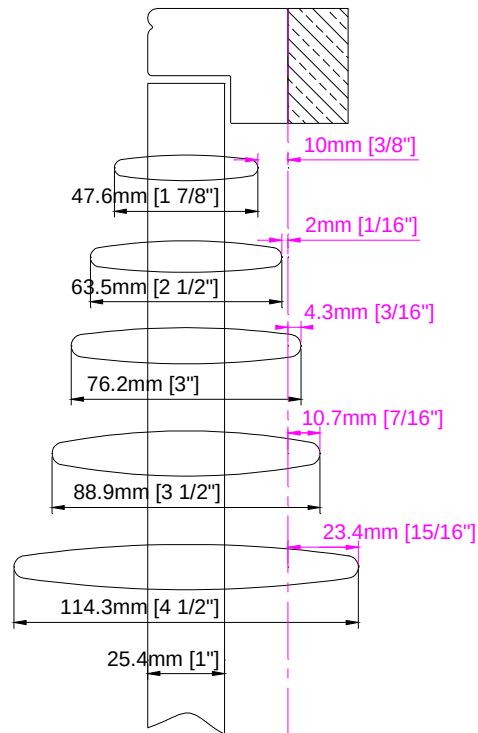
*Some surcharges apply

ND b- 2 / 5

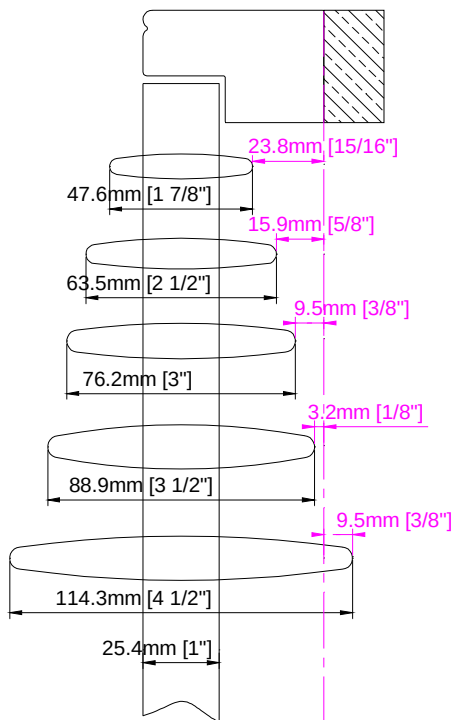
Louver Depth Clearance For Specific Frames



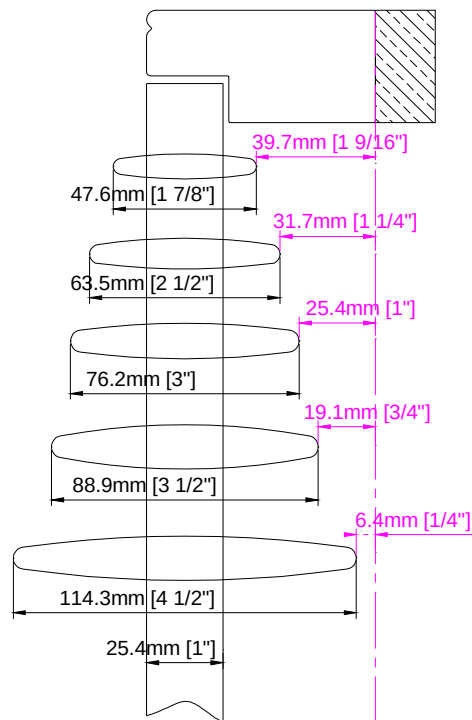
L Frame with 1/4" light block



L Frame



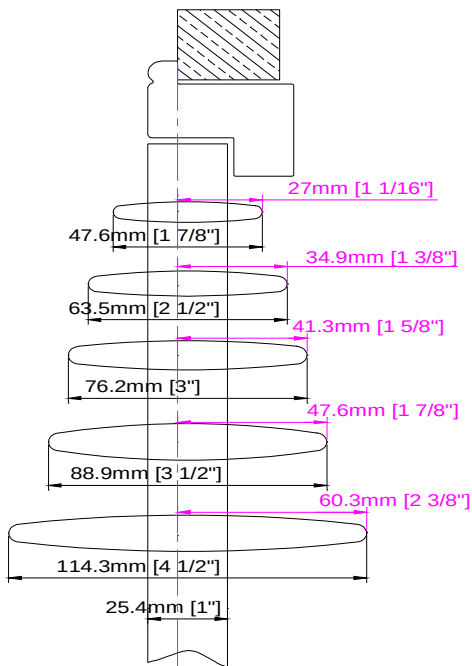
L Frame with 1/2" Buildout



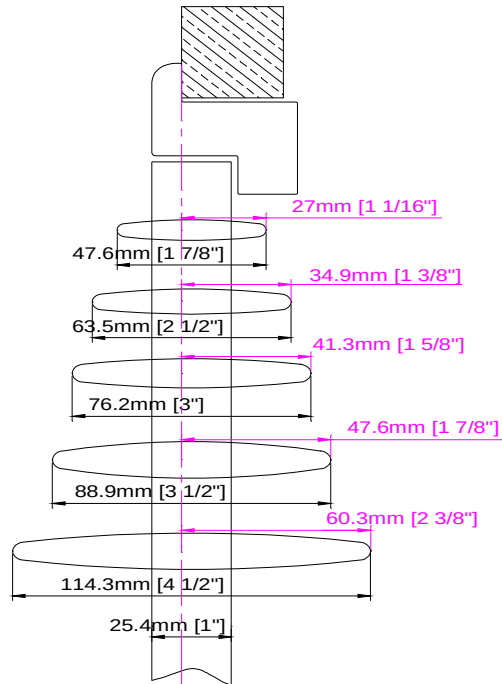
L Frame with 1" Buildout

*Some surcharges apply

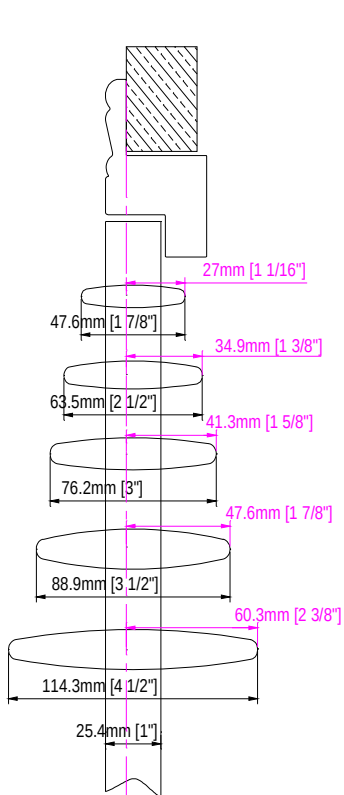
Normandy® Shutters



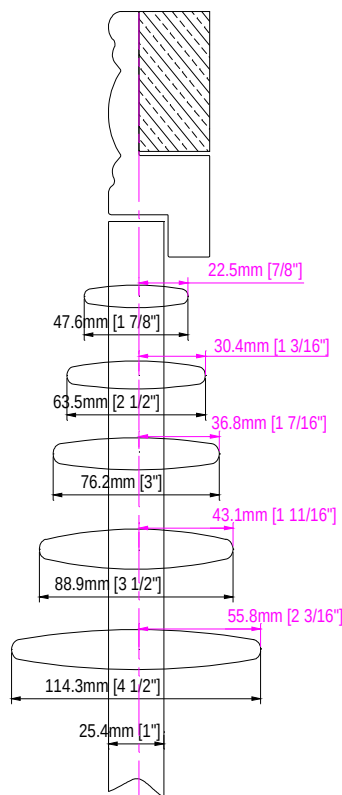
1 1/4" Beaded Z Frame



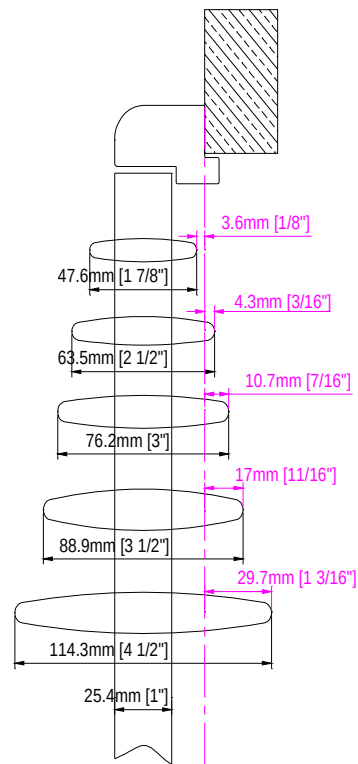
1 1/2" Bullnose Z Frame



2" Bel Air Z Frame



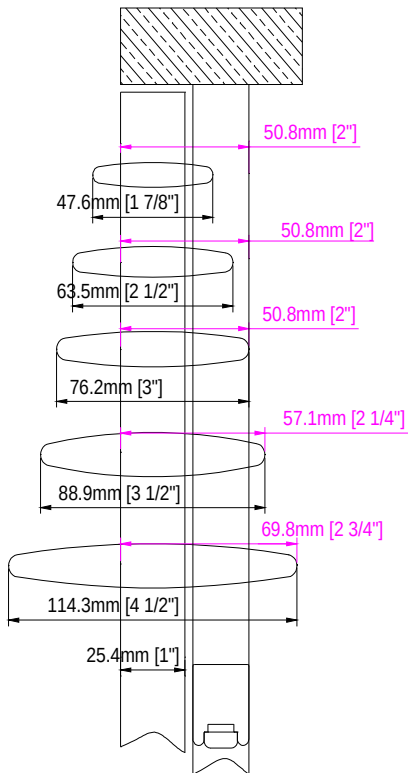
3" Crown Z Frame



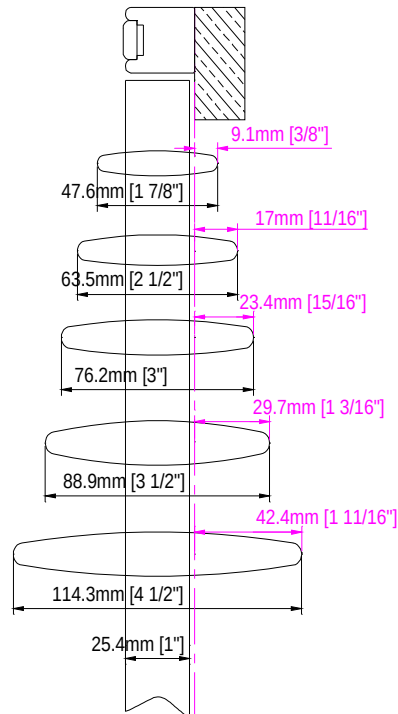
Tilt Out Z Frame

*Some surcharges apply

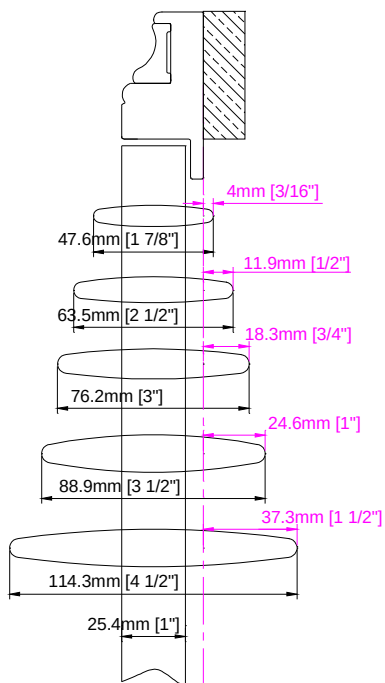
ND b- 4 / 5



Hang Strip (Behind the panel)



Hang Strip (Beside the panel)

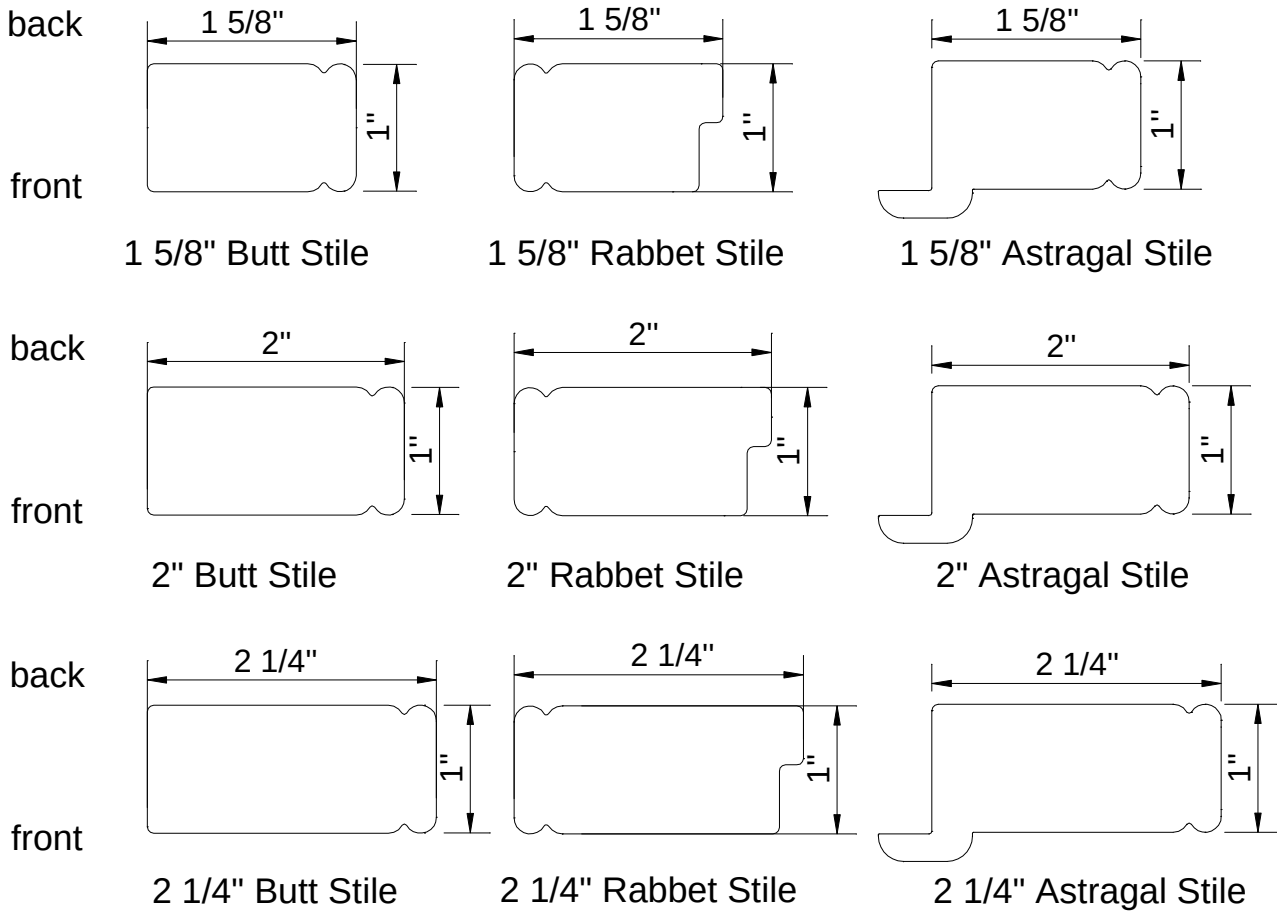


Deco Frame

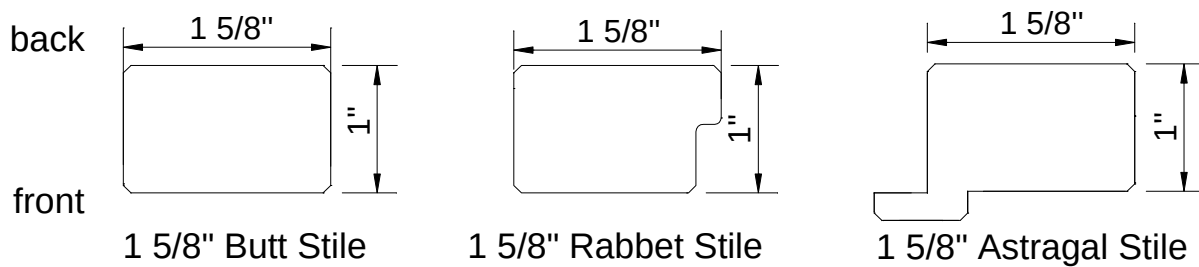
*Some surcharges apply

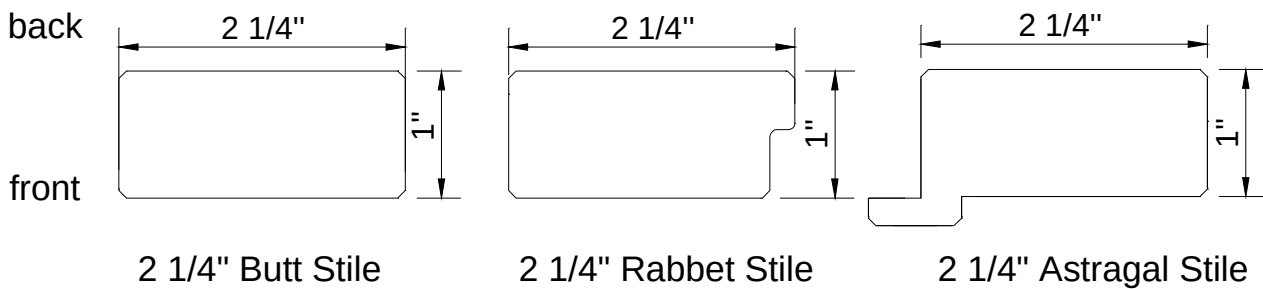
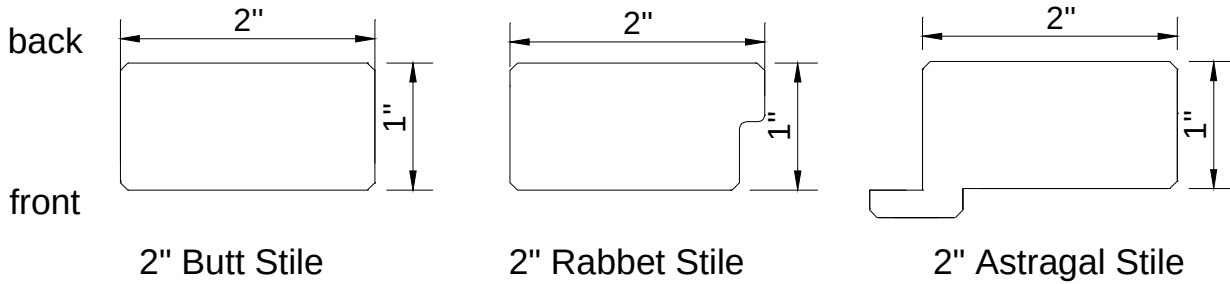
Stiles

Default: Beaded stile



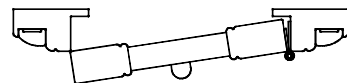
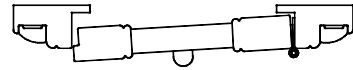
Optional: Chamfer stile





Stile between panel closure Single panel

- I) Panel width < 9"
- Rabbeted stile (Deco frame, L frame, Z frame and Hang strip beside panel)
 - 1 5/8"
 - 2" (default)
 - Butt stile only (Hang strip behind panel)
 - 1 5/8"
 - 2" (default)
- II) Panel width ≥ 9"
- Butt stile only
 - 1 5/8" (Optional for panel width ≤ 12")
 - 2" (default)

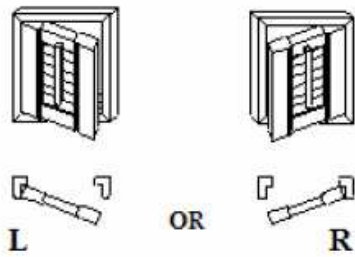


2 panels or more

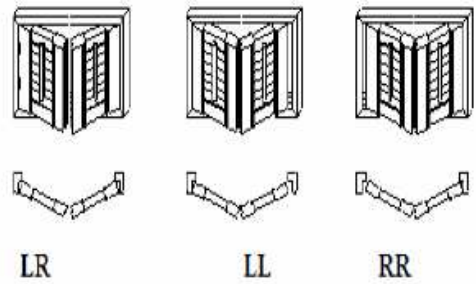
- Astragal stile
 - 1 5/8" (Optional for panel width ≤ 12")
 - 2" (default)
- Rabbet stile
 - 1 5/8" (Optional for panel width ≤ 12")
 - 2"

Panel Configurations

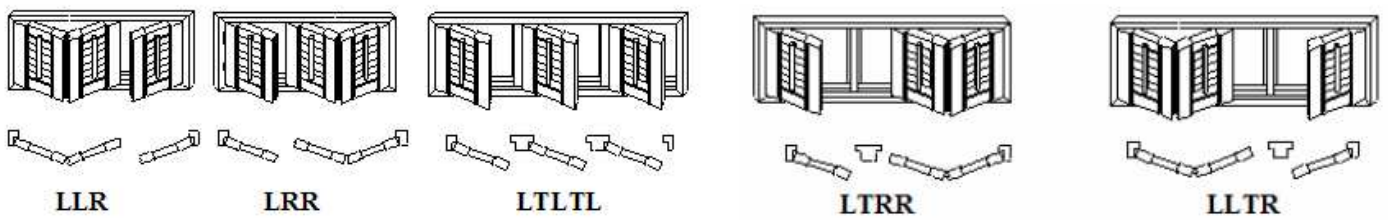
Single Panel



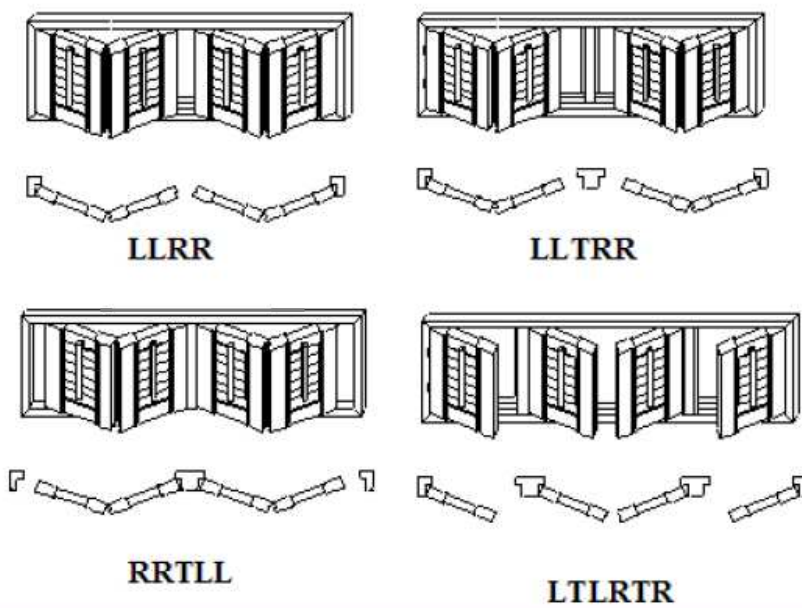
2 Panels



3 Panels

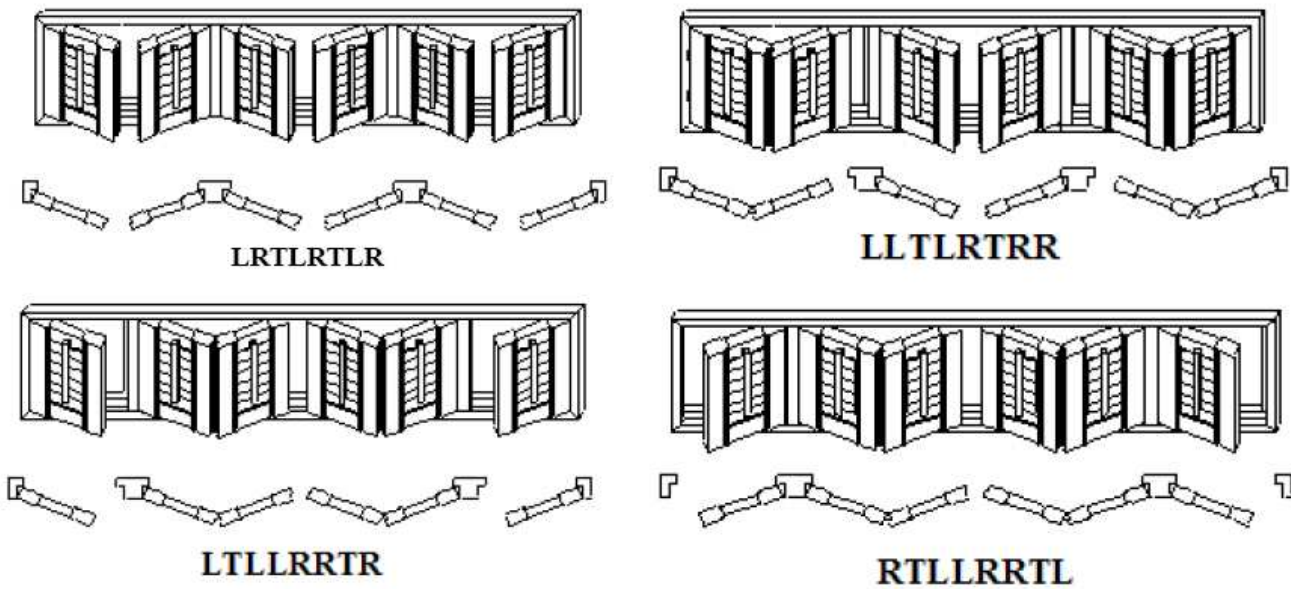


4 Panels



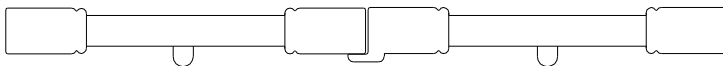
Normandy® Shutters

6 Panels

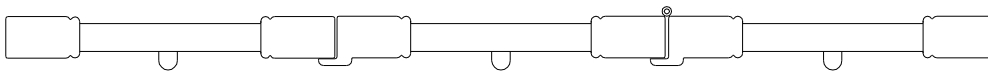


Panel Configurations – Astragal Stile

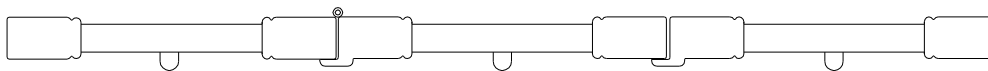
○ 2 Panels – LR



○ 3 Panels - LRR



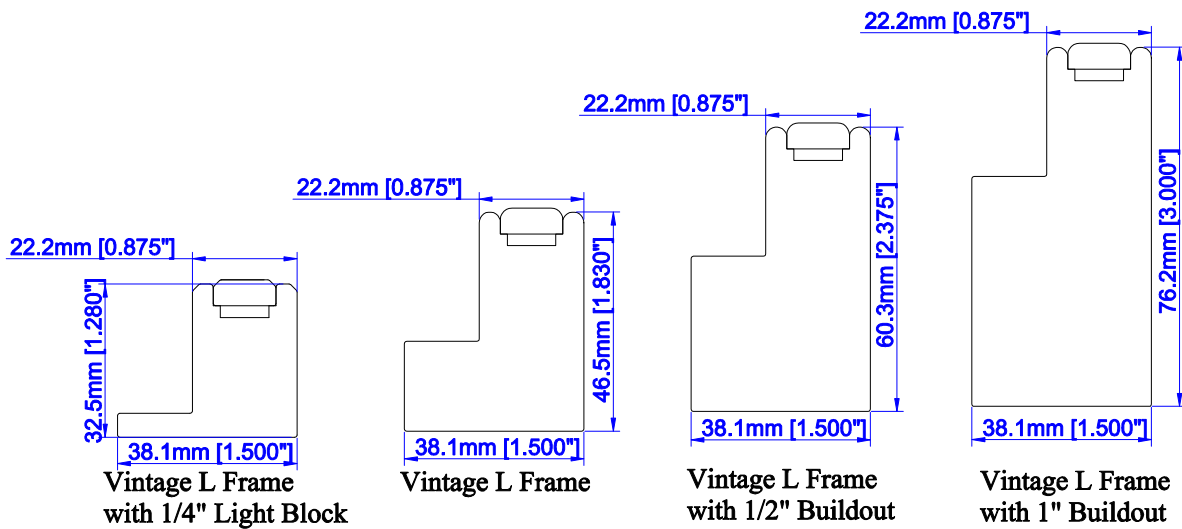
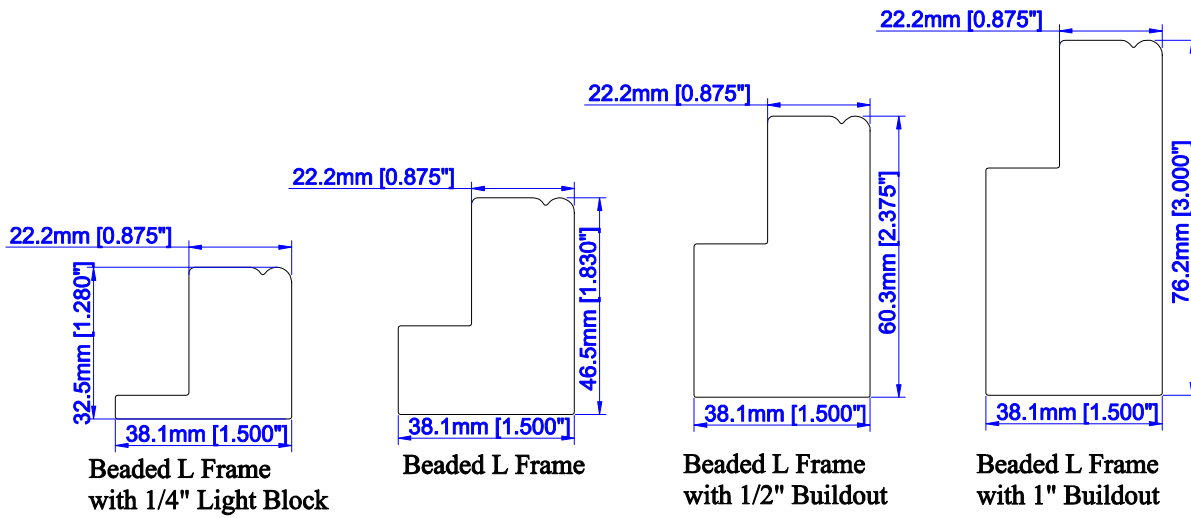
○ 3 Panels – LLR



○ 4 Panels - LLRR

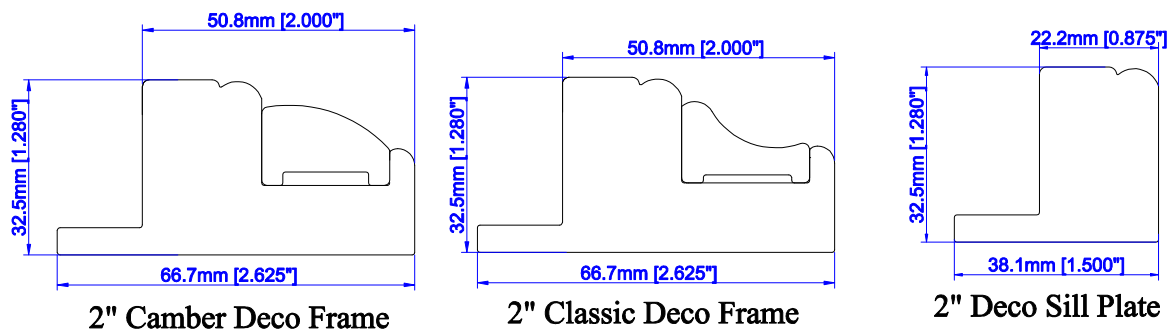


Frame Styles

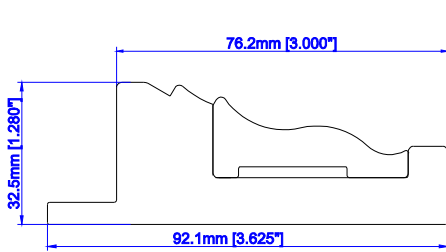


Also available

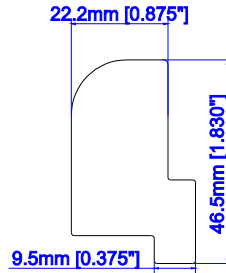
- Custom extensions* up to 2" for L frames



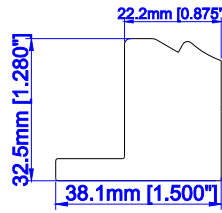
Frame Styles



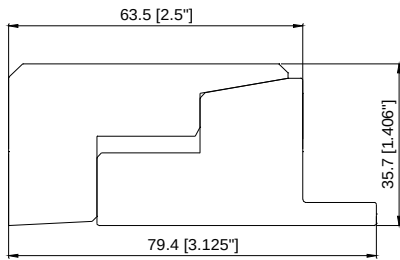
3" Ridge Deco Frame



Tilt Out Z Frame



Ridge Deco Sill Plate

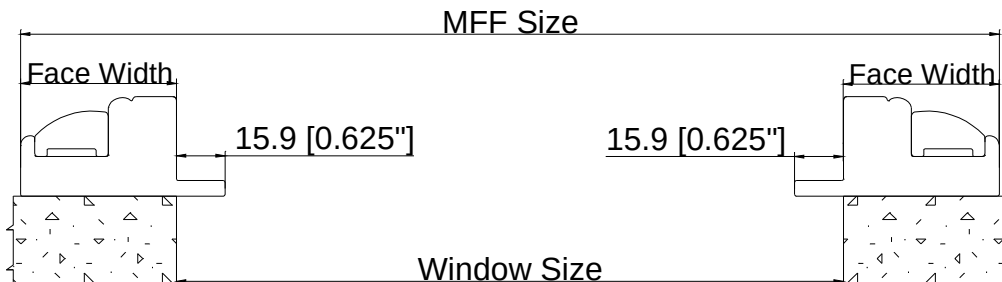


2.5" Mission Deco Frame

There are 2 options for the Deco frame installation when Window Size is given (Take Camber Deco frame as an example)

1. Default: light block part install into the window opening;

The window size is the exact size of the window opening;
 The MFF size is the measurement of the entire shutter, including the frames;
 The MFF size = Window Size + Face Width

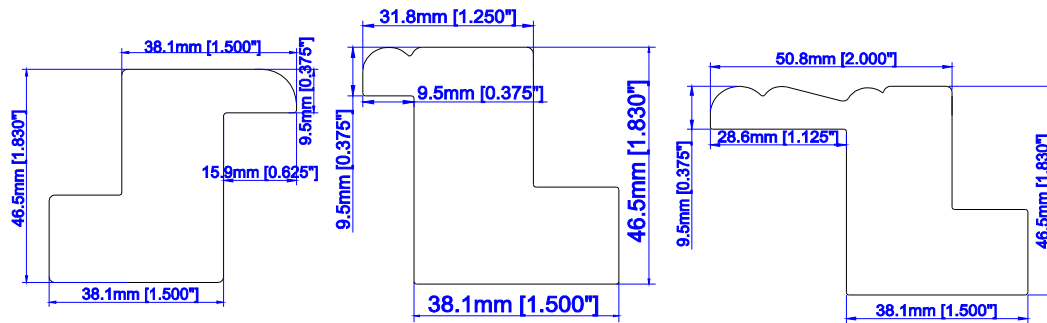


2. Optional: light block part flush with window opening.

The window size is the exact size of the window opening;
 The MFF size is the measurement of the entire shutter, including the frames;
 The MFF size = Window Size + Frame Back Width



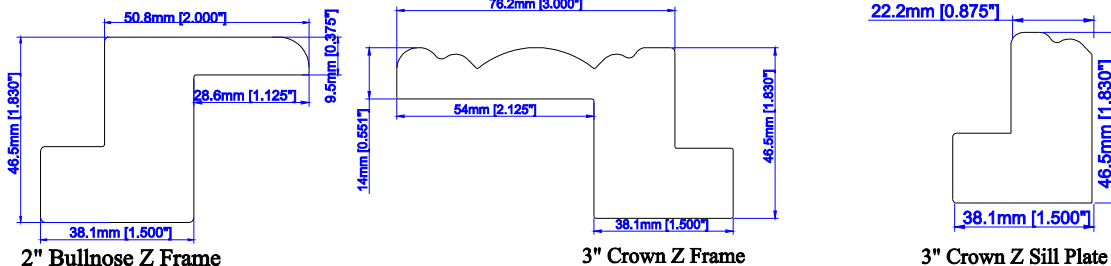
Frame Styles



1 1/2" Bullnose Z Frame

1 1/4" Beaded Z Frame

2" Bel Air Z Frame



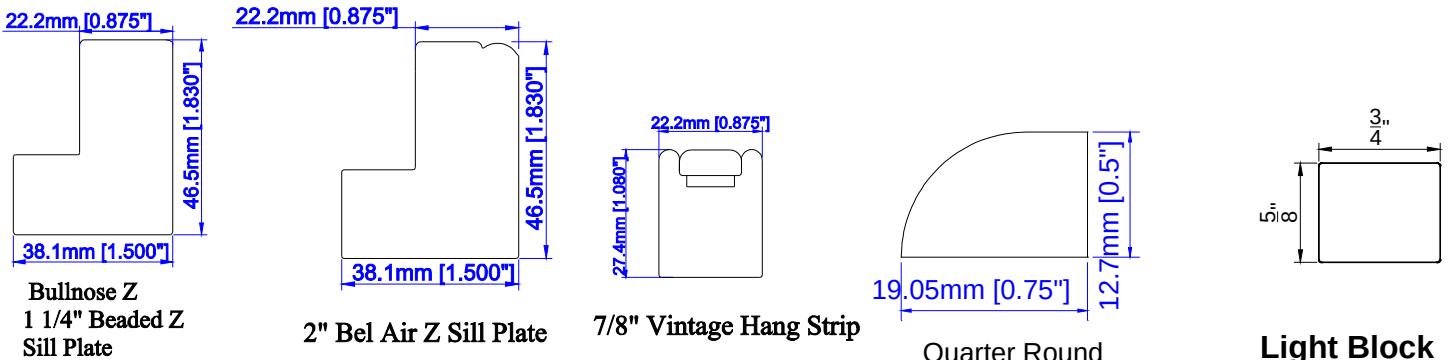
2" Bullnose Z Frame

3" Crown Z Frame

3" Crown Z Sill Plate

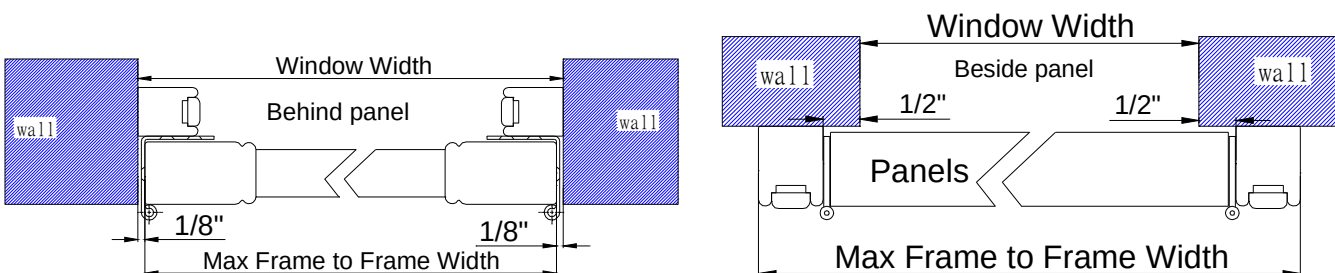
Also available

- Custom extensions* up to 1" for Z frames
- 1/2", 1" and 1 1/2" standard extensions* and custom extensions* up to 2" for Deco frames
- Custom extensions* up to 2" for hang strips (OM)



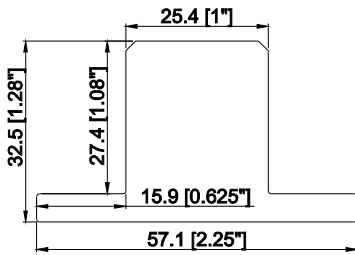
Quarter Round

Light Block

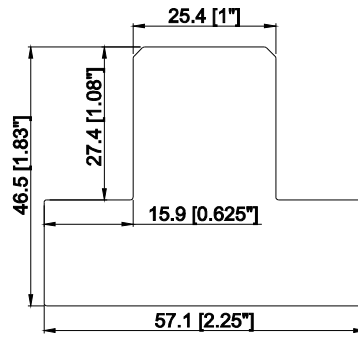


Note: Outside mounted hang strips, behind the panel, is not available for all 4 sides ("all around") or 3 sides inverted.

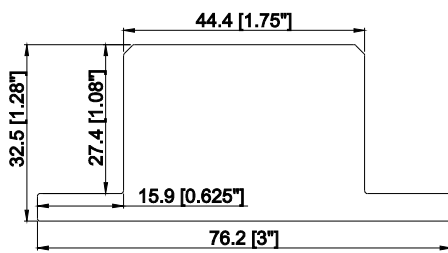
Frame Styles



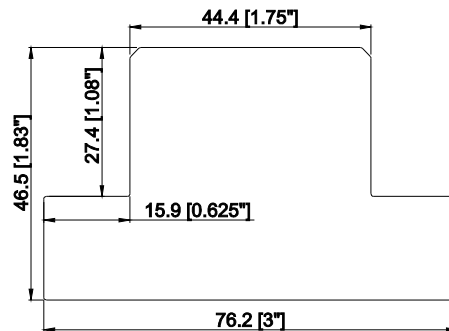
1" Chamfer T Post with 1/4" Light Block (Deco)



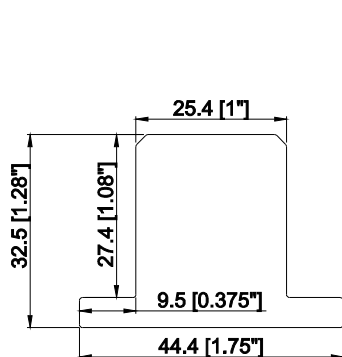
1" Chamfer T Post with 3/4" Light Block (L/Z/HS)



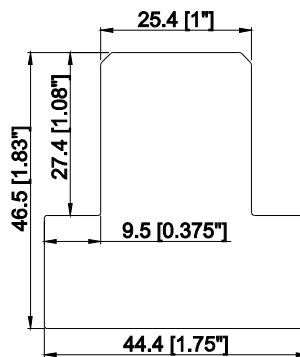
1.75" Chamfer T Post with 1/4" Light Block (Deco)



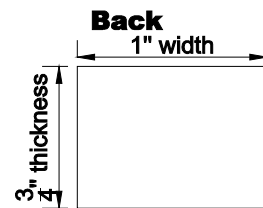
1.75" Chamfer T Post with 3/4" Light Block (L/Z/HS)



1" Chamfer T Post with 1/4" Thick x 3/8" Wide Light Block (Deco)



1" Chamfer T Post with 3/4" Thick x 3/8" Wide Light Block (L/Z/HS)



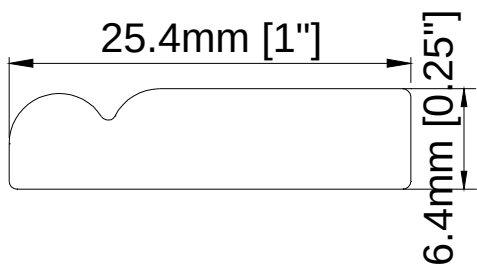
Back
1" width
Front
3/4" thickness
1" x 3/4" Post (No light block)

Note: 1" Chamfer T Post with 3/4" Light Block, 1 3/4" Chamfer T Post with 3/4" Light Block, 1" Chamfer T Post with 3/4" Thick x 3/8" Wide Light Block and Custom Chamfer T Post with 3/4" Thick Light Block are not available for all Deco frames and L frames with 1/4" Light Block.

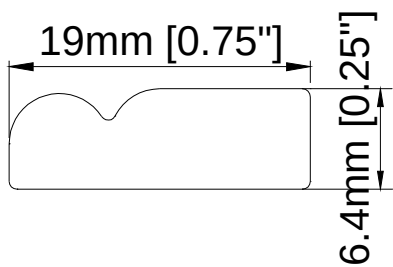
- Custom depth* up to 2" / width* up to 7½ for T Posts
 - Size limits (7 1/2")
 - That a surcharge applies and customers should refer to their shutter pricing
 - Any limitations

Note: The 1"x3/4" Post (No Light Block) is only available for Direct Mount. And no hardware will be supplied to install it on the existing window opening.

Frame Styles

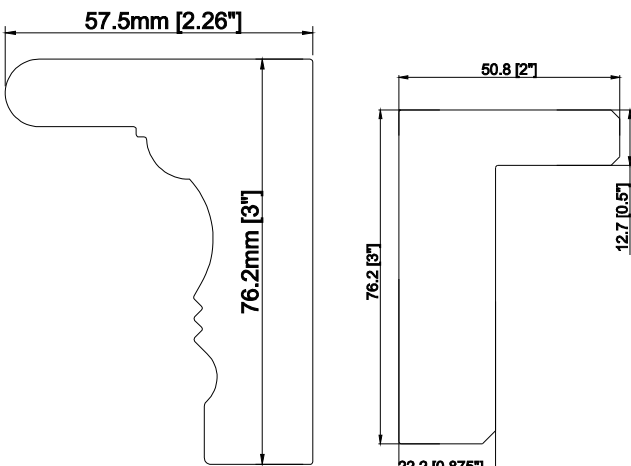


Scribe Trim 1



Scribe Trim 2

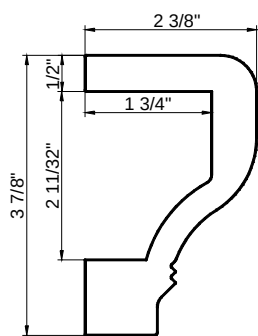
Sill Frames- For use where there is no existing sill



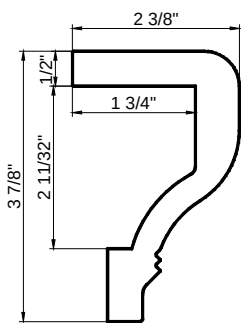
3" Deco Sill Frame

Mission sill Frame

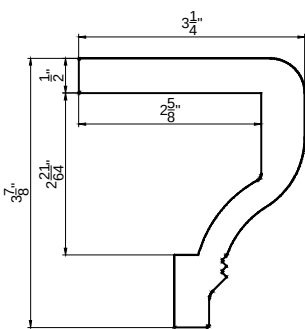
Sill caps-Used to cover existing sills



Designer Sill Cap (L)



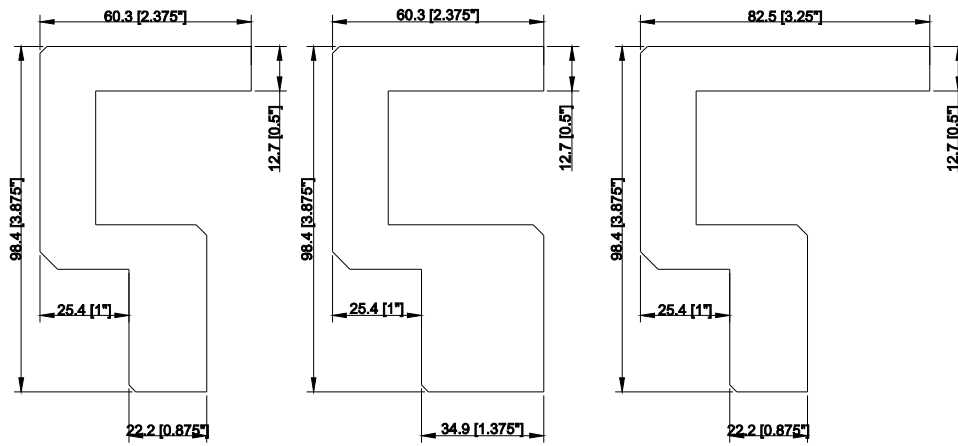
Designer Sill Cap (D/HS)



Designer Sill Cap (Z)

*Some surcharges apply

Frame Styles



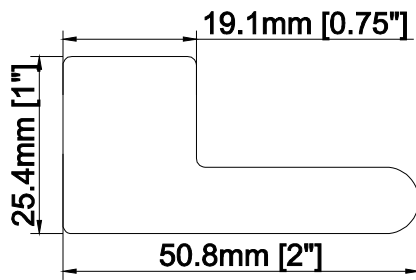
Mission Sill Cap (D/HS)

Mission Sill Cap (L)

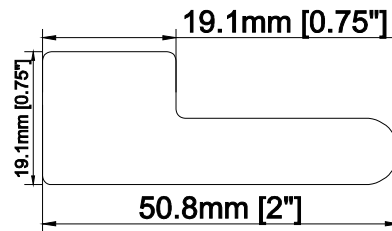
Mission Sill Cap (Z)

Note: For Mission Sill Cap, the extended size at each side needs to be 1" at least.

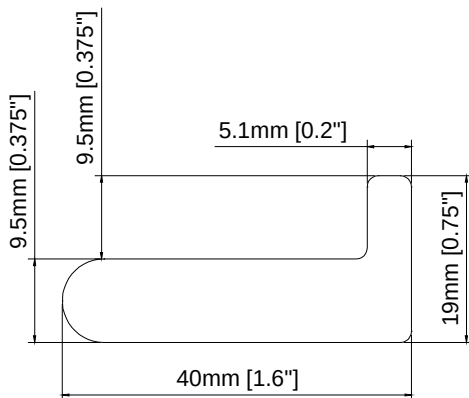
Sill Plates- Used with existing sills



**2" Sill Plate with 5/8"
Light Block (L/Z)**

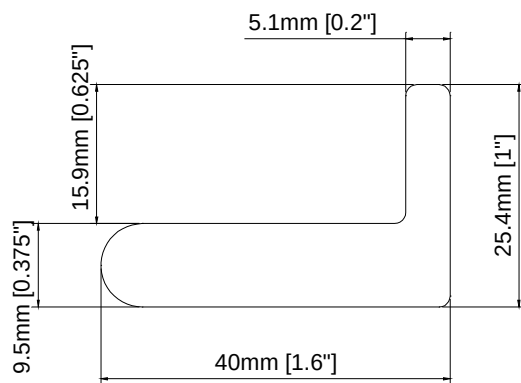


**2" Sill Plate with 3/8"
Light Block (L/Z)**



1.6" Sill Plate with 3/8" Light Block

(All Deco & L frames with 1/4" light block)

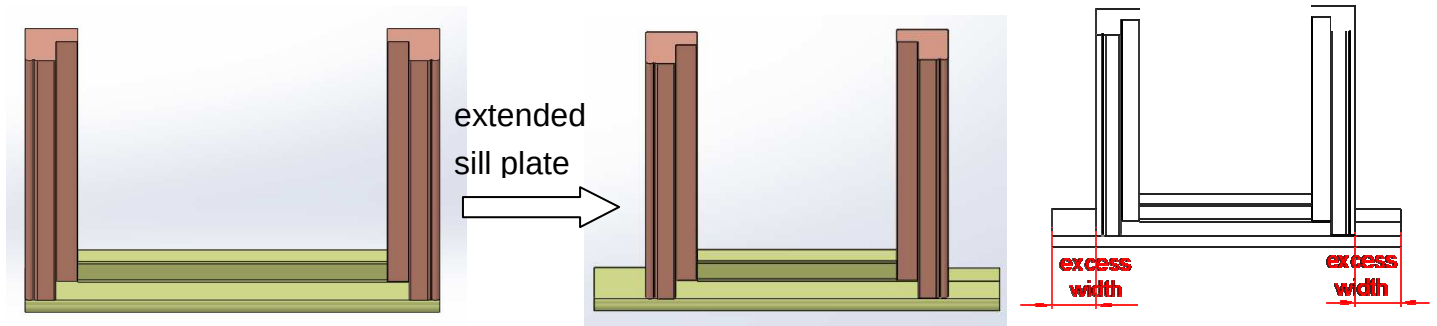


1.6" Sill Plate with 5/8" Light Block

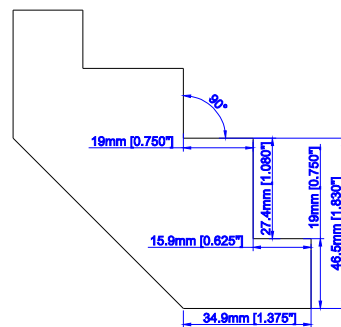
(All Deco & L frames with 1/4" light block)

Frame Styles

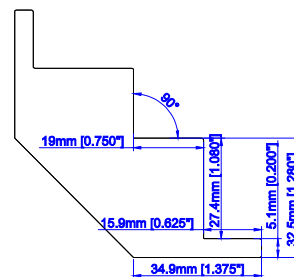
For 1.6" / 2" sill plate with 5/8" (or 3/8") light block, when an extended sill plate is requested, the excess width will be divided evenly on both sides, take the following drawing as an example for reference:



Standard Corner & Bay Posts

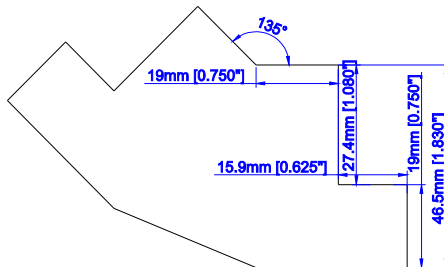


Corner Post with 3/4"
Light Block (L/Z/HS)

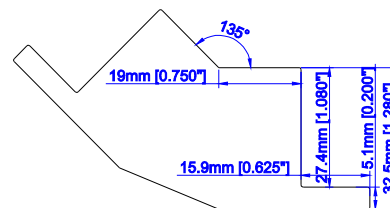


Corner Post with 1/4"
Light Block (Deco)

*Notch out corner post required if ordering Z Frame with solid drywall corner.



Bay Post with 3/4"
Light Block (L/Z/HS)



Bay Post with 1/4"
Light Block (Deco)

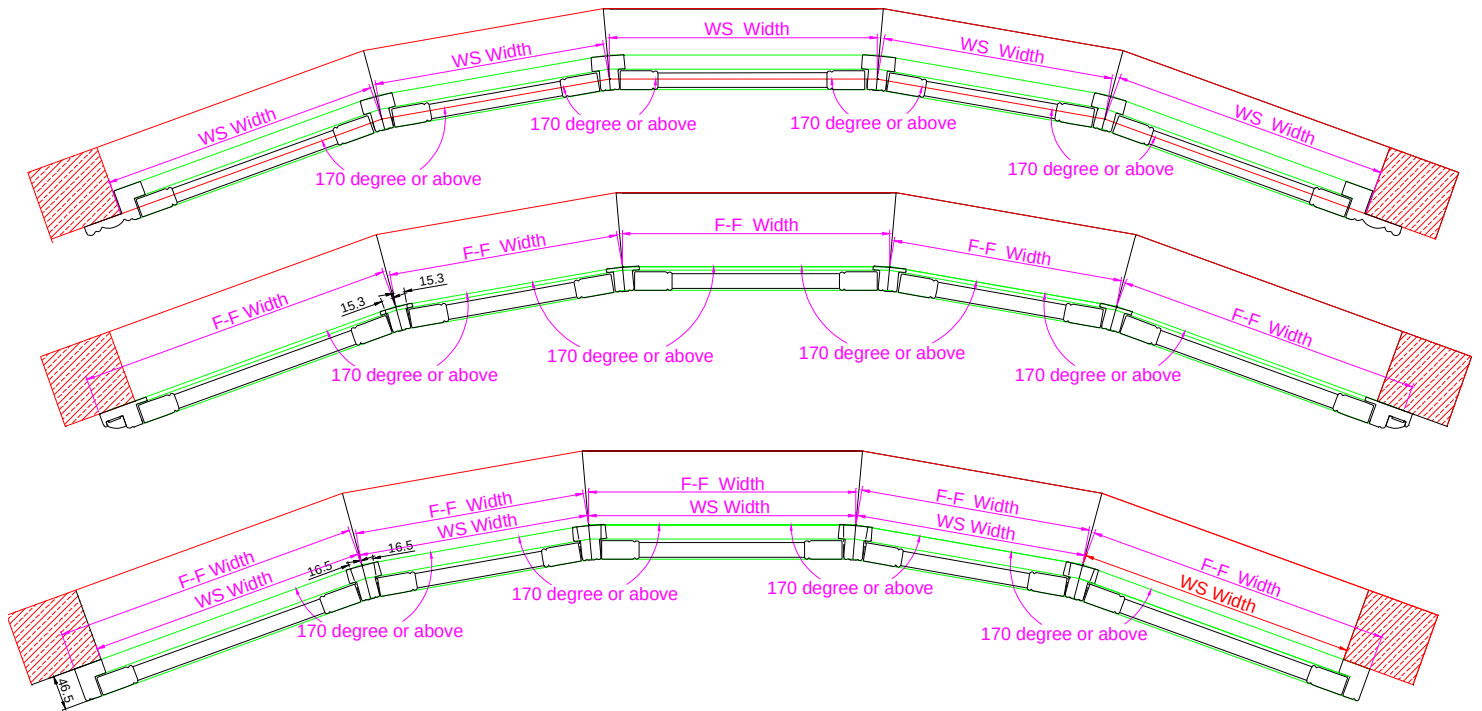
Also available

- Custom angle bay posts *

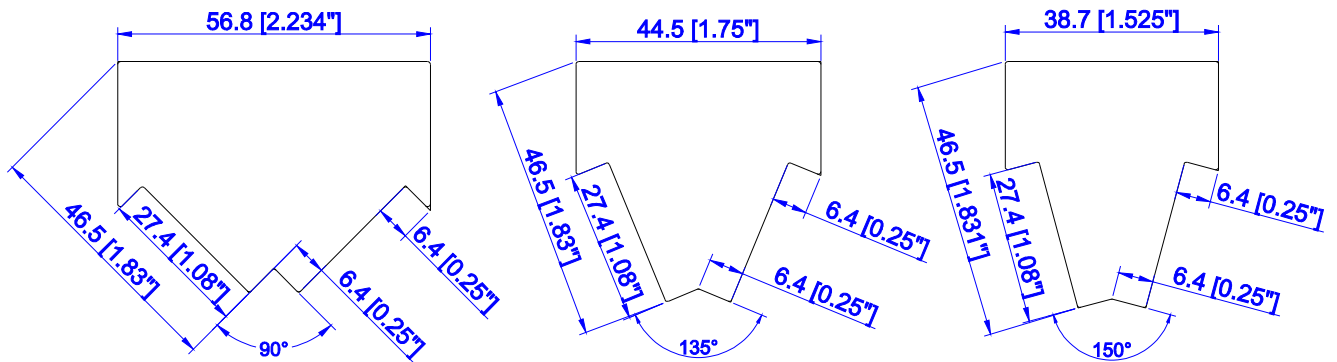
Frame Styles

For bay window at 170 degree or above,

- 1" T Post with 3/8" light block will be used as Bay post;
- For Hang Strip (Behind Panel), bay post will be used.



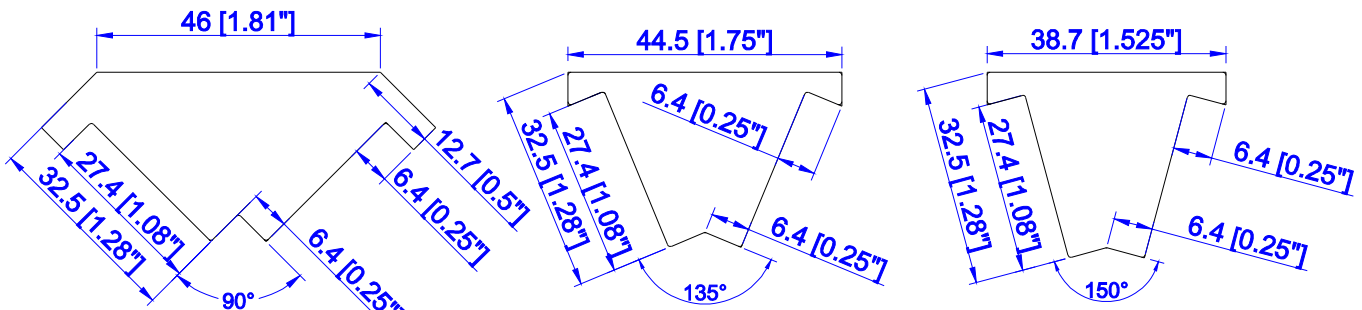
Slim Fit Corner & Bay Posts



Slim Fit Corner Post for 90 Degree (L/Z/HS)

Slim Fit Bay Post for 135 Degree (L/Z/HS)

Slim Fit Bay Post for 150 Degree (L/Z/HS)



Slim Fit Corner Post for 90 Degree (D)

Slim Fit Bay Post for 135 Degree (D)

Slim Fit Bay Post for 150 Degree (D)

Note: Due to size and weight limitations, Slim Fit Posts may only be hinged on one side and with a single panel (no bi-folded panels). Please consider this when determining panel configuration.

- Slim Fit Posts are available in 90, 135 and 150 Degrees only.

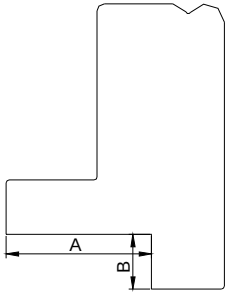
Frame Styles

In order to better match corner windows / bay windows:

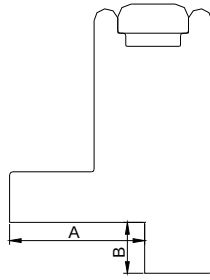
- **For non sample orders, please note that corner windows must have the same:**
 - Post Type (Standard or Slim Fit)
 - Color
 - Hinge Options
 - Louver Size
 - Frame Type
 - Mount Type
 - Extension
 - Sill Options
 - T-Post Options
 - Height
 - Stile Type and Profile
 - Tilt Type
 - Divider Rail Options
 - Double-Hung Options
 - Split Tilt Options
 - PerfectTilt Options

- **For non sample orders, please note that bay windows must have the same:**
 - Post Type (Standard or Slim Fit)
 - Color
 - Hinge Options
 - Louver Size
 - Frame Type
 - Mount Type
 - Extension
 - Sill Options
 - T-Post Options
 - Height
 - Stile Type and Profile
 - Tilt Type
 - Divider Rail Options
 - Double-Hung Options
 - Split Tilt Options
 - PerfectTilt Options

Frame Styles

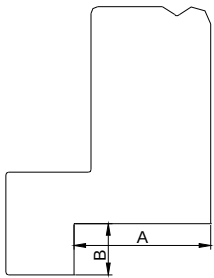


Notch Out
Beaded L Frame (A) *

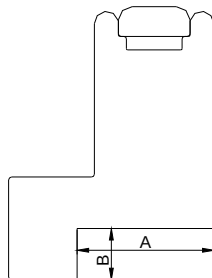


Notch Out
Vintage L Frame (A) *

Limitation for the Notch Out Size:
 $A \leq 1"$; $B \leq 1/2"$

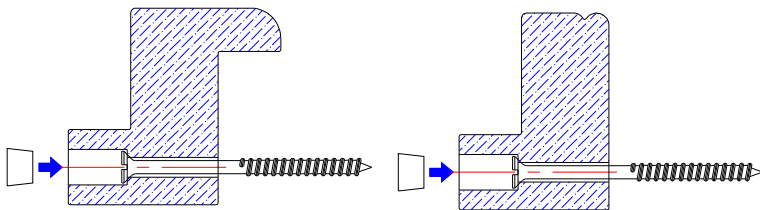


Notch Out
Beaded L Frame (B) *



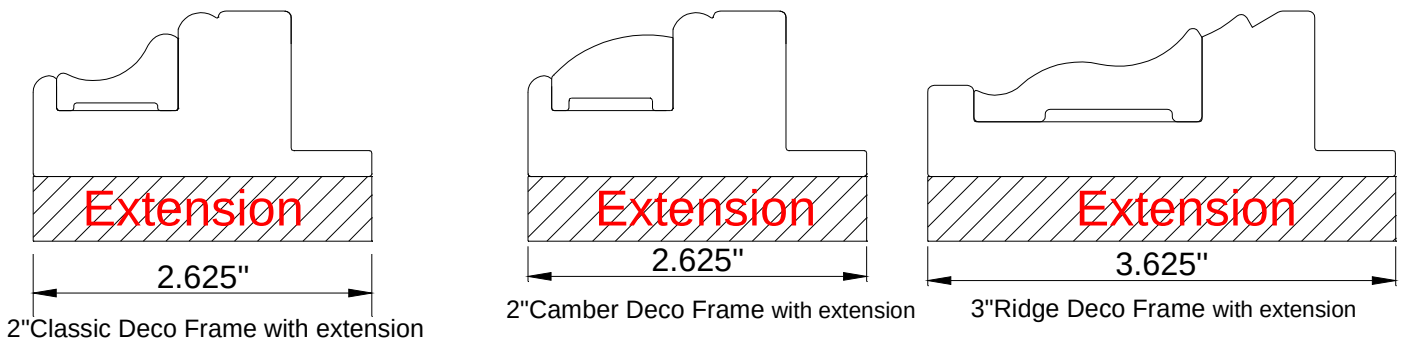
Notch Out
Vintage L Frame (B)

Pre-drill L Frames (Inside Mount Only) or Z frames:



Please note that the prefilled hole and the plug will not be a 100% complete fit, but that would not affect the performance of the screw. Shutters can't be remade solely because of this issue.

Extension for Deco Frame* (Default to have full extension):



Note: Custom extensions* are available for Deco Frames, L frames & Hang strips (OM, beside the panel) & T posts, when the extension thickness is more than 2", the extension will not be attached to the frames and/or T posts.

*Some surcharges apply

Frame Styles

Custom extensions* are available for the lips of all Z frames except Tilt out Z frames, when the extension thickness is more than 1", the extension will not be attached to the Z frames.

● Standard Filler Strips (width x depth):

1/8" x 9/16"	3/16" x 1 3/4"	1 1/4" x 3/8"
1/8" x 3/4"	3/16" x 2 1/2"	1" x 1 3/16"
1/8" x 1"	1 3/16" x 1 9/16"	1" x 1 9/16"
1/8" x 1 3/16"	1 9/16" x 1 9/16"	1" x 2"
1/8" x 2 3/8"	13/16" x 13/16"	1" x 2 3/8"
3/16" x 9/16"	3/4" x 3/4"	1" x 2 15/16"
3/16" x 1"	1" x 3/4"	2 7/16" x 7/16"

Note:

1. The width or depth of Custom filler strip / Standard filler strip can't be less than 1/8".
2. The width or depth of Custom filler strip / Standard filler strip can't be greater than 7 7/8".
3. The width and depth of Custom filler strip / Standard filler strip can't be greater than 3 15/16" at the same time.

Default principle:

Regarding shutter with horizontal T-post type issue: Horizontal T-Post type will always be the Chamfer T post unless otherwise specified.

General Product Specifications

<u>Description</u>	<u>Minimum</u>	<u>Maximum</u>
Single Panel Width (Regular)	6"	30" for 1.875" louver, 36" for 2.5" & 3" louver, 42" for 3.5" & 4.5" louvers
Single Panel Width (Bifold)	6"	26"
Single Panel Height	10"	132"
Shutter Width (Without T Post)	N/A	4x single panel width (bifold) + frame
Shutter Width (With T Post)	N/A	Unlimited (For orders shipped by sea, frames longer than 142" will be spliced in the middle. For orders shipped by air, frames longer than 120" will be spliced in the middle.)
Shutter Height	N/A	132" + frame

Occasional finger joint line may be visible in the frames and stiles.

Divider rail is required for panel heights over 78".

84" panel height limit without divider rail is for shutter types below:

- Normandy French Door with OR without Cut-out (including type E&F), which should be installed on existing door, solid glass or sidelights instead of window opening.
- Normandy Bypass and bi-fold track shutters.

Standard divider rail size is 3", Custom Size (3"~7.875") Divider Rail is available with surcharge. Please note the standard top/bottom rail size should be within 2.75"~6.5". For regular shutters, the step for the rail size is 1/8".

Material: Solid Color: Made from Hardwood; Stain Color: Made from Phoenix wood.

Shutters to be installed where there is no bottom supports are not available.

Shutters without bottom supports may sag and deform, including:

1. Shutters with a bottom gap over 0.1";
2. Shutters with no frame or existing bottom sill to support the panel (or shutter has bottom frame/existing sill, but the panel is frequently left in the open position).

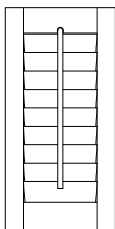
Side by Side / Match Line Number:

When **Side by Side** is selected, it is solely to match top or bottom rails and the number of hinges will also be matched when the panel heights are within +/- 2". When selected, the option to match with another line number or previous work order is solely for matching the top or bottom rails. Neither option will be used to match any other specifications.

Tilt Rods

Standard Tilt Rod

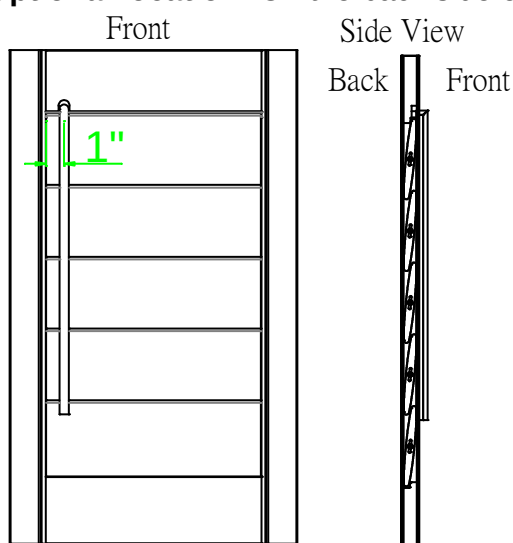
Default Location: In the middle of the panel on front side (default).



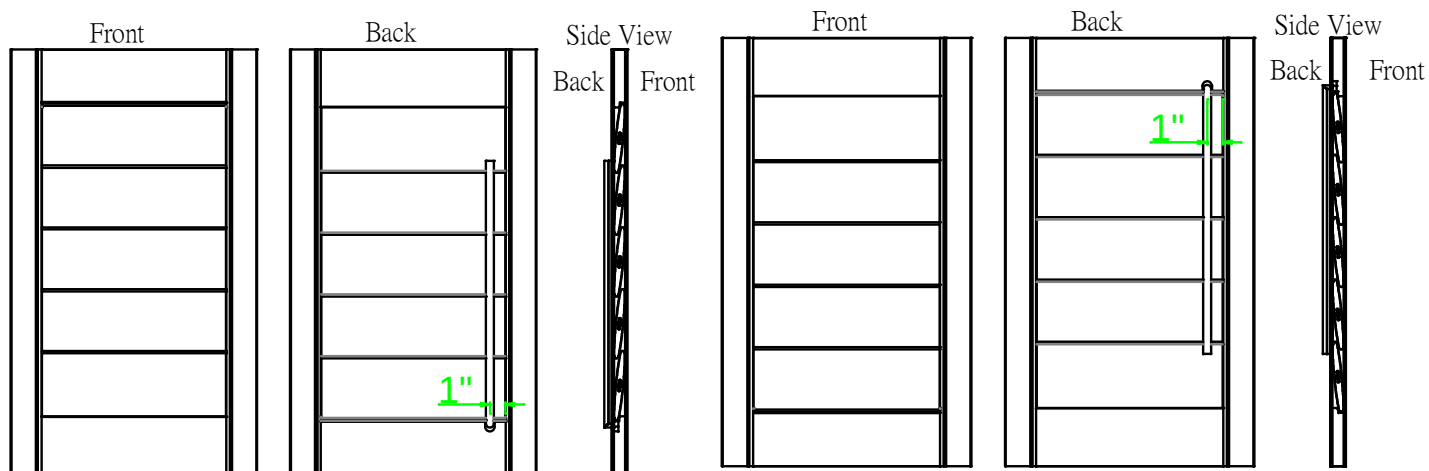
Offset Tilt Rod*

Default Location: On the front side of the panel and at 1" from the louver ends on the hinged side.

Optional location: On the back side of the panel, 1" from the end of the louvers on the hinged side.

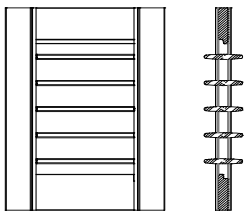


Offset tilt rod on the front side of panel,
the louvers dress up from front view



Offset tilt rod on the back side of panel,
the louvers dress up from front view

Offset tilt rod on the back side of panel,
the louvers dress down from front view

InvisibleTilt™ *

Louver tilting system embedded within panel stile. It is not available for $1\frac{7}{8}$ " louver. The tilting system is split when the number of louvers exceeds the limits as listed below:

Louver Size	Maximum Possible Louvers	Panel Height Range
2½"	24	54 ¾" – 57 ¼"
3"	19	54 ¼" – 57"
3½"	16	54 ¾" – 58"
4½"	12	54 ¾" – 60"

For even numbers of louvers, the number of louvers operated by the top and the bottom tilting system will be the same;

For odd numbers of louvers, the number of louvers operated by the bottom tilting system will be one more than the number of louvers operated by the top tilting system;

To ensure split tilt is included, always specify on the order.

Split Tilt

When selected, the louvers are split evenly by default unless a custom location is requested.

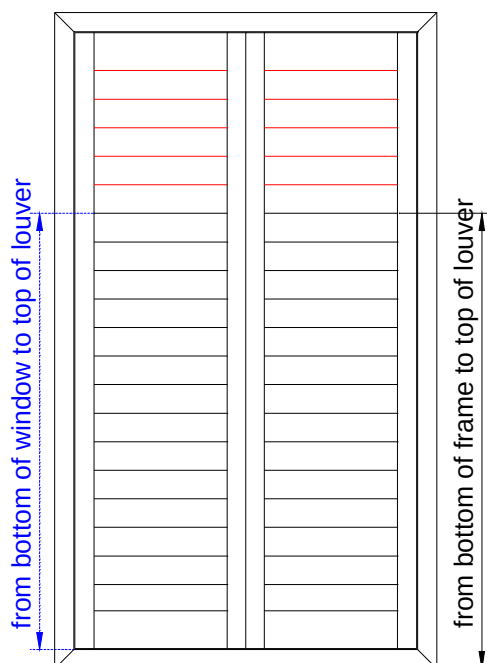
When the customer selects a custom location, the measurement will be as follows (please note that the louvers are in the closed position when measured):

Window size:

— Measure from the bottom of the window to the top of the louver

Max frame-to-frame:

— Measure from the bottom of the frame (or panel if there is no bottom frame, sill or sill cap) to the top of the louver.



Split tilt allowable deviations

Louver Size	Allowable $\pm$ Deviation
1 7/8"	3/4"
2 1/2"	1"
3"	1 1/4"
3 1/2"	1 1/2"
4 1/2"	2"

Split Location:

— Per the customer's specification. However, the location of the top louver of the bottom split section may be adjusted to achieve more balanced top and bottom rails when **allow deviations** is selected during order entry. If necessary the location will be moved up from the desired height to ensure any obstruction is covered. The amount of the deviation from the desired height will depend on the size of the louver selected.

If the exact location of the split is critical, customer may select **exact location** in the order entry system when placing the order. This may result in significantly different top and bottom rails. When exact location is selected, a popup will reveal the resulting top and bottom rails measurements from which the customer may either:

1. Accept the listed top and bottom rail sizes
2. Modify the split location (sometimes a slight adjustment in the exact split location may achieve desired top and bottom rail sizes)
3. Change to **allow deviations** so the system will calculate the split location

Note: Top and bottom rail sizes may change if the current line item is being matched with another line item. Carefully review all top and bottom rail sizes and split locations to ensure a desirable look is achieved

- Please note that Tilt options, Split Tilt, Divider Rail, Double Hung, and PerfectTilt™ motorization are not available for panels with one louver. In addition, there is no mouse hole on the rails.

Divider Rails

A divider rail is an extra rail in the middle of the panel.
Creates two sections in the panel so louvers can be operated independently.
Strengthens the panel and enhances structural integrity.

Divider rail is required for panel heights over 78".

Custom Size(3"~7.875") Divider Rail is available with surcharge.

84" panel height limit without divider rail is for shutter types below:

- Normandy French Door with OR without Cut-out (including type E&F), which should be installed on existing door, solid glass or sidelights instead of window opening.
- Normandy Bypass and bi-fold track shutters

Specifications

Standard size: 3"

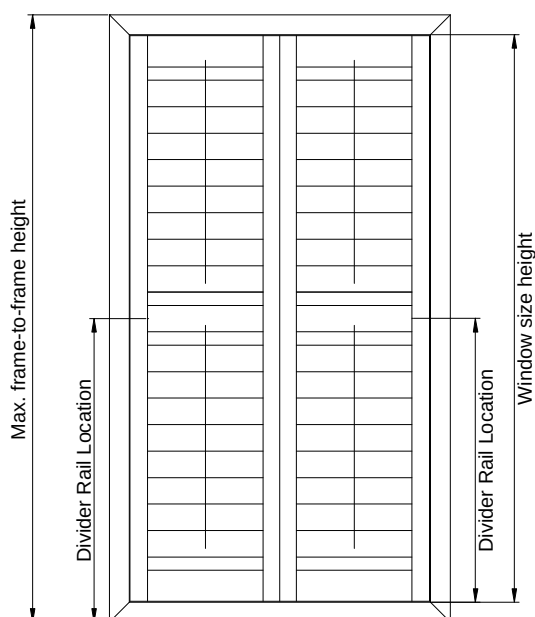
Default location: center of the panel unless specified otherwise.

If the Divider Rail Location must be accurate, "Exact Location" must be selected or specified when placing an order.

Measurements:

- Window size:
 - Measure from the bottom of the window to the center of the divider rail.
- Max frame-to-frame:
 - Measure from the bottom of the frame to the center of the divider rail.
- Position:
 - Per customer's specification. However, the divider rail may be adjusted to a lower or higher position, within our allowable deviation, to ensure reinforcement and aesthetics of the shutter.

When placing the order, please clearly specify if the divider rail position needs to line up with that of other shutter divider rails in the same room. Please also refer to the table below for the allowable deviation details.
- Between 2 divider rails / 2 split tilt rod locations / 1 divider rail location and 1 split tilt location, there needs to have at least two or more louvers for non-motorization shutters.



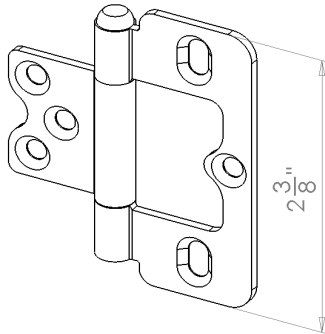
Divider rail allowable deviations

Louver Size	Allowable + Deviation
1 7/8"	3/4"
2 1/2"	1"
3"	1 1/4"
3 1/2"	1 1/2"
4 1/2"	2"

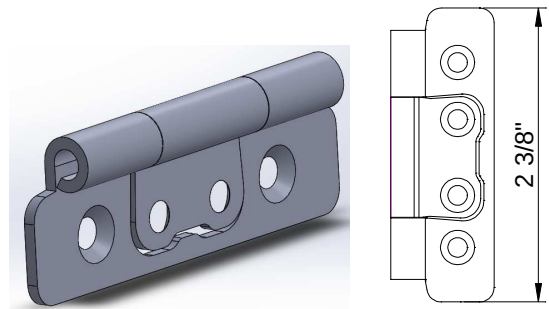
Hardware

Hinges

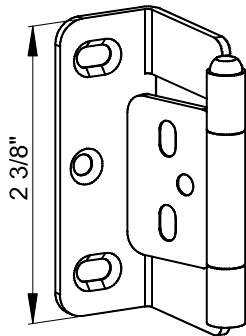
2.375" Self-Mortise Hinge



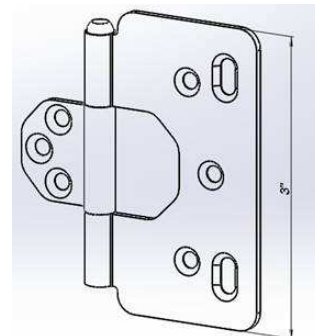
2.375" Bi-fold Hinge



2.375" L Hinge



Direct Mount 3" Hinge



Hinges are available in:

Pure White, Silk White, Bisque, Pearl, Bright Brass, Antique Brass, Black, Nickel plated, Brushed Nickel, Stainless Steel*, Crisp Linen, String, Sea Mist, Stone Gray, Brown Gray, Taupe Gray. While hinge colors of Crisp Linen, String, Sea Mist, Stone Gray, Brown Gray, Taupe Gray are not available for Direct mount 3" Hinges.

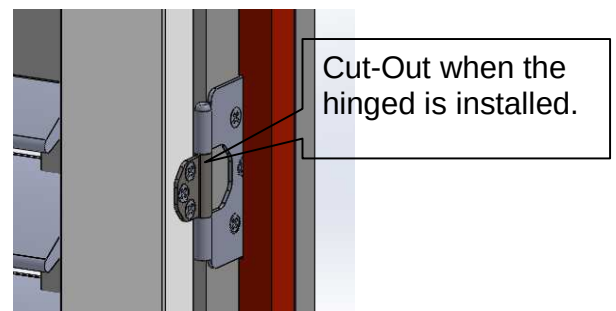
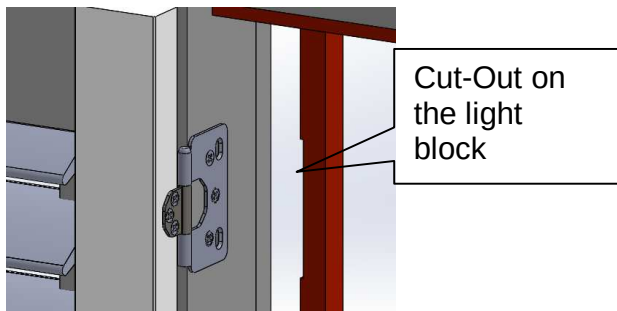
False hinges are not available.

Direct Mount 3" Hinge:

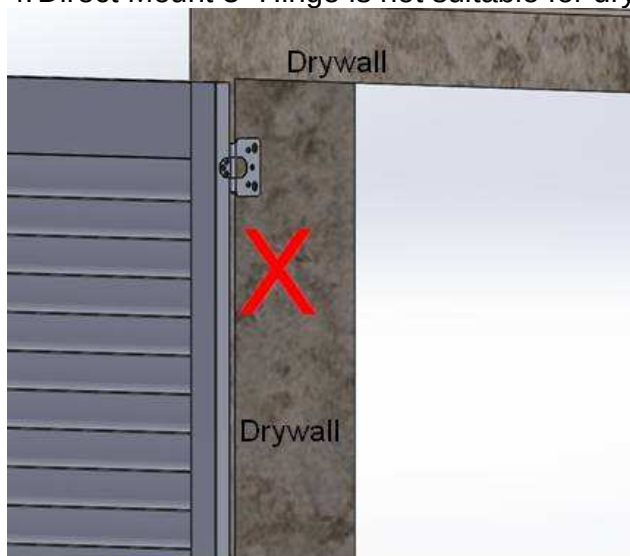
1. The hinge position of the direct mount 3" hinge will be different from that of the 2.375" ones. For shutters used both the 2.375" and the direct mount 3" hinge, the hinge position of the 2.375" ones will match that of the direct mount 3" hinge;
2. The hinge is only to install between the panel and the existing window. For other situation, 2.375" hinges will be used;

*Some surcharges apply

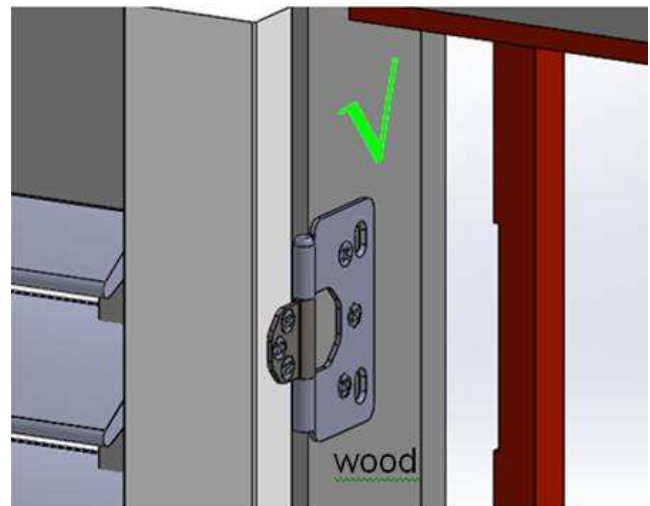
3. To avoid confliction, there will be cut-out for the light block of the hinged side. Please refer to the drawing below.



4. Direct Mount 3" Hinge is not suitable for drywall installations.



Direct Mount 3" hinge is not suitable for drywall installations.
It's not recommended.

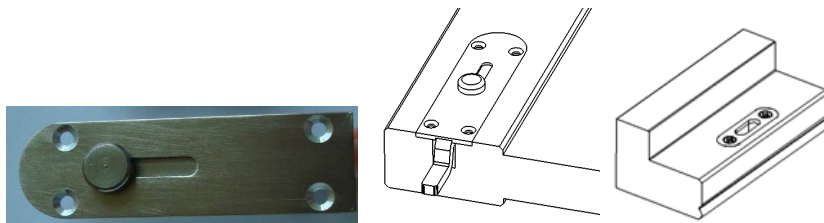


It's acceptable.

Panel Lock: (not available for 1.625" Stile)

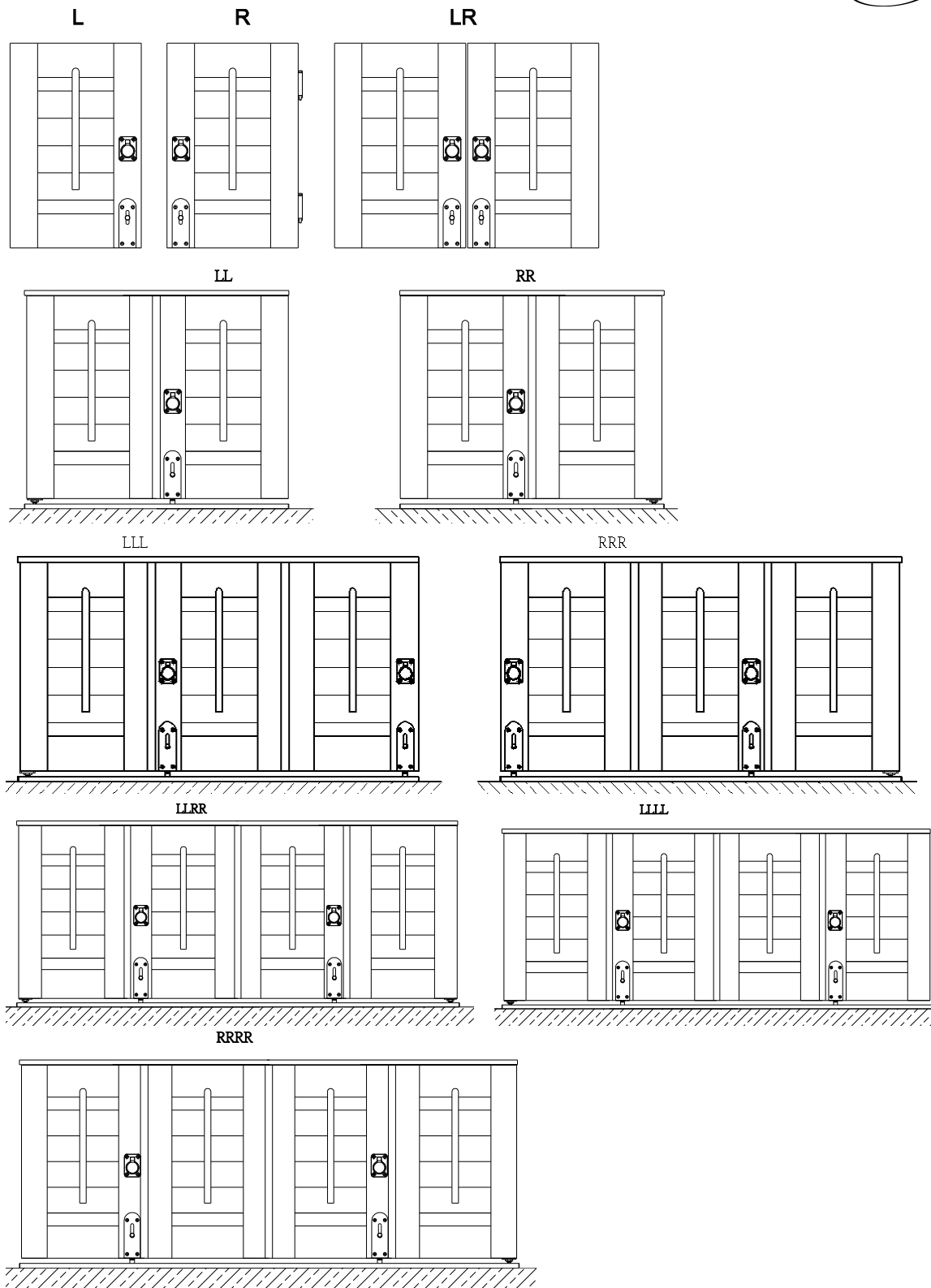
Available Material: All

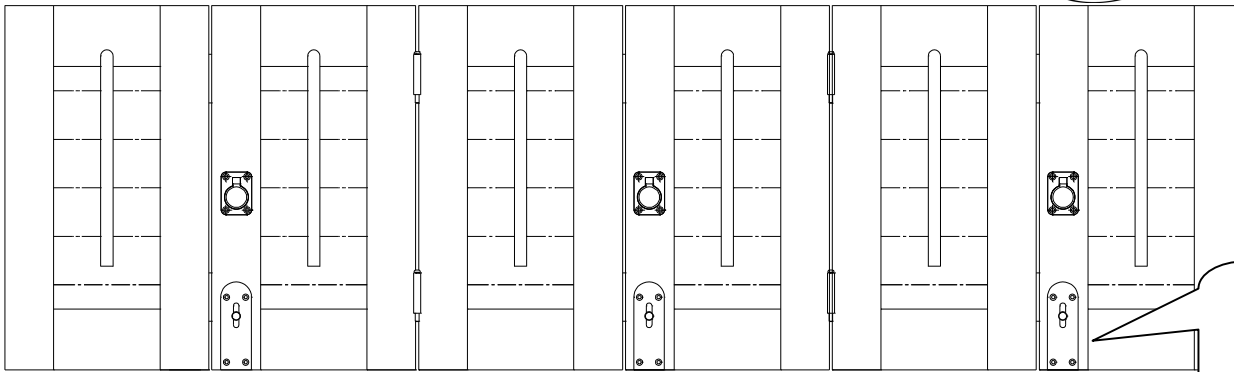
Color: Brush Nickel Plated, Pure White, Pearl, **Black.**



Position: (Only available on Bottom for Track Shutters with Bottom Track; For Regular, available on Top/Bottom/both Top and Bottom)

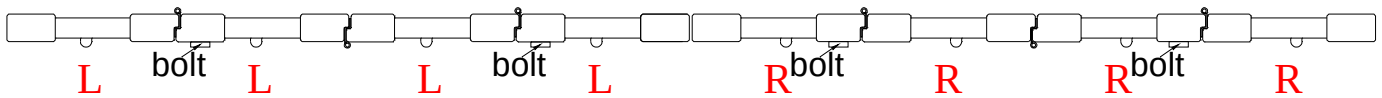
*Some surcharges apply





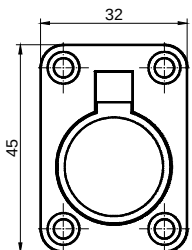
Panel Config: FFFFFFF(R)

The Panel Lock location for both track system shutters and regular shutters would be the same. The general principle is that the Panel Lock position will avoid the stile which will install the spring-loaded guide at bottom (same stile as the Wheel carrier).



Ring pull: for easy opening and closing. (Not available for 1.625" Stile)

Available in all shutter programs.



Handles (Not available for 1.625" Stile):

Available in all shutter programs.

Curved Hidden Handle & Rectangle Hidden Handle (for non Bypass track shutter)



Recessed Handle (for Bypass track shutter)



Ring pull and Handles location principle for Floating Panels:

1. When floating panel config is in pairs, it will be split into halves, the left will be for LL and the right will be for RR. So the ring pull / handles location for FF/FF, FF/FF/FF/FF will be same as that of panel config as LL/RR, LL/LL/RR/RR.
2. When floating panel config is FF, the ring pull / handles location will be same as that of panel config LL.
3. When floating panel config is FFFF/FF, the ring pull / handles location will be same as that of panel config LLLL/RR.
4. When floating panel config is singular, it will be split into 3 sections, the left and mid sections are for LL and the right section is for RR. So the ring pull / handles location for FF/FF/FF, FF/FF/FF/FF/FF will be same as that of LL/LL/RR, LL/LL/LL/RR/RR.

Please note the ring pull / handles color will match with the hinge color.

When shutter order with Invisible hinge, the ring pull / handles color will match with the normal hinge color.

Please see the matching chart below:

Hinge color	Pure White	Silk White	Bisque	Pearl	Black	Nickel Plated	Brushed Nickel	Antique Brass	Bright Brass	Stainless Steel
Ring Pull color	White	White	Bisque	Pearl	Black	Brushed Nickel	Brushed Nickel	Antique Brass	Bright Brass	Brushed Nickel
Handles color	White	White	Pearl	Pearl	Black	Brushed Nickel	Brushed Nickel	Brushed Nickel	Brushed Nickel	Brushed Nickel

Hinge color	Crisp Linen	String	Sea Mist	Stone Gray	Brown Gray	Taupe Gray
Ring Pull color	Bisque	Bisque	Brushed Nickel	Brushed Nickel	Brushed Nickel	Brushed Nickel
Handles color	Pearl	Pearl	Brushed Nickel	Brushed Nickel	Brushed Nickel	Brushed Nickel

- The ring pull color is selectable.

*Some surcharges apply

Paint colors (Solid colors) (24) :

No.	Color ID	Color Name	2.375" Hinge color (Default)	Direct Mount 3" Hinge color (Default)	T post connector color (Default)	Ring pull color (Default)	Curved Hidden Handle / Recessed Handle / Rectangle Hidden Handle color (Default)	Panel Lock (Default)	Invisible hinge color	New Top & Bottom Tracks color (Default)	Dovetail Key Color (Default)
1	001	Pure White	Pure White	Pure White	Pure White	White	White	Pure White	Pearl Chrome Plated	Bright White	White
2	002	Extra White	Silk White	Silk White	Silk White	White	White	Pure White		Bright White	White
3	003	Silk White	Silk White	Silk White	Silk White	White	White	Pure White		Bright White	White
4	004	Bright White	Silk White	Silk White	Silk White	White	White	Pure White		Bright White	White
5	006	Pearl	Pearl	Pearl	Pearl	Pearl	Pearl	Pearl		Bright White	White
6	007	Ivory Lace	Pearl	Pearl	Pearl	Pearl	Pearl	Pearl		Metallic Gray	White
7	009	Creamy	Pearl	Pearl	Pearl	Pearl	Pearl	Pearl		Bright White	White
8	012	Crisp Linen	Crisp Linen	Bisque	Bisque	Bisque	Pearl	Pearl		Metallic Gray	White
9	013	Bisque	Bisque	Bisque	Bisque	Bisque	Pearl	Pearl		Metallic Gray	White
10	017	Gray Black	Black	Black	Black	Black	Black	Black		Black	Black
11	019	String	String	Bisque	Bisque	Bisque	Pearl	Pearl		Metallic Gray	Transparent
12	032	Sea Mist	Sea Mist	Nickel Plated	Nickel Plated	Brushed Nickel	Brushed Nickel	Brush Nickel plated		Metallic Gray	White
13	046	Ice	Silk White	Silk White	Silk White	White	White	Pure White		Bright White	White
14	053	Clay	Nickel Plated	Nickel Plated	Nickel Plated	Brushed Nickel	Brushed Nickel	Brush Nickel plated		Bronze	Transparent
15	063	Decorator's White	Silk White	Silk White	Silk White	White	White	Pure White		Bright White	White
16	066	Winchester White 2010	Pearl	Pearl	Pearl	Pearl	Pearl	Pearl		Metallic Gray	White
17	076	Aura White	Silk White	Silk White	Silk White	White	White	Pure White		Bright White	White
18	080	Taupe Gray	Taupe Gray	Nickel Plated	Nickel Plated	Brushed Nickel	Brushed Nickel	Brush Nickel plated		Metallic Gray	White
19	836	Classic Black	Black	Black	Black	Black	Black	Black		Black	Black
20	090	TS White	Pure White	Pure White	Pure White	White	White	Pure White		Bright White	White

*Some surcharges apply

21	600	True White	Silk White	Silk White	Silk White	White	White	Pure White		Bright White	White
22	601	Chiffon	Silk White	Silk White	Silk White	White	White	Pure White		Bright White	White
23	603	Natural Linen	Pearl	Pearl	Pearl	Pearl	Pearl	Pearl		Metallic Gray	White
24	609	Chateau Brown	Nickel Plated	Nickel Plated	Nickel Plated	Brushed Nickel	Brushed Nickel	Brush Nickel plated		Metallic Gray	Transparent

Stain colors (18) :

No.	Color ID	Color Name	2.375" Hinge color (Default)	Direct Mount 3" Hinge color (Default)	T post Connector color (Default)	Ring Pull color (Default)	Curved Hidden Handle / Recessed Handle / Rectangle Hidden Handle color (Default)	Panel Lock (Default)	Invisible hinge color	New Top & Bottom Tracks color (Default)	Dovetail Key Color (Default)
1	108	Rustic Gray	Nickel Plated	Nickel Plated	Nickel Plated	Brushed Nickel	Brushed Nickel	Brush Nickel plated	Pearl Chrome Plated	Bronze	Transparent
2	110	Limed White	Pearl	Pearl	Pearl	Pearl	Pearl	Pearl		Bronze	White
3	200	Natural	Antique Brass	Antique Brass	Nickel Plated	Antique Brass	Brushed Nickel	Brush Nickel plated		Bronze	Coffee
4	202	Golden Oak	Antique Brass	Antique Brass	Nickel Plated	Antique Brass	Brushed Nickel	Brush Nickel plated		Bronze	Coffee
5	205	Goldenrod	Antique Brass	Antique Brass	Nickel Plated	Antique Brass	Brushed Nickel	Brush Nickel plated		Bronze	Coffee
6	221	Black Walnut	Antique Brass	Antique Brass	Nickel Plated	Antique Brass	Brushed Nickel	Brush Nickel plated		Bronze	Black
7	227	Red Oak	Antique Brass	Antique Brass	Nickel Plated	Antique Brass	Brushed Nickel	Brush Nickel plated		Bronze	Coffee
8	229	Rich Walnut	Antique Brass	Antique Brass	Nickel Plated	Antique Brass	Brushed Nickel	Brush Nickel plated		Bronze	Coffee
9	230	Old Teak	Antique Brass	Antique Brass	Nickel Plated	Antique Brass	Brushed Nickel	Brush Nickel plated		Bronze	Coffee
10	237	Wenge	Antique Brass	Antique Brass	Nickel Plated	Antique Brass	Brushed Nickel	Brush Nickel plated		Bronze	Black
11	242	Auburn	Antique Brass	Antique Brass	Nickel Plated	Antique Brass	Brushed Nickel	Brush Nickel plated		Bronze	Coffee
12	246	Matte Black	Black	Black	Black	Black	Black	Black		Black	Black

13	248	Pretzel	Antique Brass	Antique Brass	Nickel Plated	Antique Brass	Brushed Nickel	Brush Nickel plated		Bronze	Coffee
14	250	Sumatra	Antique Brass	Antique Brass	Nickel Plated	Antique Brass	Brushed Nickel	Brush Nickel plated		Bronze	Coffee
15	252	Toffee	Antique Brass	Antique Brass	Nickel Plated	Antique Brass	Brushed Nickel	Brush Nickel plated		Bronze	Black
16	254	Driftwood	Nickel Plated	Nickel Plated	Nickel Plated	Brushed Nickel	Brushed Nickel	Brush Nickel plated		Metallic Gray	White
17	255	Silver Gray	Nickel Plated	Nickel Plated	Nickel Plated	Brushed Nickel	Brushed Nickel	Brush Nickel plated		Metallic Gray	Transparent
18	862	French Oak	Antique Brass	Antique Brass	Nickel Plated	Antique Brass	Brushed Nickel	Brush Nickel plated		Bronze	Coffee

OSMO colors (6) :

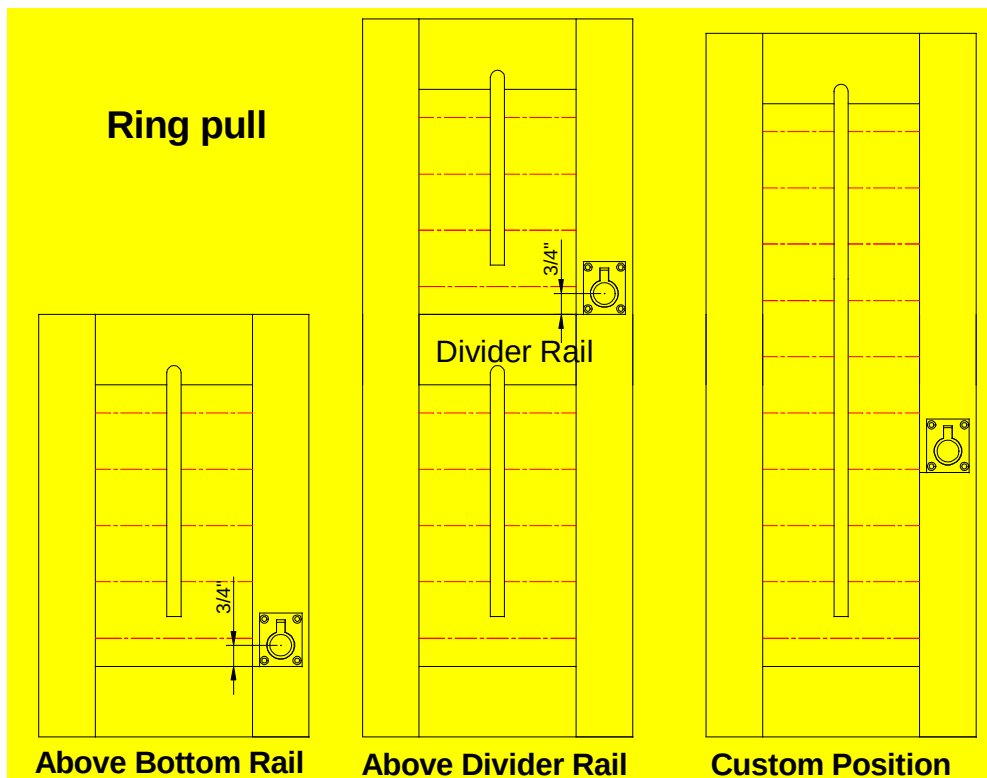
No.	Color ID	Color Name	2.375" Hinge color (Default)	Direct Mount 3" Hinge color (Default)	T post Connector color (Default)	Ring Pull color (Default)	Curved Hidden Handle / Recessed Handle / Rectangle Hidden Handle color (Default)	Panel Lock (Default)	Invisible hinge color	New Top & Bottom Tracks color (Default)	Dovetail Key Color (Default)
1	1003	White Matte	Silk White	Silk White	Silk White	White	White	Pure White	Pearl Chrome Plated	Metallic Gray	White
2	1109	Silk Gray	Nickel Plated	Nickel Plated	Nickel Plated	Brushed Nickel	Brushed Nickel	Brush Nickel plated		Bronze	Transparent
3	1111	Granite Gray	Antique Brass	Antique Brass	Nickel Plated	Antique Brass	Brushed Nickel	Brush Nickel plated		Bronze	Transparent
4	1203	Black	Antique Brass	Antique Brass	Nickel Plated	Antique Brass	Brushed Nickel	Brush Nickel plated		Bronze	Transparent
5	1204	Ebony	Antique Brass	Antique Brass	Nickel Plated	Antique Brass	Brushed Nickel	Brush Nickel plated		Bronze	Transparent
6	1502	Teak	Antique Brass	Antique Brass	Nickel Plated	Antique Brass	Brushed Nickel	Brush Nickel plated		Bronze	Coffee

** The 017 Gray Black, 246 Matte Black, 836 Classic Black and 609 Chateau Brown Colors will have a surcharge.

** All OSMO Colors will have a surcharge.

- When the bottom track colors are bright white and metallic gray, the default spring loaded guide color will be white; When the bottom track colors are bronze and black, the default spring loaded guide color will be coffee.
- Please note that the color options for pin/tension gears are White and Coffee. For custom color orders please select either White or Coffee when placing an order.
- For custom color orders, the supplied dovetail keys will match the selected pin/tension gear color (White or Coffee).
- For custom color orders, please select the desired T-post connector color from the available options when placing an order.
- For custom color orders with Invisible Hinge, but without Bifold hinge, please select the desired magnet catch (round magnet catch or sill magnet catch or 8.5x25mm magnet catch) color from the available options when placing an order.

Position: For different panel configuration, please refer to the location of Panel Lock to show on which stile the pull will be fixed.

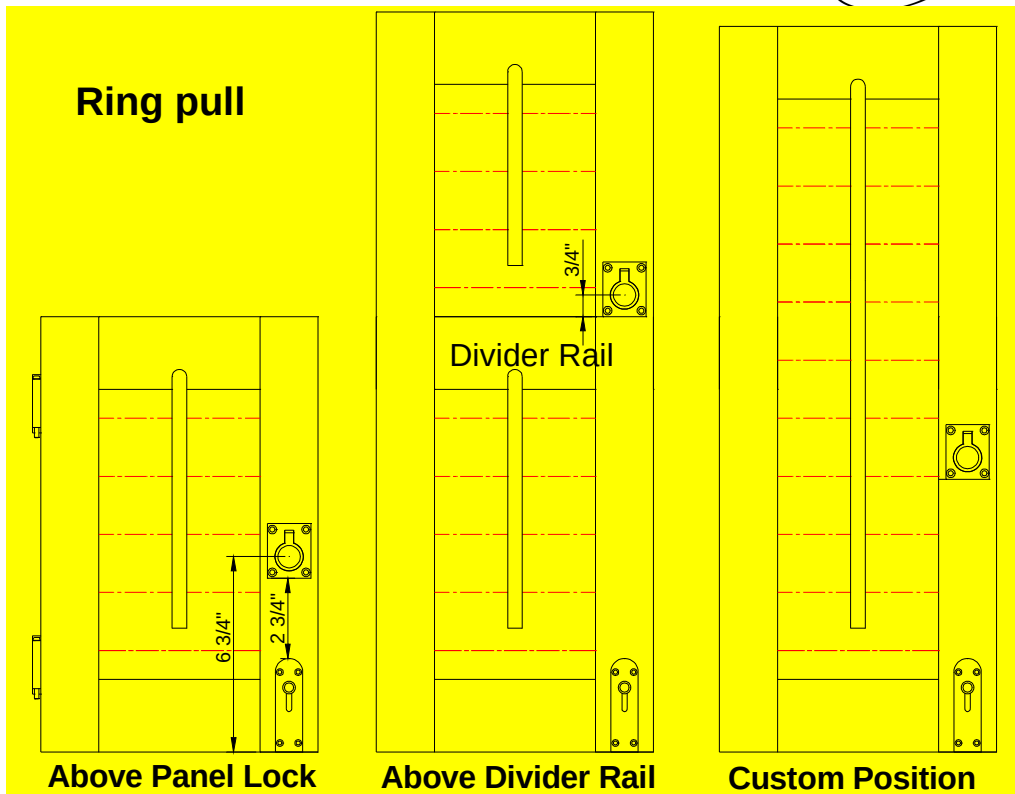


Position:

- A. Above Bottom Rail
- B. Above Divider Rail
- C. Custom Position

If both Panel Lock & ring pull are specified, the distance between them is 2 3/4", and the ring pull location is 6 3/4" from the bottom of panel.

When there is no mid rail, ring pull will be just on the top of Panel Lock (2 3/4" above Panel Lock) as default location; if there are mid rails, bottom of ring pull lines up with top of the 1st mid rail as default location.



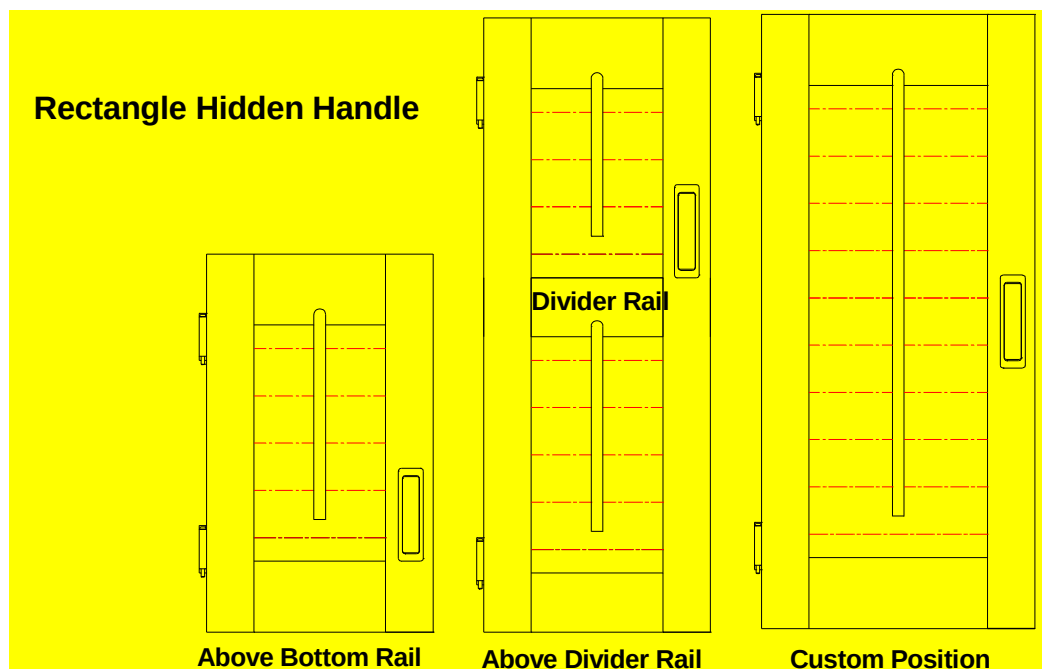
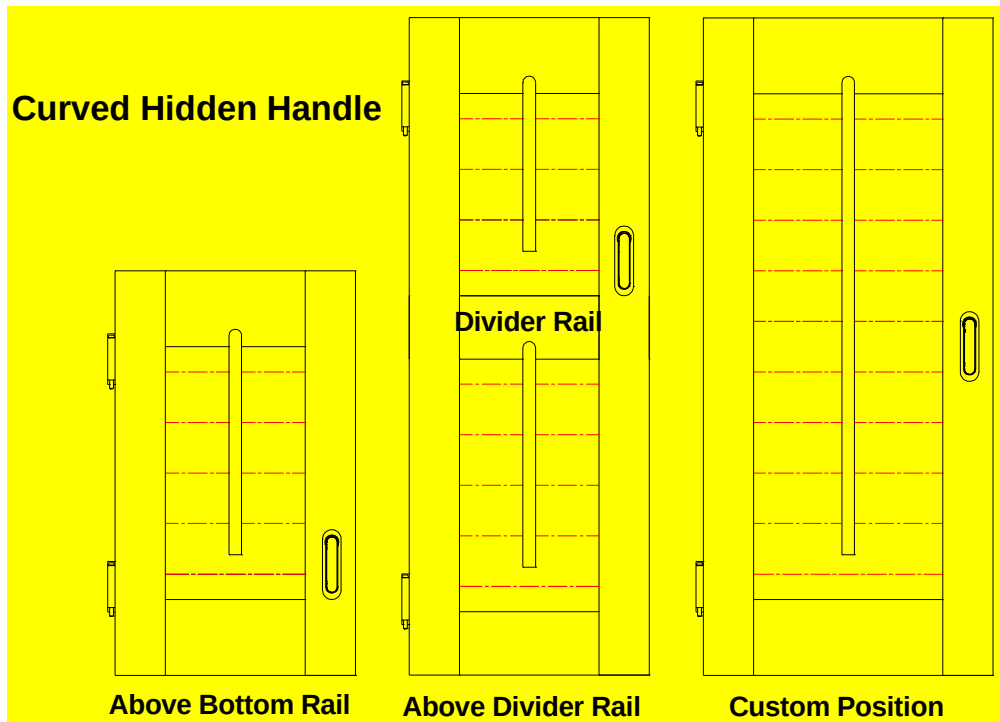
Position:

- A. 2 3/4" Above Panel Lock
- B. Above Divider Rail
- C. Custom Position

Measurements:

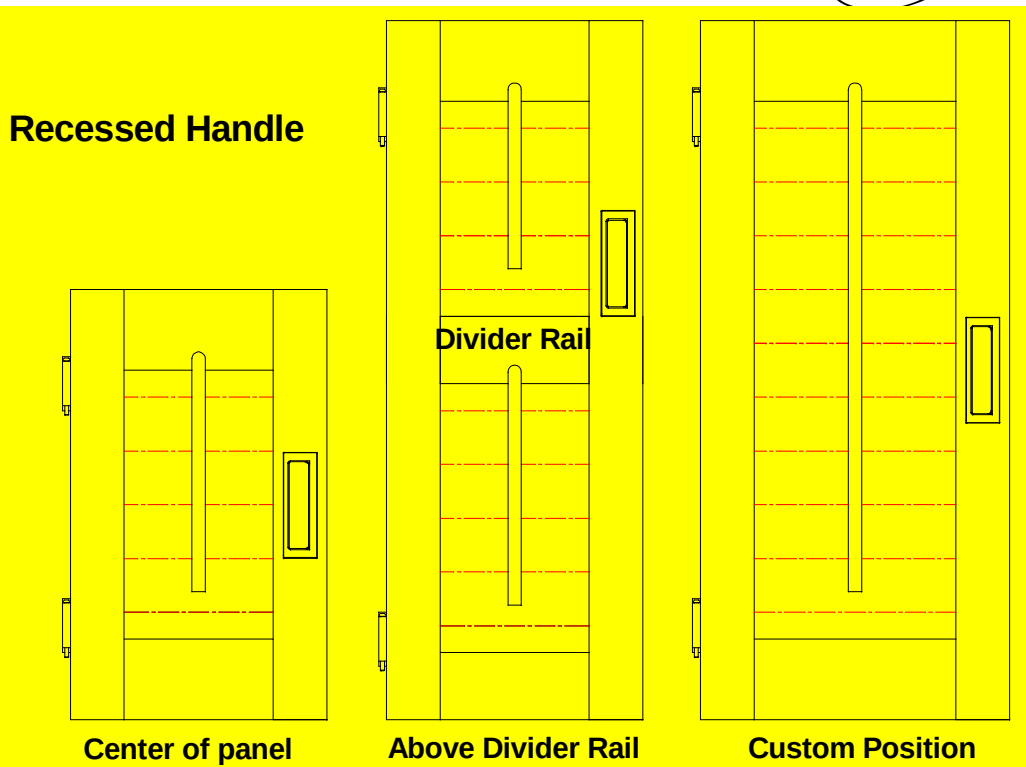
- Window size:
 - Measure from the bottom of the window to the center of the ring pull.
- Max frame-to-frame:
 - Measure from the bottom of the frame to the center of the ring pull.

Position: For different panel configuration, please refer to the location of Panel Lock to show on which stile the Handles will be fixed.



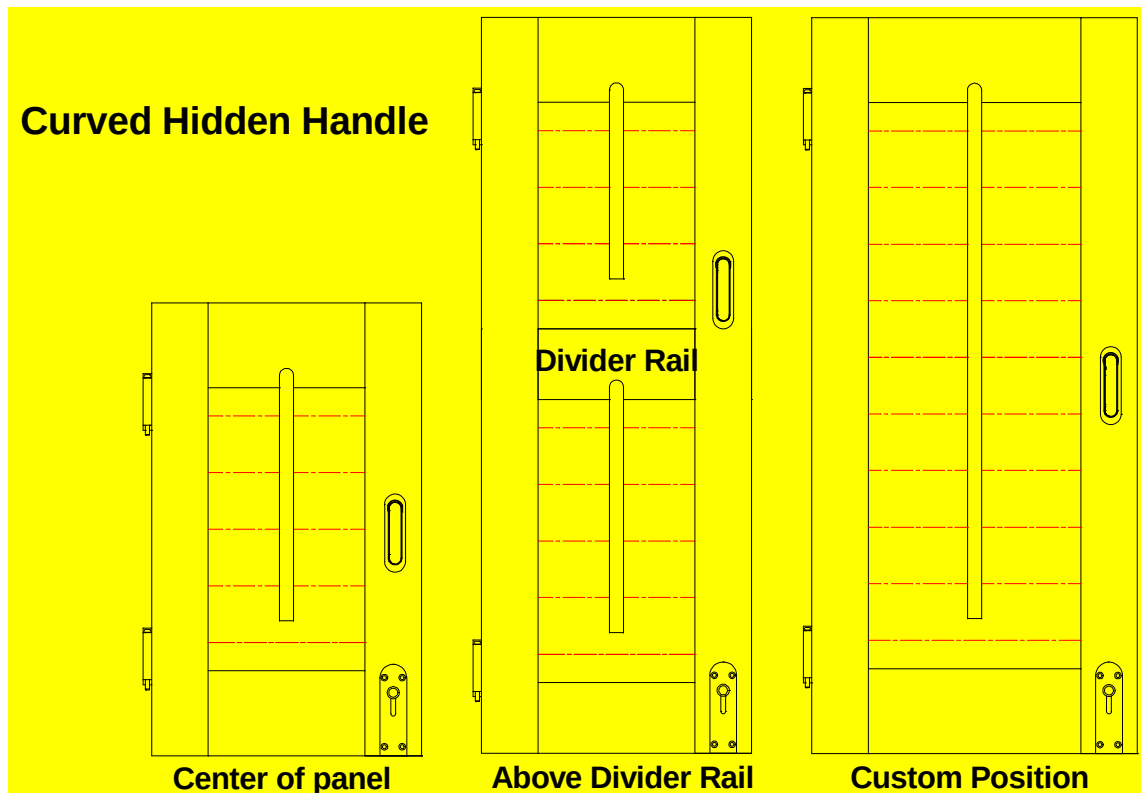
Position:

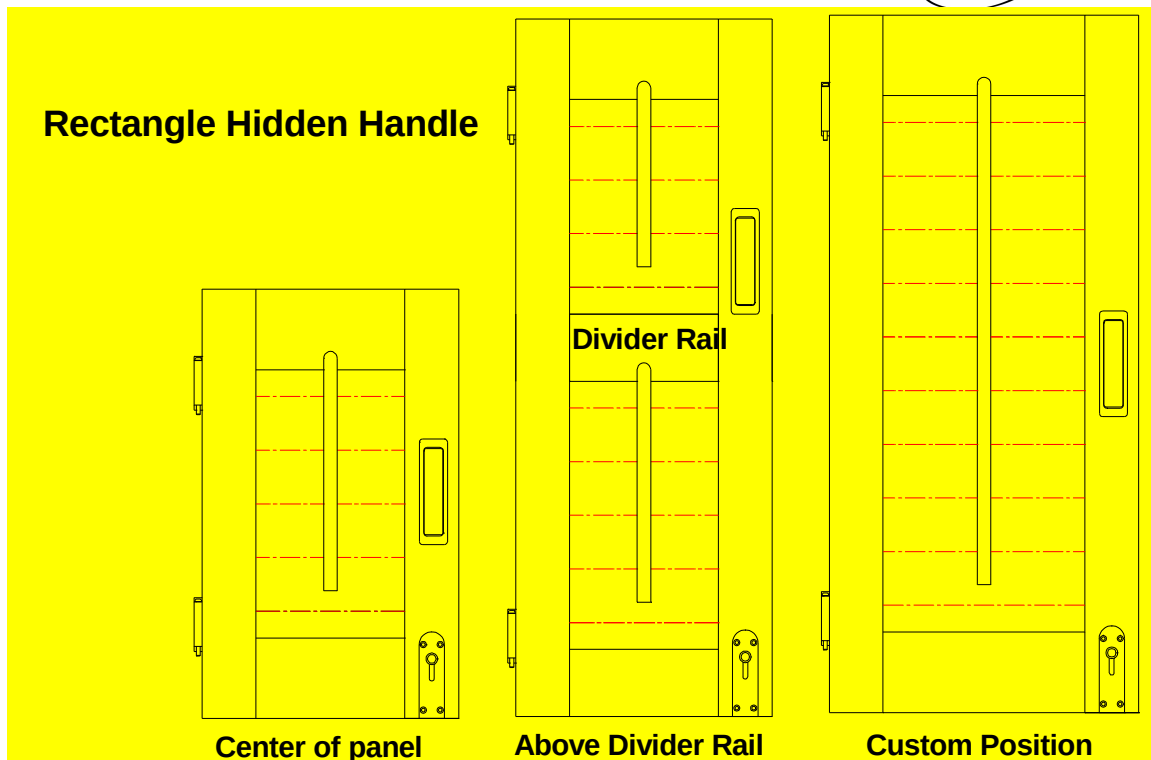
- A. Above Bottom Rail
- B. Above Divider Rail
- C. Custom Position



Position:

- A. Center of panel
- B. Above Divider Rail
- C. Custom Position





Position:

- A. Handle at center of panel
- B. Handle Above Divider Rail
- C. Custom Location

Measurements:

- Window size:
 - Measure from the bottom of the window to the center of the Handles.
- Max frame-to-frame:
 - Measure from the bottom of the frame to the center of the Handles.

Invisible Hinge*:



1. Invisible hinge is not available for Multifold panels.
2. Invisible hinge can only be between the panel & frame/T post, connections from panel to panel will use a regular hinge.
3. Color Option: Only available for Pearl Chrome Plated.

Frame Options	Hinge	Notes
2" Camber Deco Frame	Y	/
2" Classic Deco Frame	Y	/
2.5" Mission Deco Frame	N	/
3" Ridge Deco Frame	N	/
Vintage L Frame W / 1/4" Light Block	N	/
Vintage L	N	/
Vintage L Frame w/ 1/2" Buildout	N	/
Vintage L Frame w/ 1" Buildout	N	/
Beaded L w 1/4"light block	Y	/
Beaded L	Y	/
Beaded L Frame w/ 1/2" Buildout	Y	/
Beaded L Frame w/ 1" Buildout	Y	/
Tilt out Z Frame	N	/
1.25" Beaded Z Frame	Y	/
1.5" Bullnose Z Frame	Y	/
2" Bullnose Z Frame	Y	/
2" Bel Air Z Frame	Y	/
3" Crown Z Frame	N	/
7/8" Vintage Hang Strip	N	/
1" Chamfer T-Post w/ 1/4" Light Block(for Deco)	Y	Only available with hinge on one side, for example, LTL is available, but RTL is not.
1" Chamfer T-Post w/3/4" Light Block(for L/Z/HS)	Y	
1" Chamfer T-Post w/ 1/4" thick 3/8" wide Light Block(for Deco)	Y	
1" Chamfer T-Post w/3/4" thick 3/8" wide Light Block(for L/Z/HS)	Y	
1 3/4" Chamfer T-Post w/1/4" Light Block(for Deco)	Y	Available with hinge on BOTH sides, for example, RTL.
1 3/4" Chamfer T-Post w/3/4" Light Block(for L/Z/HS)	Y	
Corner Post w/1/4" Light Block(for Deco)	Y	/
Corner Post w/3/4" Light Block(for L/Z/HS)	Y	/

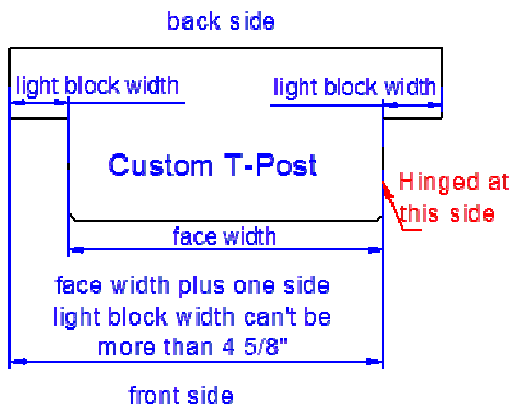
Bay Post w/1/4" Light Block(for Deco)	Y	/
Bay Post w/3/4" Light Block(for L/Z/HS)	Y	/

Invisible Hinge Qty	
Panel Height (H)	Hinge Qty
H<=48"	2
48"<H<=78"	3
78"<H<=96"	4
96"<H<=120"	5
H>120"	6

Preinstallation:

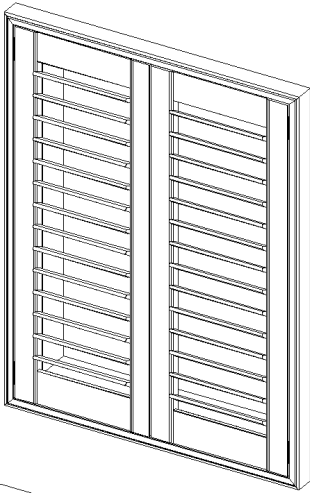
Panels and frames will be packaged individually, and the hinge is preinstalled onto the frame/post, with the screws for the other half part on the panels in the hardware box.

Note: If Custom T-posts need to install Invisible Hinge, the size of Custom T-post face width plus one side light block width can't be more than 4 5/8"; Please see drawing for reference.



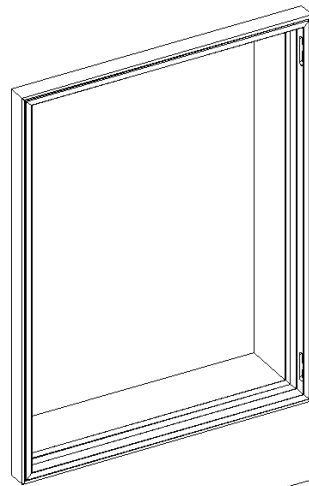
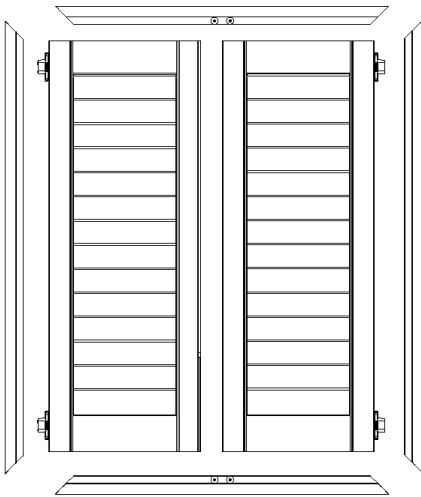
Installation Instructions for the Invisible hinges

Take L Frame as an example:

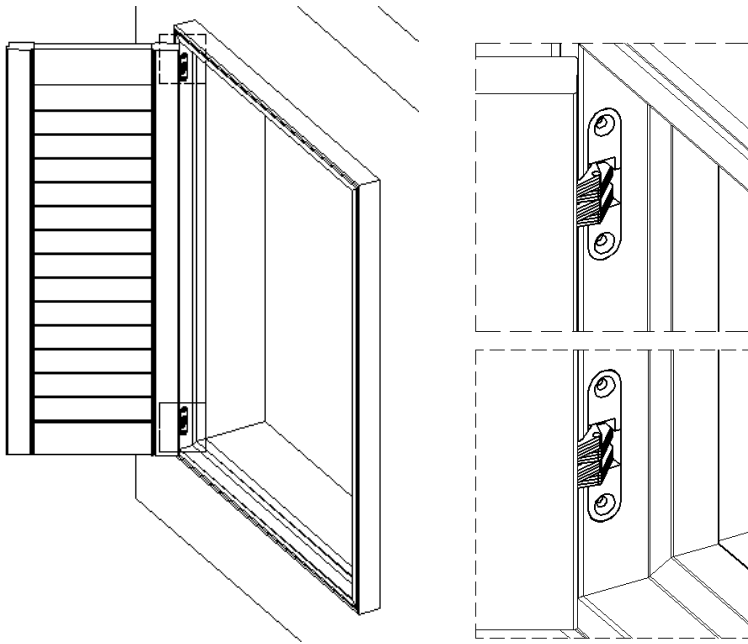


1). Remove the corresponding panels and frames from their packaging and match the panels with the correct frames.

Install the frames on the corresponding windows and make sure they are square (diagonals are equal)



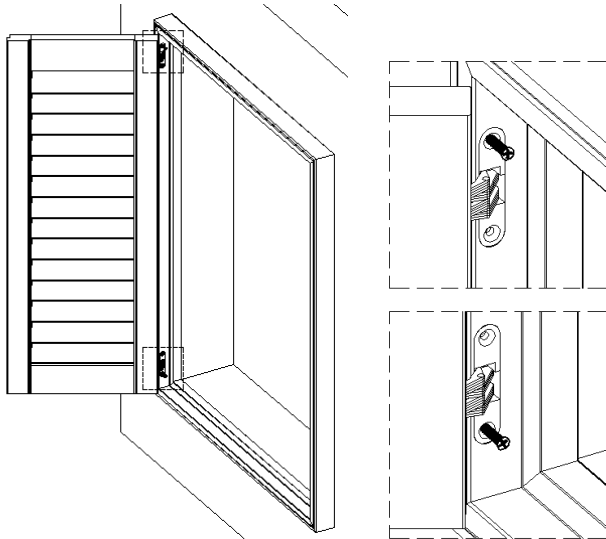
2). Please insert **ALL of** the Invisible hinges on the panel into the hinge holes on the corresponding frames prior to trying to secure any single hinge. This may require the assistance of a second person to hold the panel to prevent the hinges from sliding out of their slots while the screws are secured (Do not let the panels hang from just one hinge).



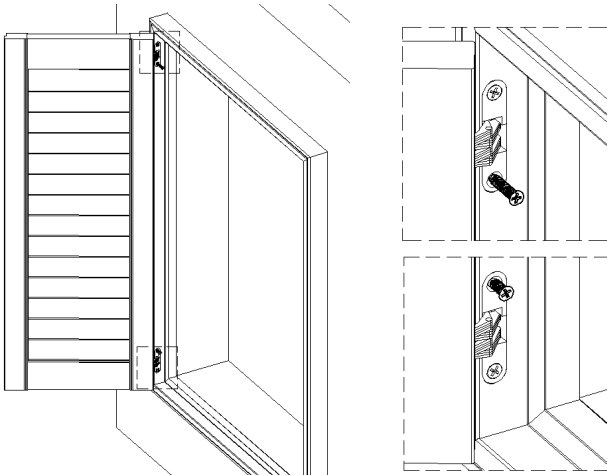
3). To properly secure the panels, first please tighten the upper screw of the uppermost top hinge and the lower screw of the lowest hinge. Close the panel to check whether the shutter closure is normal and the gap is uniform.

Note:

- a) When locking the screw, again it is important to prevent the invisible hinges from sliding out of their slots.
- b) Only the screws provided in the hardware box may be used and they cannot be replaced by other screws.



4). When the closure gap is uniform, secure the remaining screws.

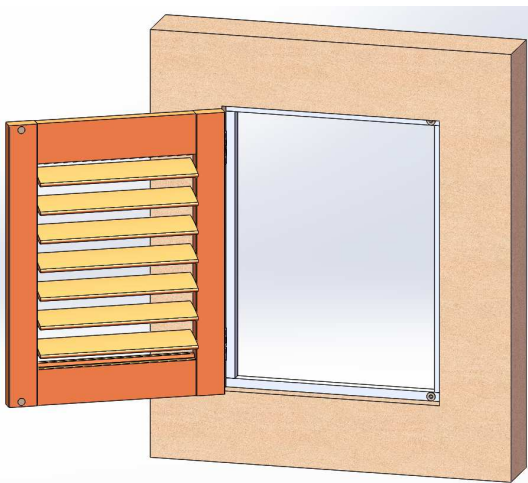


5). Install the remaining panels in the same manner.

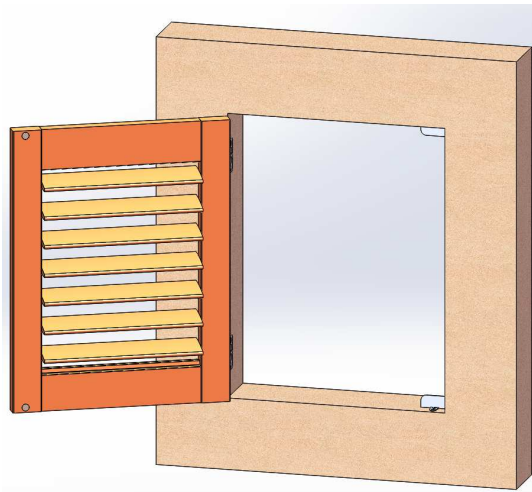


Mount Options for Direct Mount:

Light block



L magnet catch



*Some surcharges apply

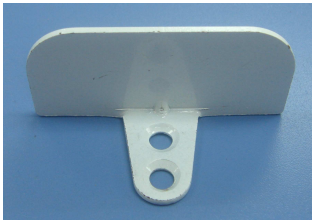


12.5mm Magnet with screw hole (recessed into panels)



Round Magnet Catch (installed on light blocks)

L Magnet Catch (sill magnet catch): used for catching the magnets when ordering panels without light blocks.



Round Magnet Catch and L Magnet Catch are available in:
Pure White, Silk White, Bisque, Pearl, Bright Brass, Antique Brass, Black, Nickel plated, Crisp Linen, String, Sea Mist, Stone Gray, Brown Gray, Taupe Gray.

Light block and L magnet catch are available in one of the following situation.

- Direct mount without top & bottom sill frame
- 2-sided Hang strip without top & bottom sill frame
- 3-sided Hang strip without bottom sill frame
- 3-sided inverted Hang strip without top sill frame

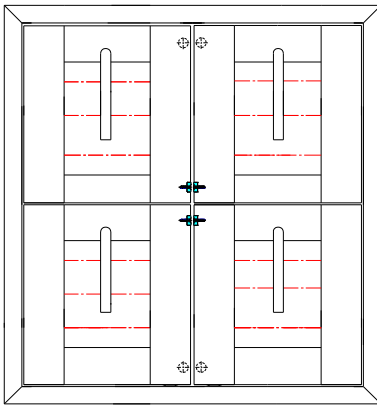
Side Magnet: Available for Butt, rabbet and Astragal stile



Side Magnets will be applied in following situation:

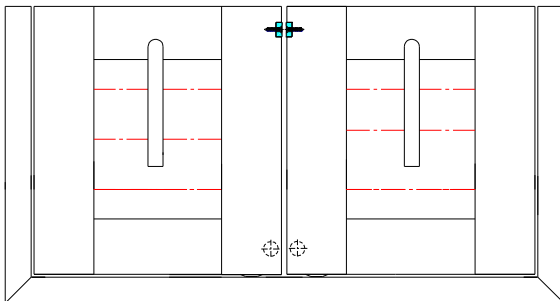
1. Double Hung shutter but without horizontal T post:

Side magnets will be applied to the meeting stiles of the L and R panel. (Default)



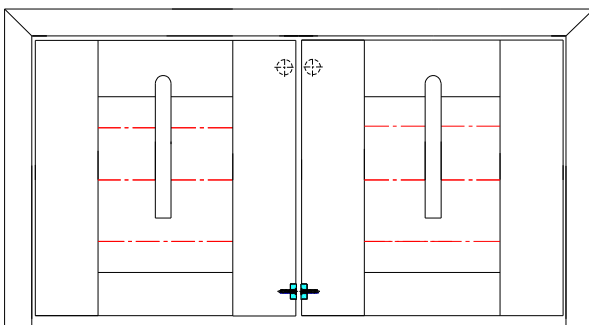
2. Café Shutter

Side magnets will be applied to the meeting stiles of the L and R panel. (Default)



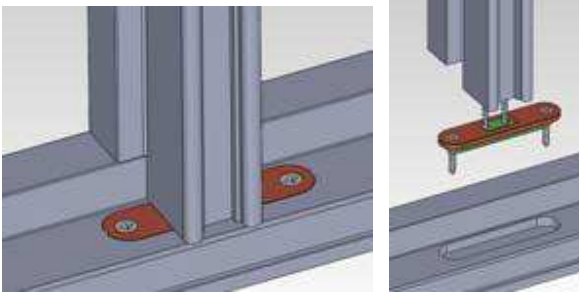
3. No frame or light block or L-shape striker plate on either top or bottom, or if larger than standard top or bottom gap is requested.

Side magnets will be applied to the meeting stiles of the L and R panel. (Optional)



4. Note: if panels only with side magnets and there is no standard magnet on the back of panels, panels may open for external factors.

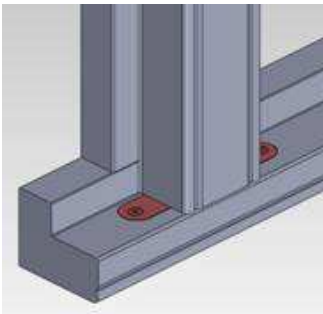
T-Post Connector



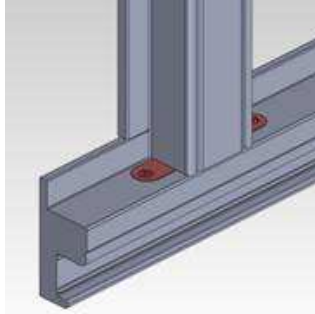
Available colors: Pure White; Silk White; Pearl; Bisque; Black; Nickel-Plated.
Available for all types of T Posts (custom-width T posts excluded).

T Post Connector can be used with the following Frames.

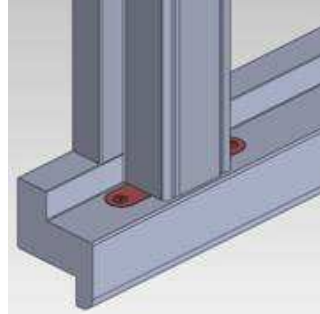
L Frame:



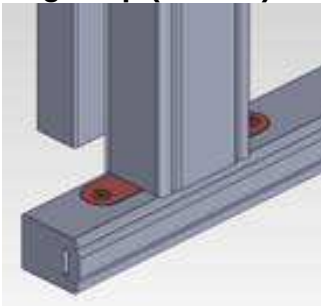
Deco Frame:



Z Frame:



Hang Strip (Beside)

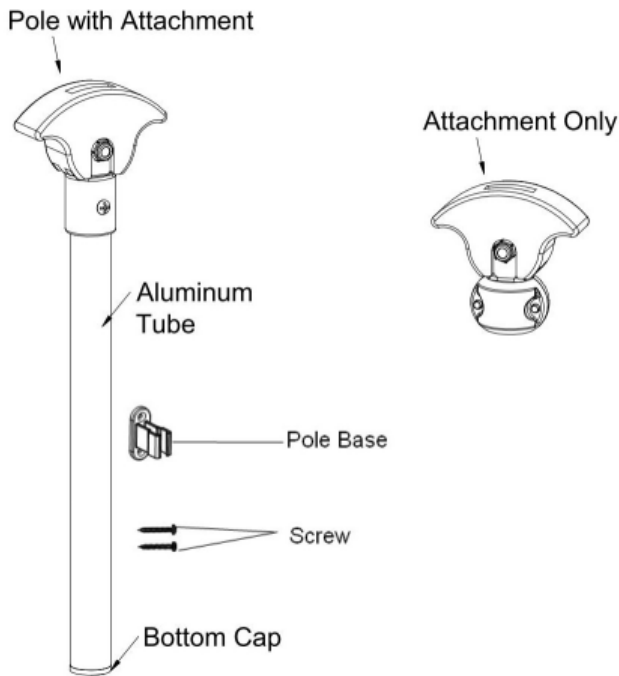


Note:

1. For Horizontal T Post, Hang Strip (Behind) and shutter with 1.6"/2" Sill, T Post connector is not available and T Post will be installed by screws instead;
2. If there is a panel lock which may interfere with the T-Post connector, all the T-Posts for the entire shutter will be installed with screws and the frames to be pre-drilled, even in cases where the panel lock may only interfere with one T-Post;
3. The screw color is nickel-plated regardless of the color of the T Post connector;
4. T post connectors can only be installed on a straight piece, and screws will be used instead for T Post installed on a curved or raked piece.

Optional Poles:

1. Terminology



Component

NO.	Name	Note
1	Pole with Attachment	Aluminum Tube, Pole Base and Bottom Cap included *Surcharge applies.
2	Attachment only	• Compatible with extension poles found in most home improvement stores. • Screw thread (3/4") and thread pitch (1/5") *Surcharge applies.

2. Available Pole Length

Two lengths available:

- 36"
- 60"

Bi-fold 180™

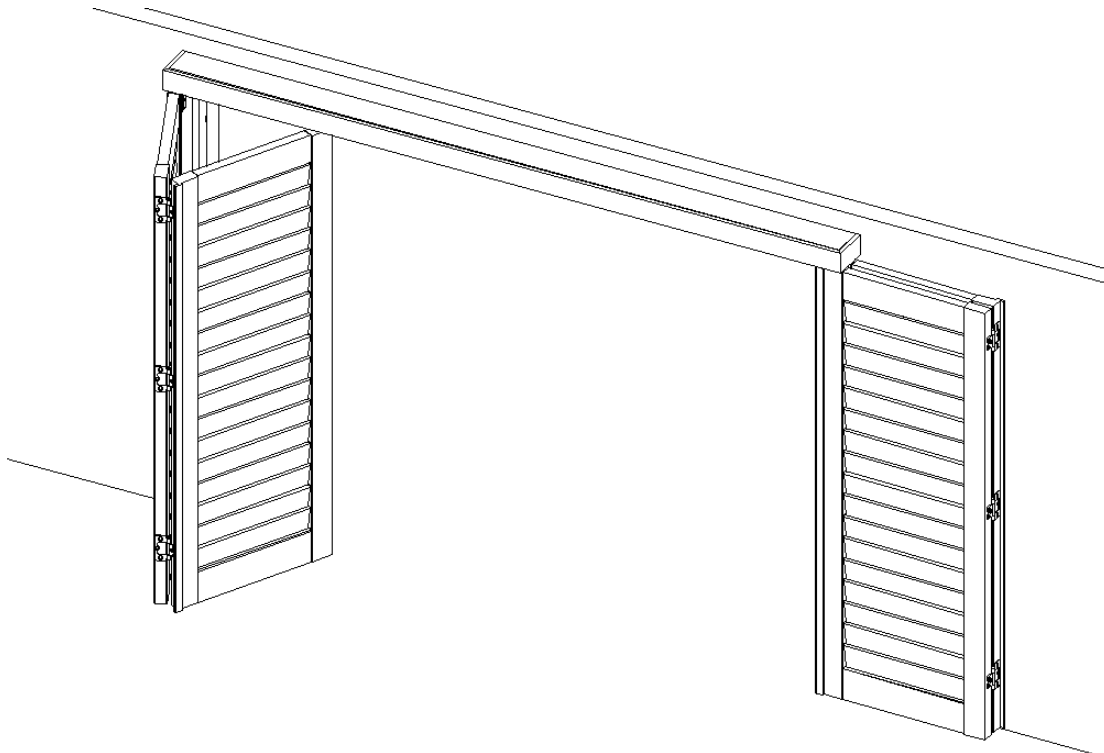
Guidelines and Specs:

<u>Description</u>	<u>Minimum</u>	<u>Maximum</u>
Single-Panel Width (Bi-fold Track)	6"	26"
Single-Panel Width (LLL/RRR)	6"	20"
Single-Panel Height	10"	132"

Outside Mount

Max frame-to-frame width: window width + 3 1/2"

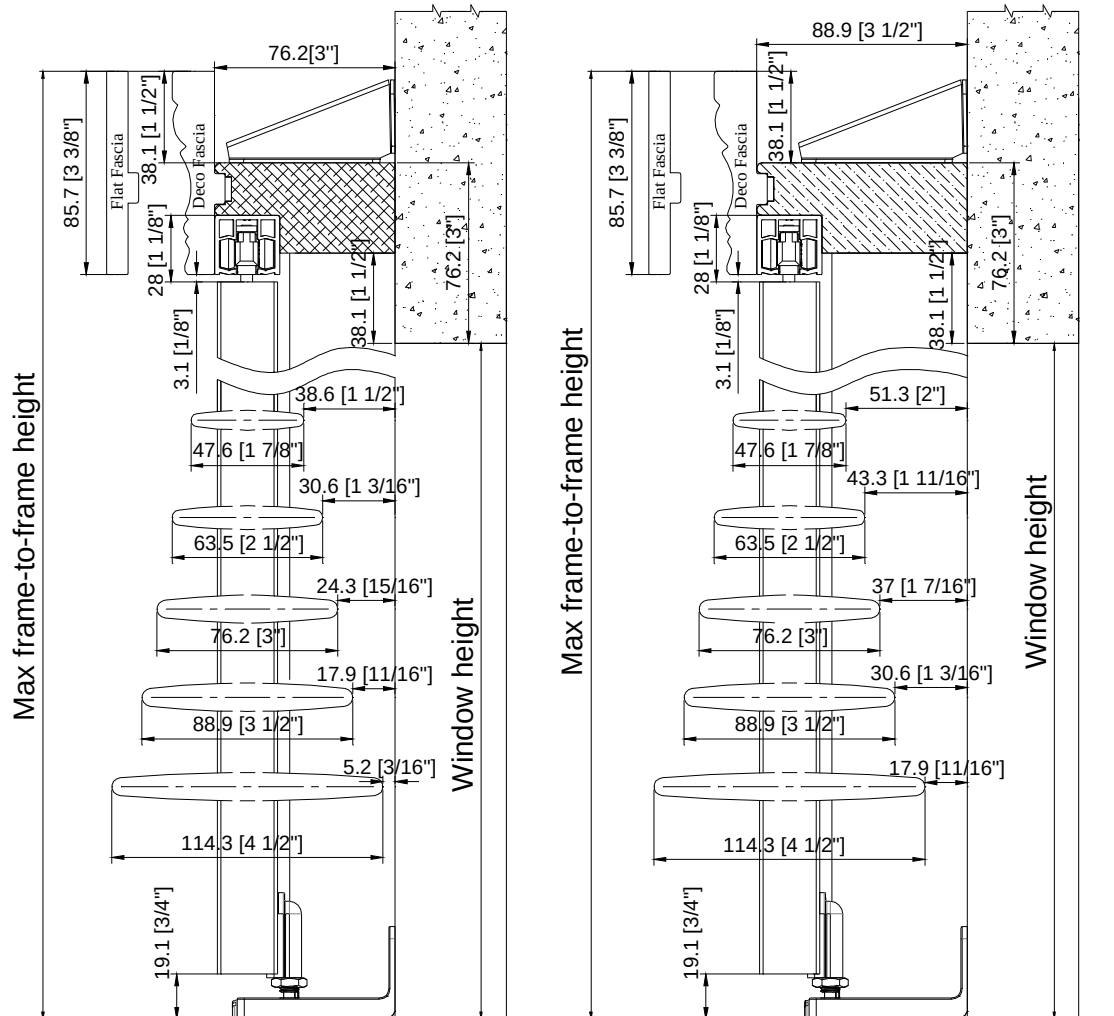
Max frame-to-frame height: window height + 4 1/2"



Track is pre-installed on the header; standard 1 track only.

Mounting Options: Outside Mount only

Side View

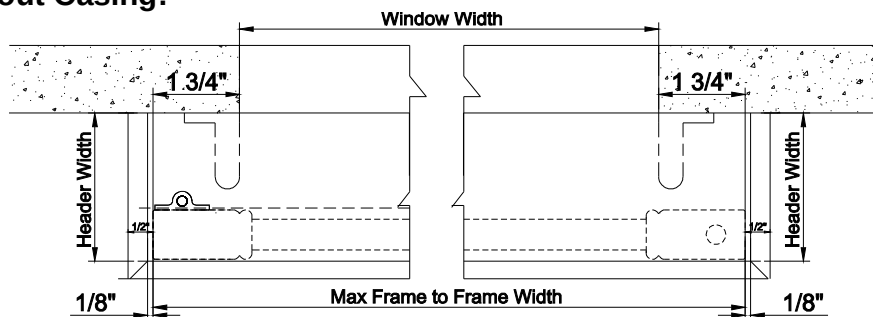


Top View

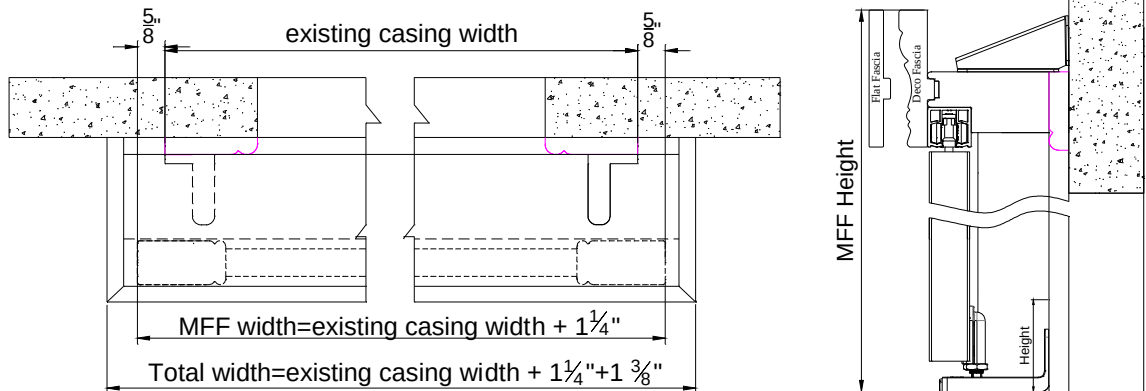
Without Casing:

Note:

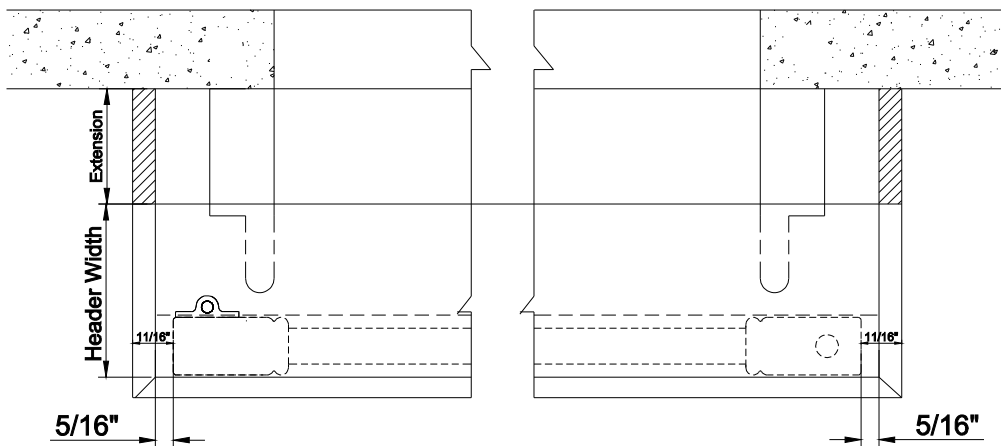
- ◇ Fascia return will protrude the panel about 1/2" for each side.
- ◇ The header will protrude the panel about 1/8" for each side.
- ◇ Header, light blocks, and bottom pivot bracket(s) must be installed on a flat surface.



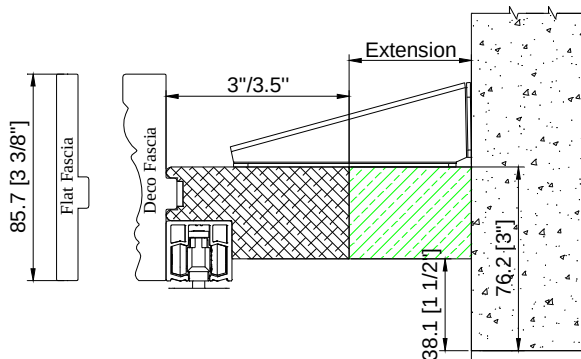
With Existing Casing:



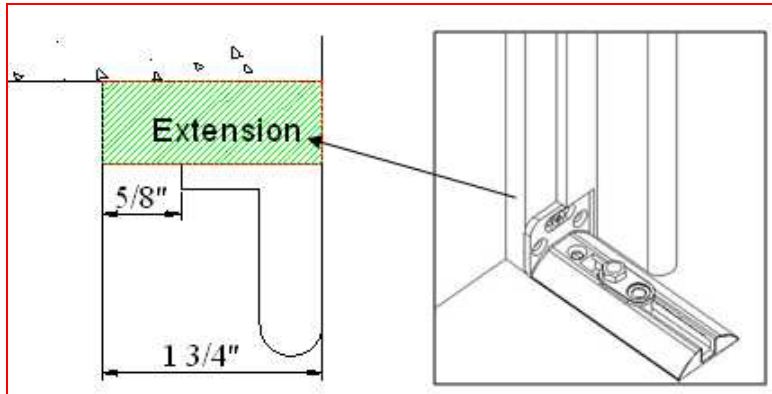
- ✧ Header Extension: Optional
- ✧ Custom extension up to 2" for Header
- ✧ When extension is needed, the header will protrude the panel 5/16" for each side and the fascia return will protrude the panel 11/16" for each side.



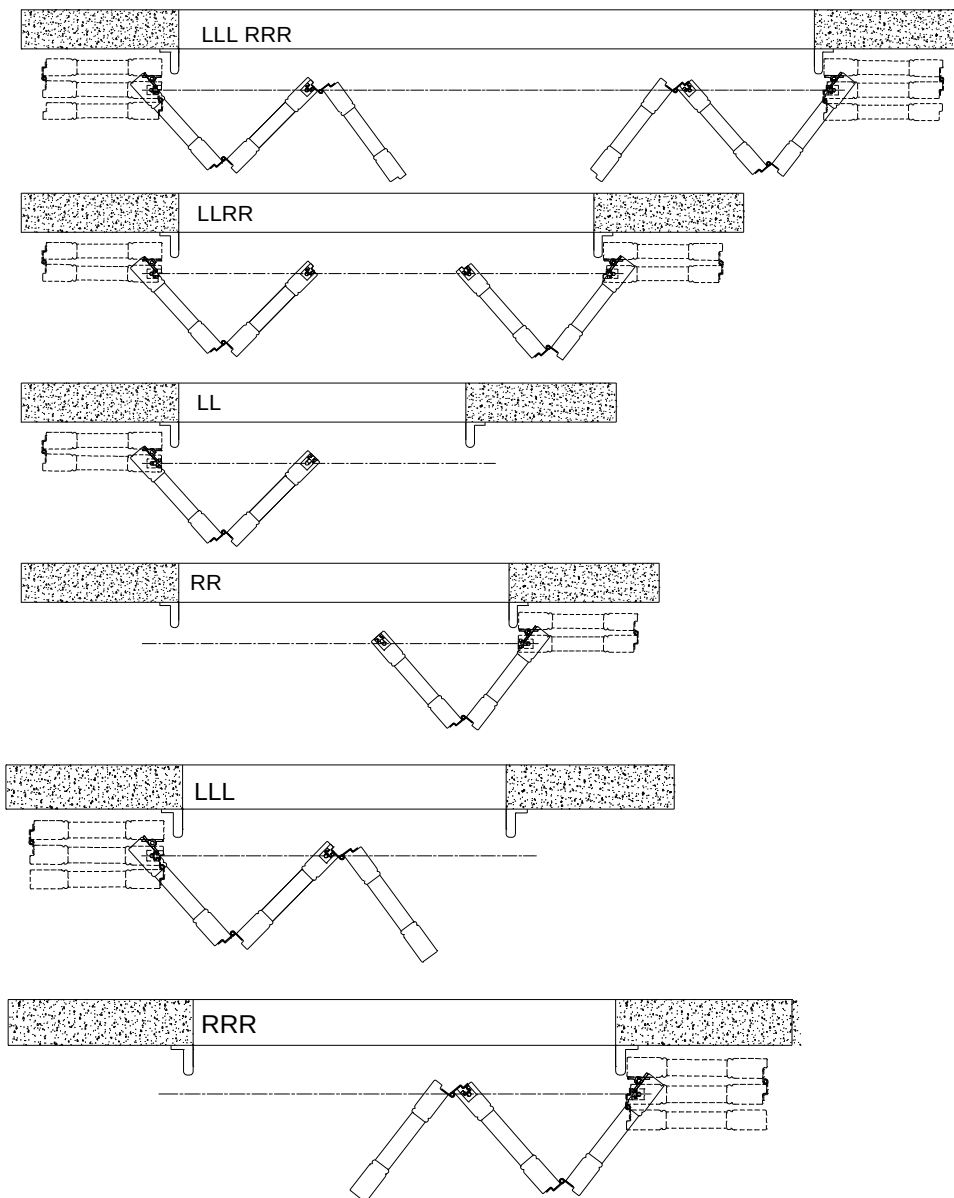
- ✧ The extension will be pre-attached.

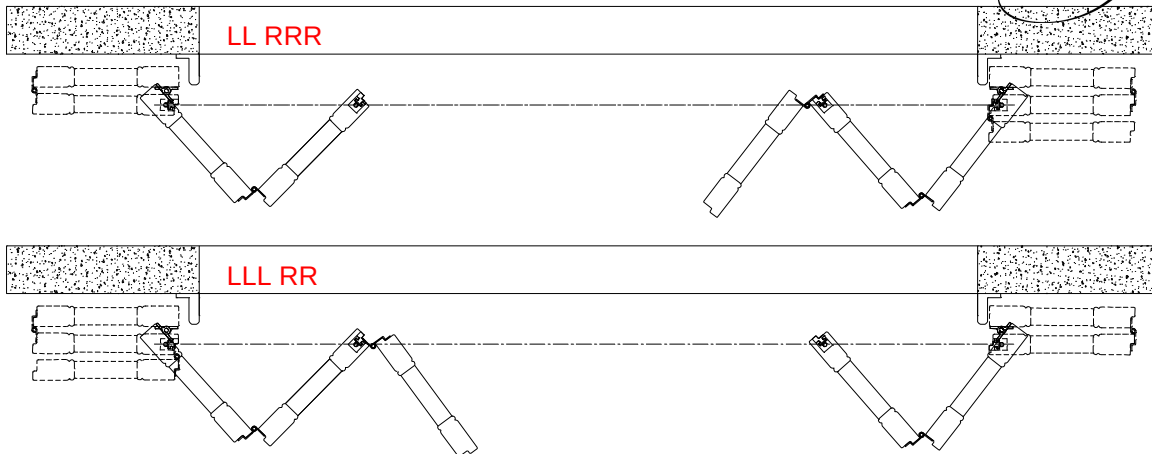


For 180 Bi-fold, if Bottom Pivot L Bracket is required, the extension for the L shape light block will be 1 3/4" in width. Please see picture below:



Panel Configuration : Only 2", 2 1/4" rabbet and butt stiles are available. Center stile is not available.





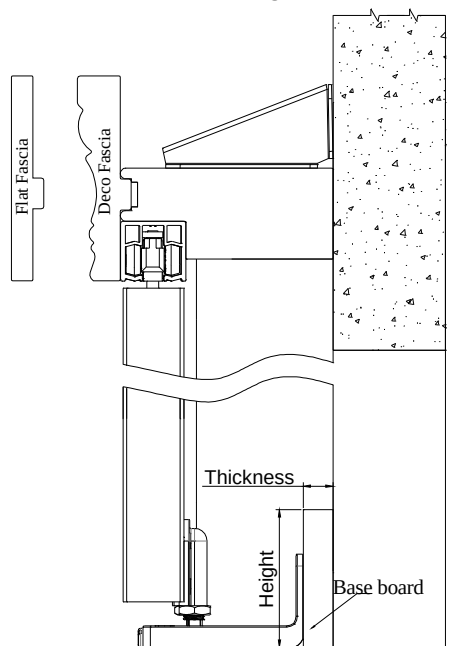
Notes:

If there is no baseboard to the wall, 3" header can be used.

If there is baseboard to the wall, then it should be as below:

1. If base board thickness is less or equal to 5/8", 3 1/2" header can be used.
2. If base board thickness is over 5/8", it is suggest using the 3" header with build out. And the build out thickness=base board thickness-1/8".

Please see the drawing below:



180° Bi-fold Track Shutters Framing Materials

Type	Header Dimension	Top Fascia Dimension	L Shape Light Block Strip Dimension
Header & Top Fascia & Light Block Strip	3"	3 3/8"	1 9/16"
	3 1/2"		2 1/16"

180° Bi-fold Track Shutters Hardware Box

Parts
Wrench
Hinge Pin
Triangle Bracket (based on the length of header)
2" screws for Triangle Bracket
¾" screws for Triangle Bracket
Bottom Pivot Bracket
1 ½" screws for bottom pivot bracket
2 ½" screws for Frame with pre-drilled screw holes

Bi-fold 90

Guidelines and Specs:

<u>Description</u>	<u>Minimum</u>	<u>Maximum</u>
Single-Panel Width (Bi-fold Track)	6"	26"
Single-Panel Width (Multifold track up to 4 panels)	6"	20"
Single-Panel Height	10"	132"

Outside Mount

Max frame-to-frame width: window width + 3 1/2"

Max frame-to-frame height: window height + 4 1/2"

Semi-Inside Mount

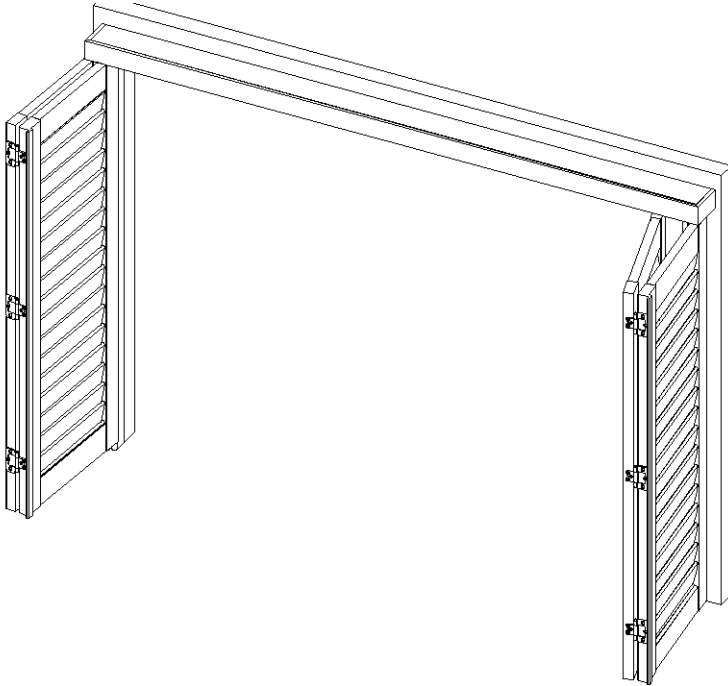
Max frame-to-frame width: window width - 1/8"

Max frame-to-frame height: window height + 1 3/8"

Inside Mount

Max frame-to-frame width: window width - 1/8"

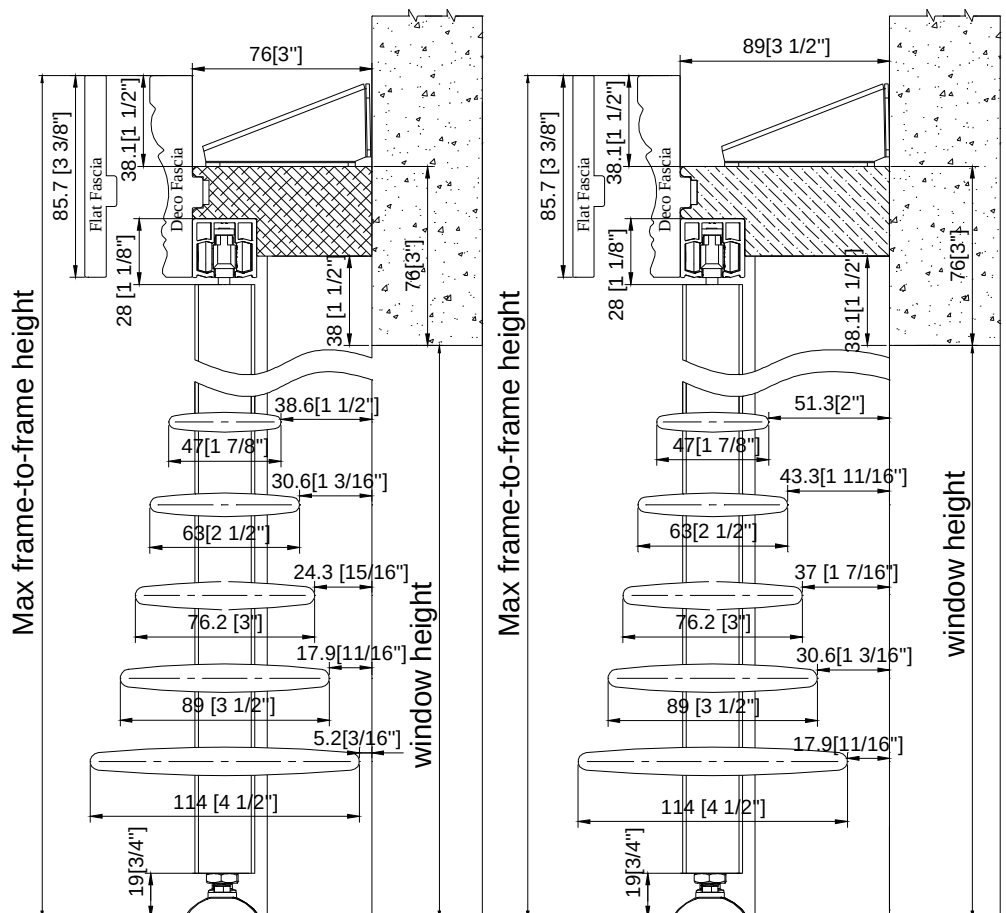
Max frame-to-frame height: window height - 1/8"



Track is pre-installed on the header; standard 1 track only.

Mounting Options: Outside Mount

Side View



Note:

-
- Technical drawings showing cross-sections of a window frame and header. The left drawing shows a side view of the frame with dimensions: Window Width, 1 3/4", 1/2", Header Width, 1/8", and Max Frame to Frame Width. The right drawing shows a top view of the header with dimensions: 1 3/4", 1/2", Header Width, and 1/8".

The technical drawings illustrate the dimensions of the MFF assembly. The top view shows the MFF width and total width, while the side view shows the MFF height.

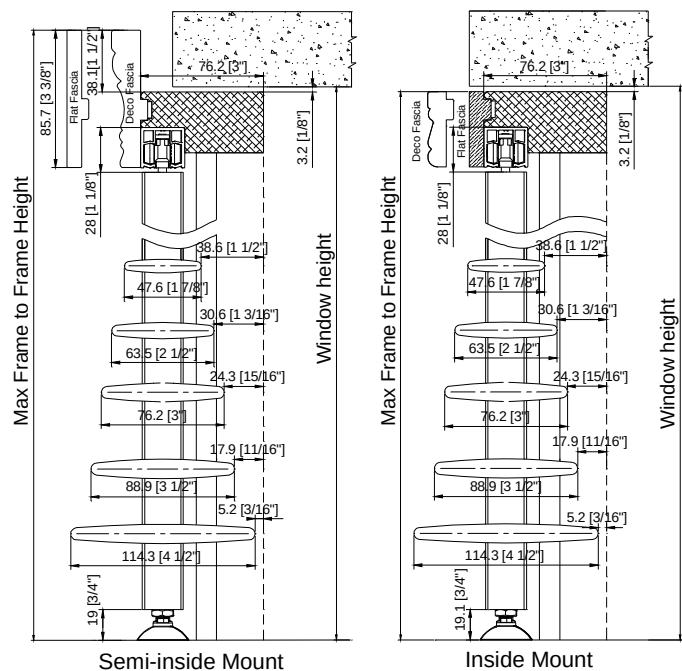
Top View Dimensions:

- existing casing width
- 5/8"
- MFF width=existing casing width + 1 1/4"
- Total width=existing casing width + 1 1/4" + 1"

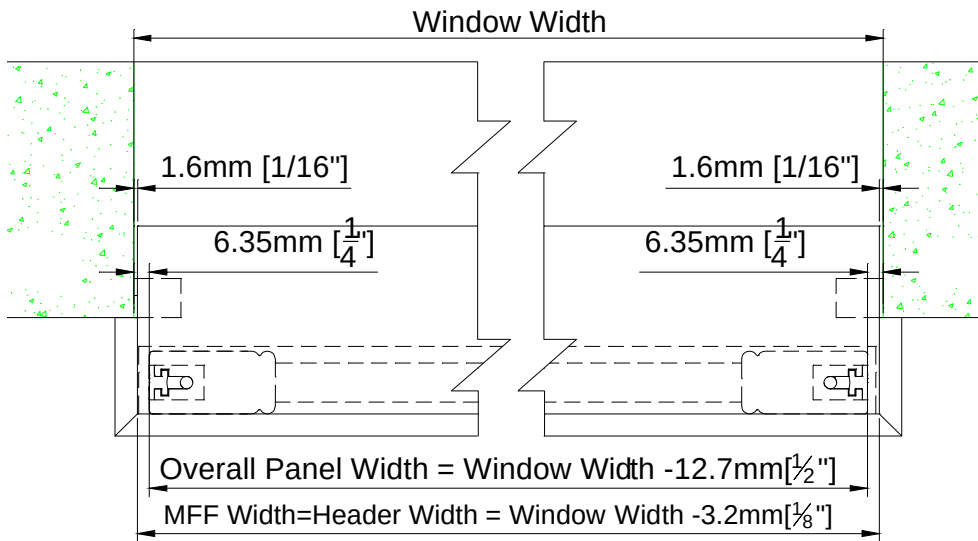
Side View Dimensions:

- MFF Height
- Flt Finish
- Body Finish

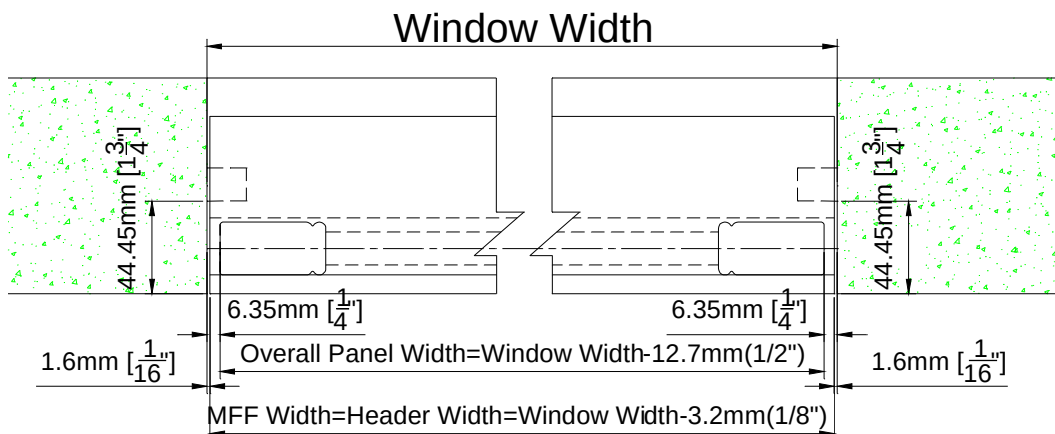
Side View



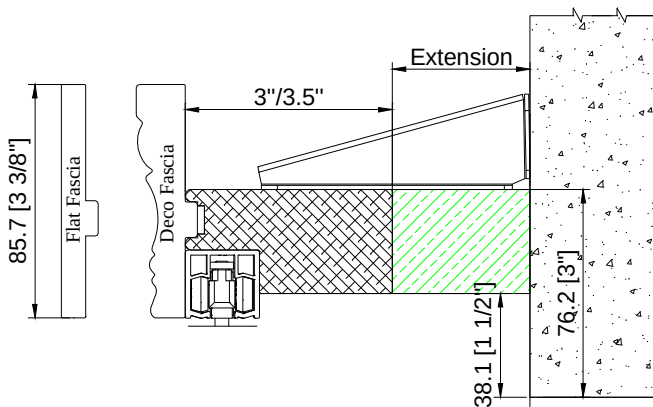
Semi-Inside Mount Top View



Inside Mount Top View

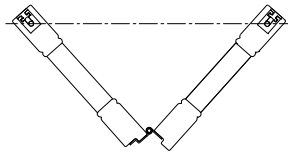


- ✧ Header Extension: Optional.
- ✧ Custom extension up to 2" for Header
- ✧ The extension will be pre-attached.

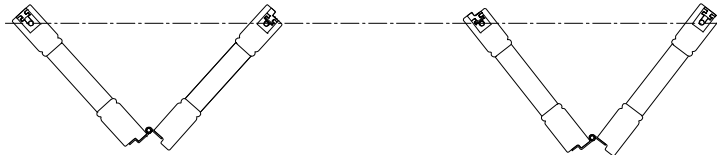


**Panel Configuration: Only 2", 2 1/4" rabbet and butt stiles are available.
Center stile is not available.**

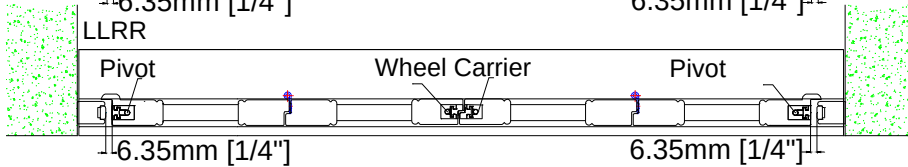
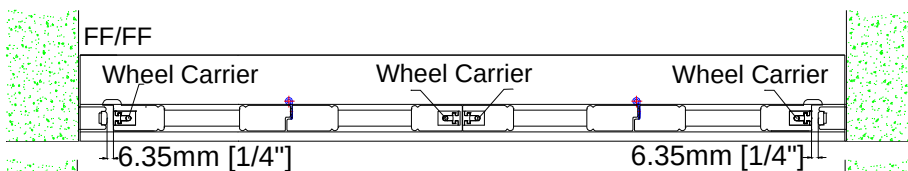
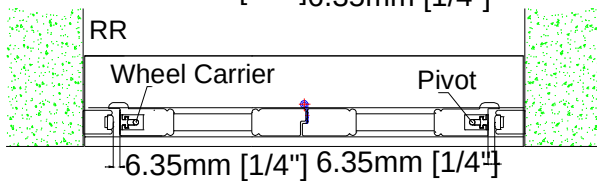
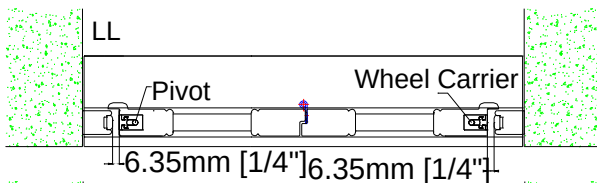
LL or RR



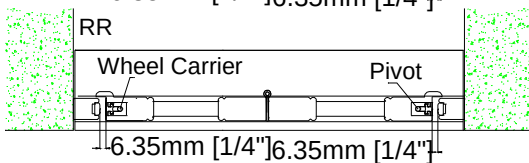
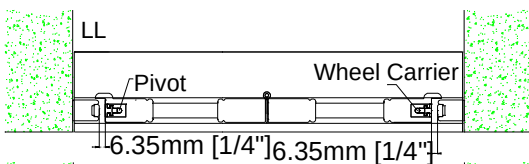
LLRR

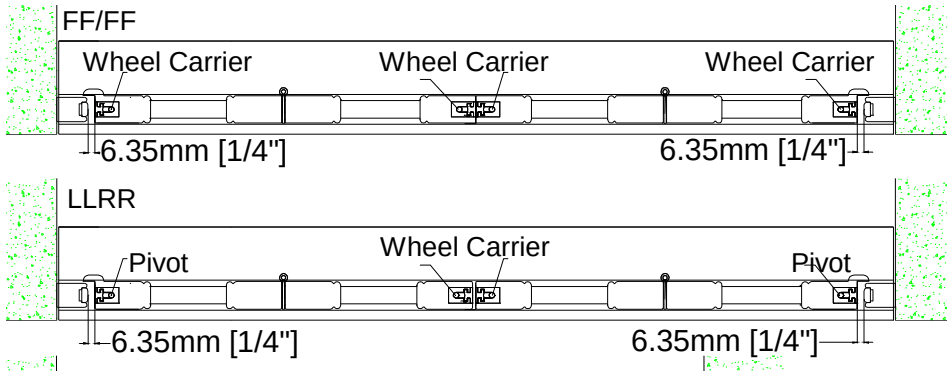


Optional (Hang Strip for Inside Mount Only)



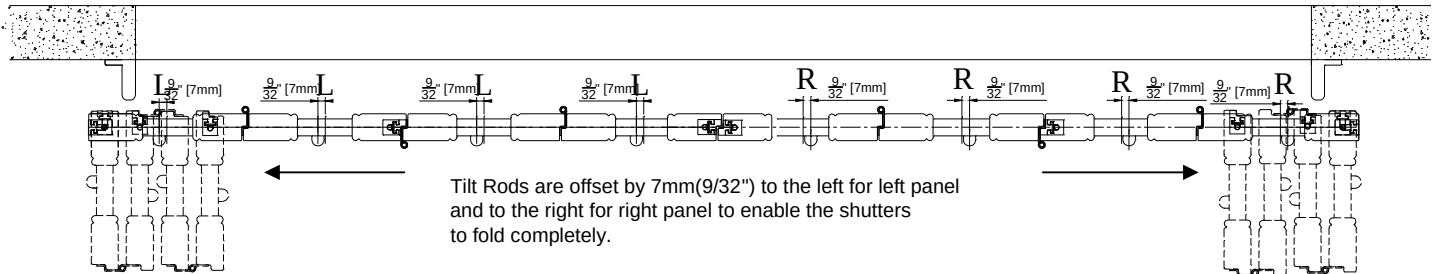
Note: When Butt/rabbet Stile is required, the stile next to the hang strip will be astragal stile.



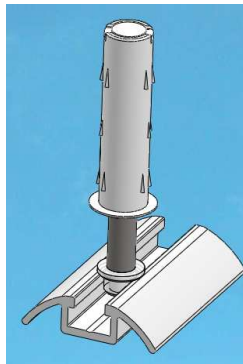
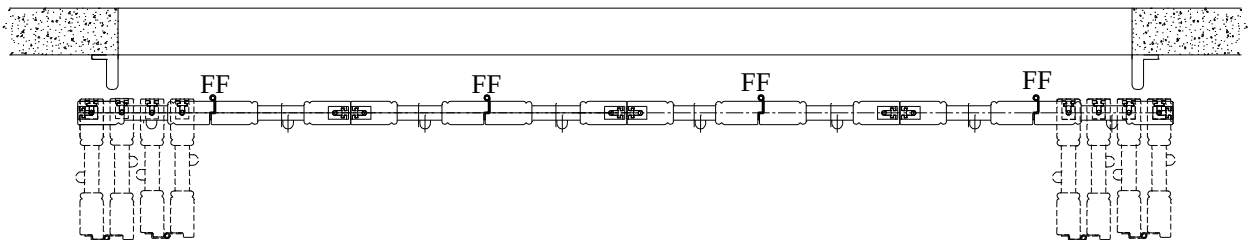


90° Multi-Fold Track Shutters (Wood only)

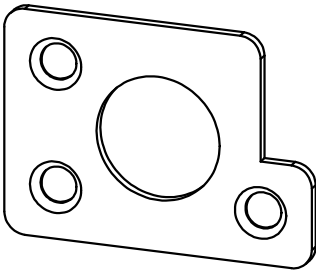
Maximum single panel width for multi-fold is 20"
 Panel Configuration: LLLL, RRRR, or LLLLRRRR



90° Floating Panels



- o Two or four panels can be hinged together, fold and slide to either direction
- o Application comes with wheel carrier, spring loaded guide and floor track
- o Can be used on large window



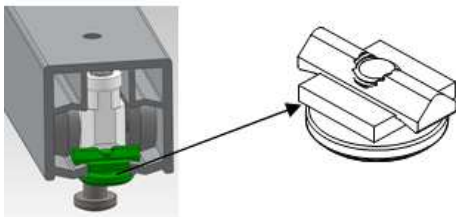
Spring loaded guide plate
(for wood panel with spring loaded guide in all Bifold track systems)

Note:

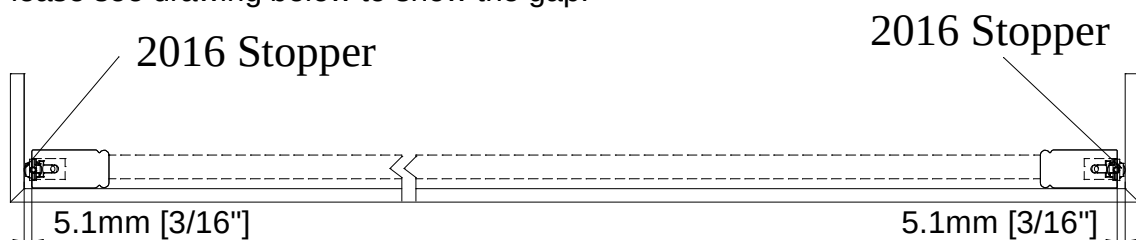
1. For 90° Floating Panels outside mount/semi-inside mount without side boards, the 2016 Stopper will be supplied by default. The gap between the fascia return / end of top track and the panel is 5mm each side.

The 2016 stopper can also be used in the center or any position of the top track as an option.

2016 Stopper



Please see drawing below to show the gap.



2. When rabbet stile required, the meeting stile of unhinged stile will be butt stile by default.

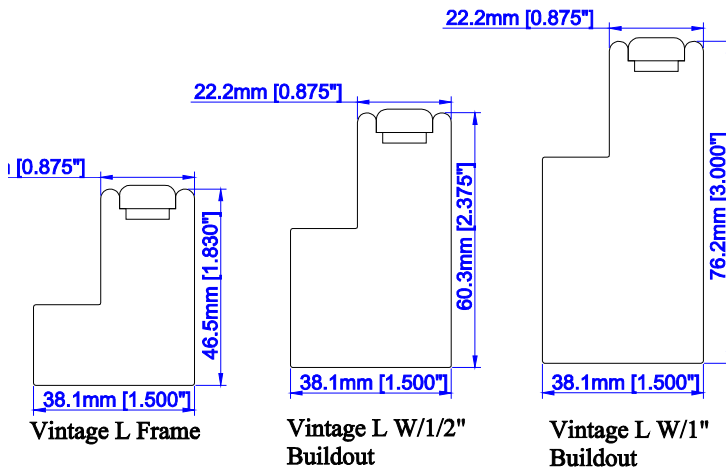
90 Degree Bi Fold Track Shutters Framing Materials

Type		Header Dimension	Top Fascia Dimension	Light Block Strip Dimension	Hang Strip Dimension
Header & Top Fascia & Light Block Strip	OM	3"	3 3/8"	1 9/16"	
		3 1/2"		2 1/16"	
	Semi-IM	3"	1 7/8"	5/8"x3/4"	1.08"x0.875"
	IM	3"		5/8"x3/4"	

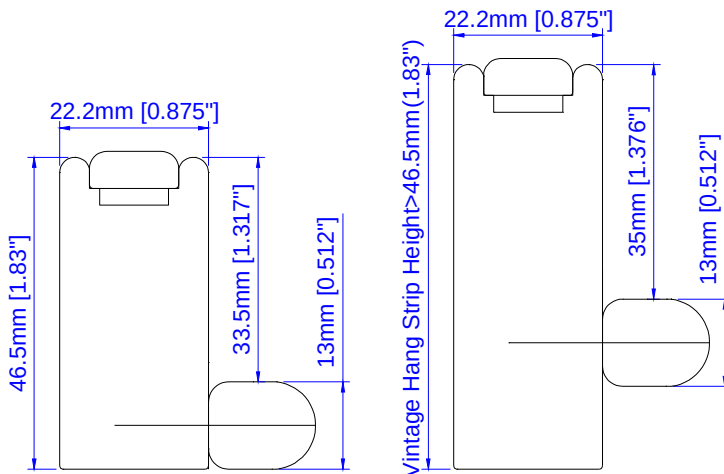
Note: It will be L shape light block strip for outside mount.

Frame Hinged Bifold

Available Stile: Rabbet Stile Only
Available Frame type

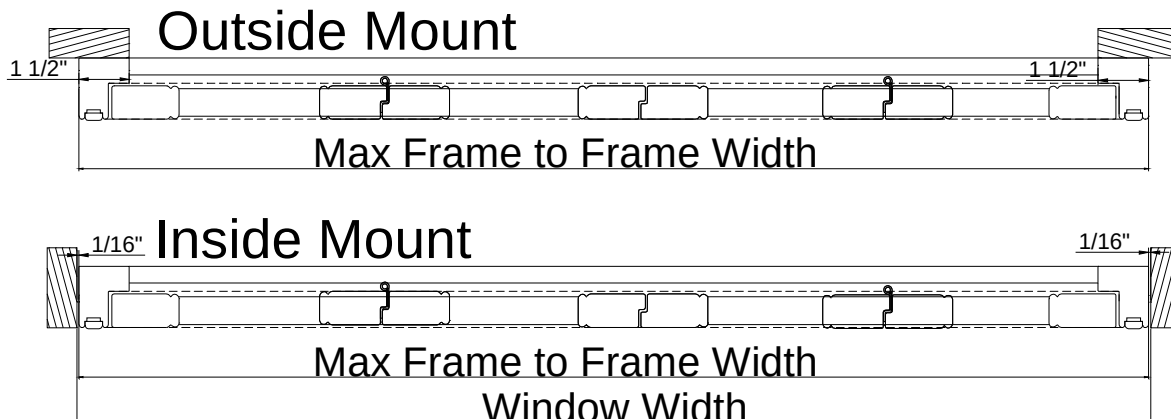


Vintage Hang Strip with 13mm light block: For single direction shutter with even panel qty, the frame on the unhinged side will be Vintage Hang Strip with 13mm [0.512"] light block, the height of the Vintage Hang Strip will match with that of the hinged side frame.



Top View

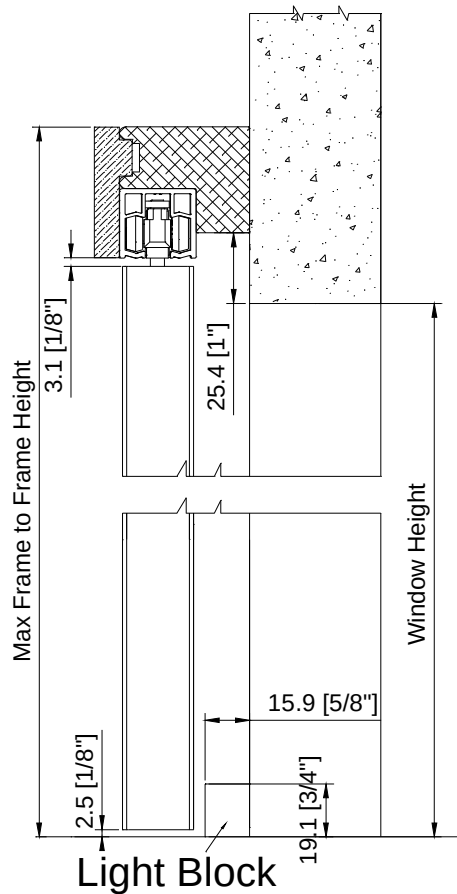
Hinged at Left or Right with wheel assembly at top;



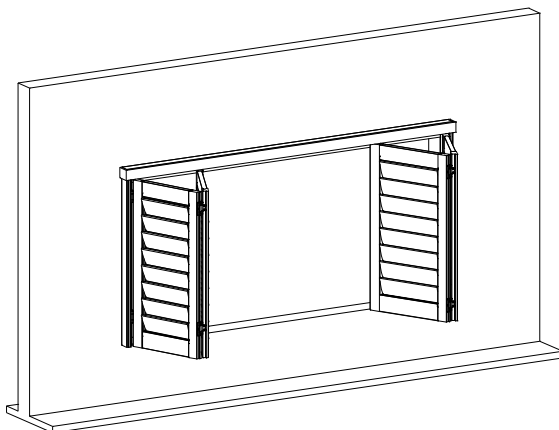
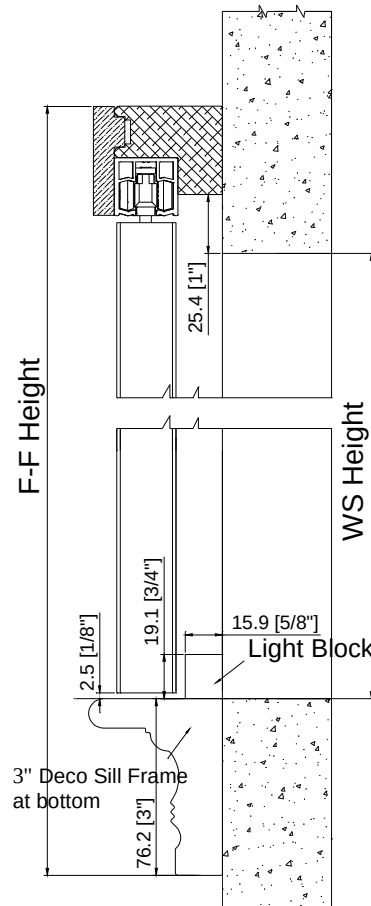
Side View

Outside Mount

3-Sides + bottom light block

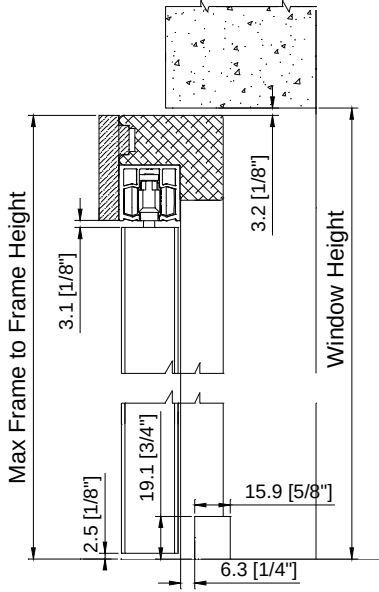


3-Sides + 3" Deco Sill Frame

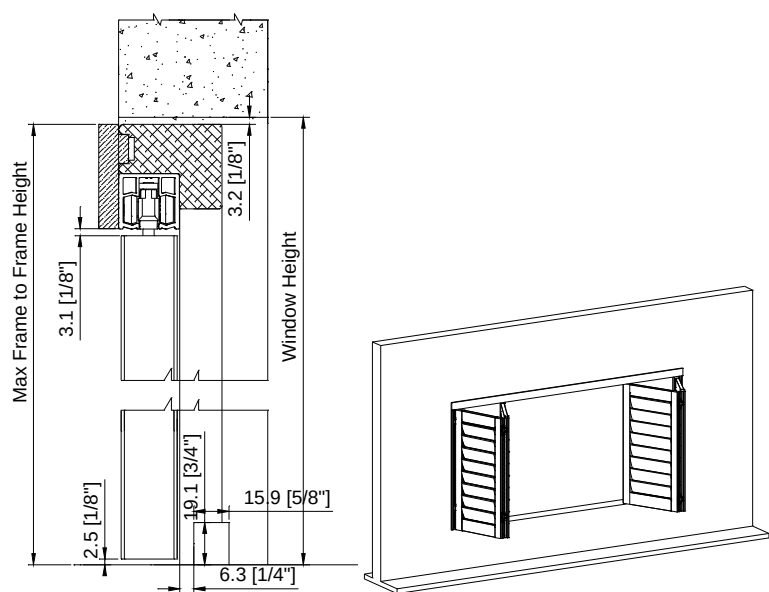


Note: For OM Frame Hinged Bifold shutter not to use as a door, if there is no existing sill at the bottom of the window opening, it is recommended to have the 3" Deco Sill Frame at the bottom.

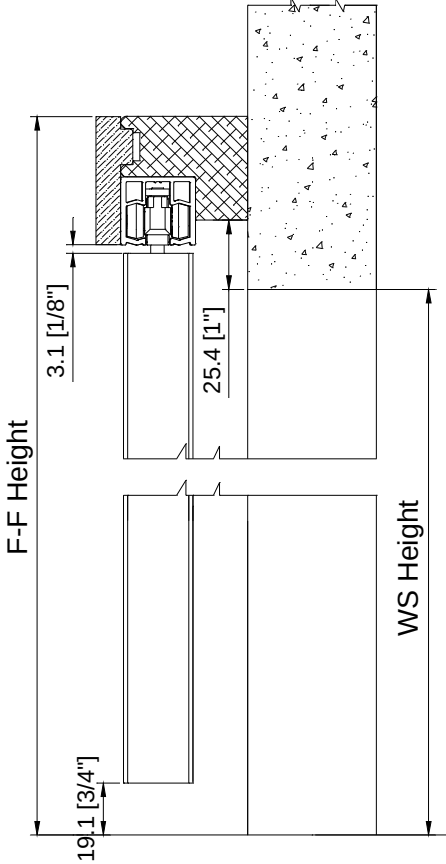
Semi-Inside



Inside Mount



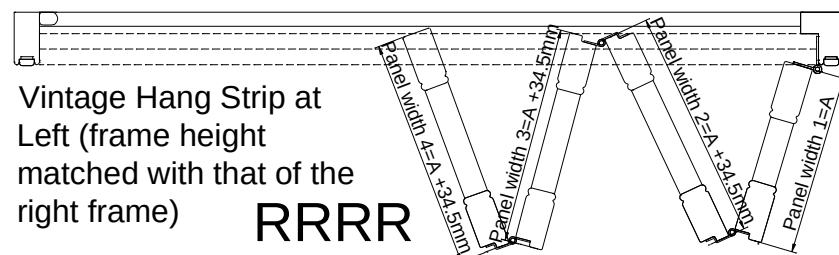
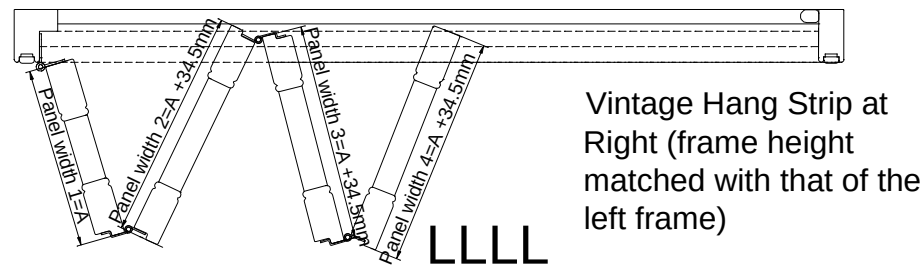
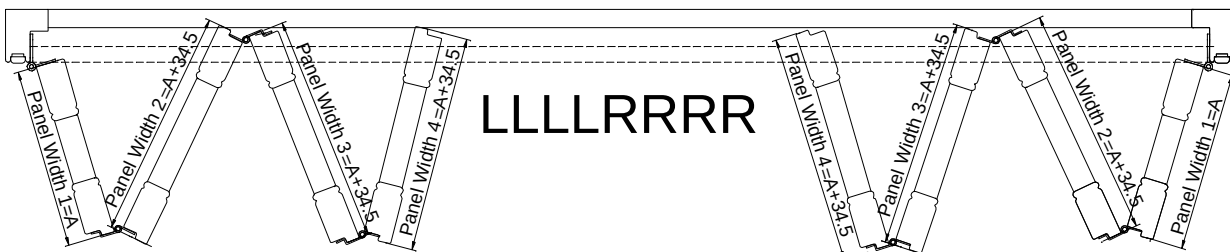
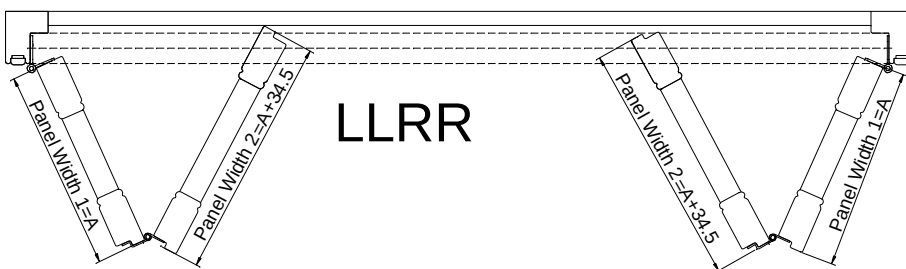
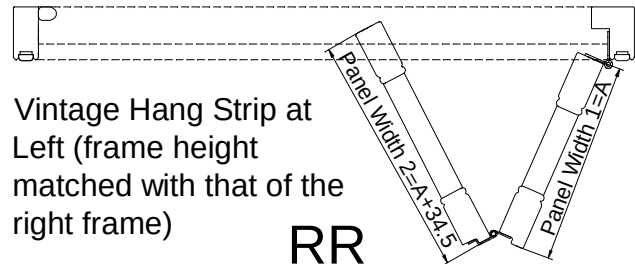
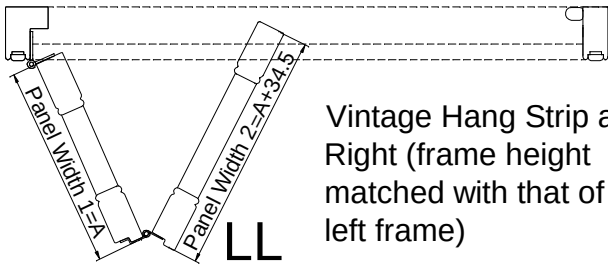
For frame hinged bifold track shutter used as door, the bottom gap will be 19mm by default. Please specify "Used As Door" on the order form.



Panel Configuration (uneven panel width)

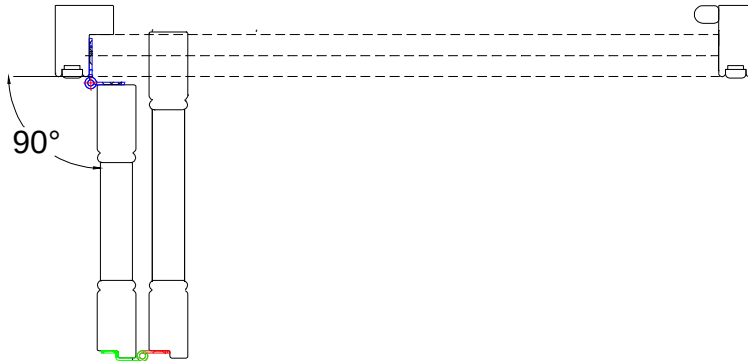
- 1) If the hinge type between panel and frame is 2 3/8" Self Mortise Hinge, the panel width1 (hinged with Self Mortise Hinge) = A, the other panels width (panel width2 / panel width3 / panel width4) = A + 34.5mm;
- 2) If the hinge type between panel and frame is Invisible Hinge, the panel width1 (hinged with Invisible Hinge) = A, the other panels width (panel width2 / panel width3 / panel width4) = A + 29.5mm;

Take the drawings of the hinge type between panel and frame is 2 3/8" Self Mortise Hinge as examples:



Notes:

1. Only available for 3-Sides + bottom light block or 3-Sides + 3" Deco Sill Frame (4-sides is not available); Panels can be folded and open at 90 degree;



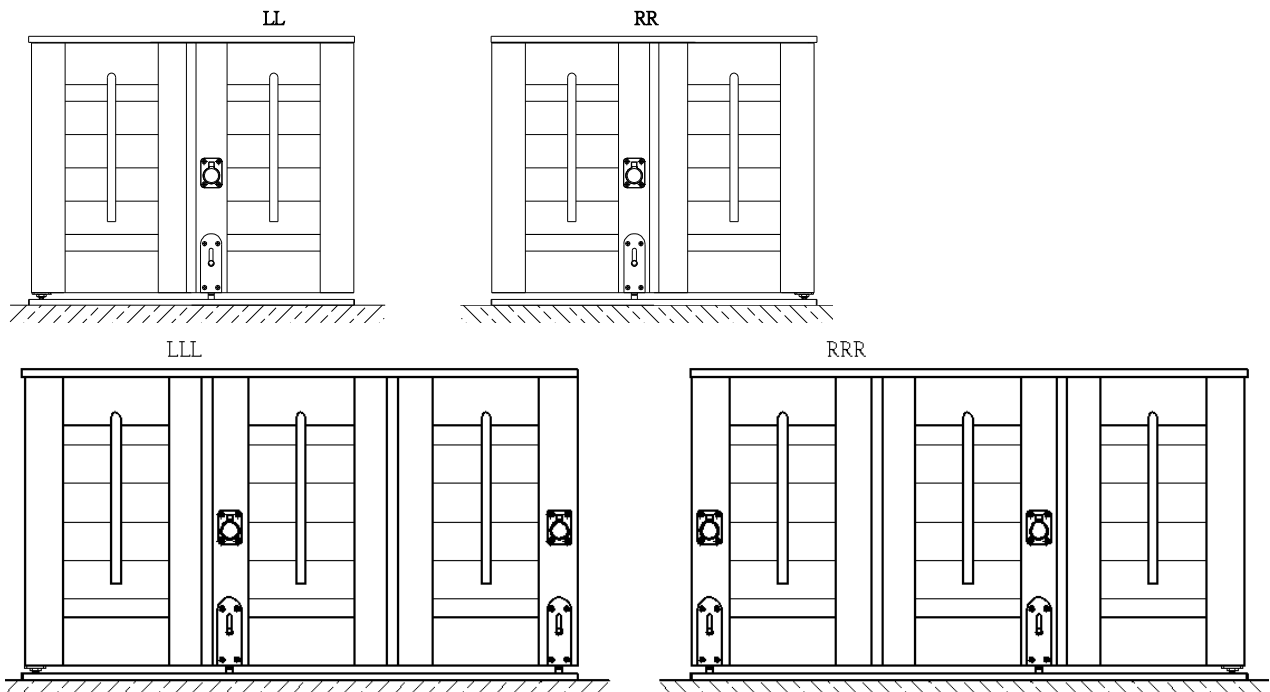
2. Frames available: Only Vintage L Frame

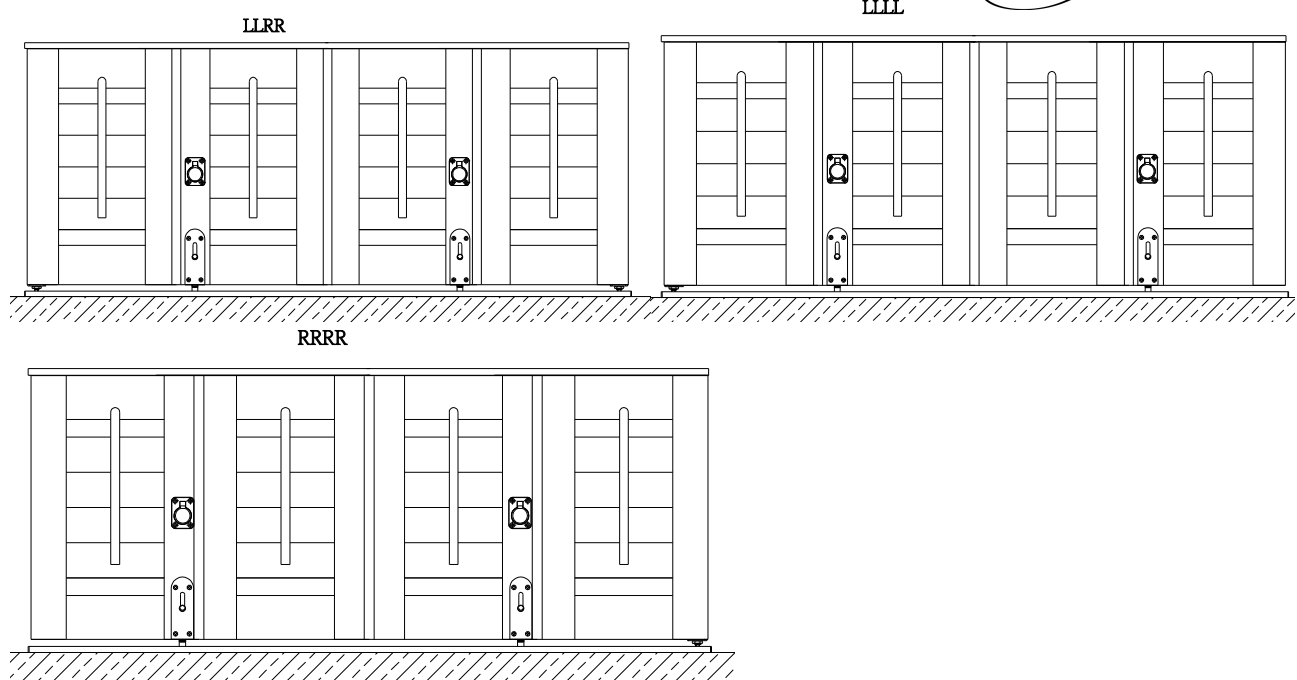
- 1) For top frame, the insert will be replaced by top fascia;
- 2) Top track will be pre-attached to the top frame;
- 3) Frame will be pre-drilled with screw holes same as regular shutters.

Ring Pull: The ring pull will be provided as an option; and for frame hinged bi-fold, the ring pull will be supplied as default; Customer need specify ring pull location on the order form. Ring pull location is measured from the floor to center of the ring pull in height direction.

Ring pull color will match the hinge color.

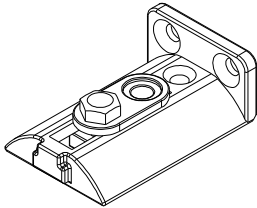
Please refer to below drawing of ring pull position for all shutter types:



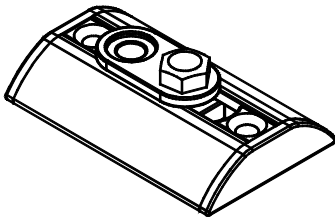


Bottom Pivot Bracket for 90 degree bifold:

For 90° Bifold Track Shutter, there will be 2 types of Bottom Pivot Brackets available.

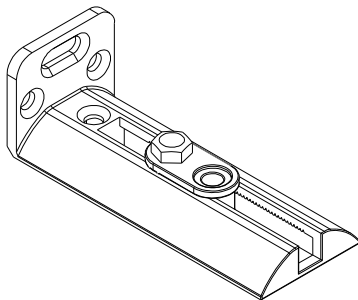


For Inside Mount, Bottom Pivot L Bracket will be supplied as default;



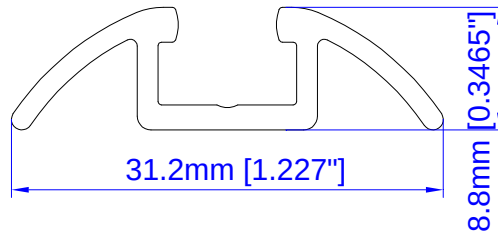
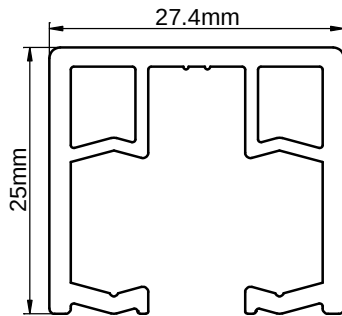
For Semi-Inside or Outside Mount, bottom pivot bracket will be supplied as default.

For 180° Bifold Track Shutter, 81mm Bottom Pivot L Bracket will be supplied as default.



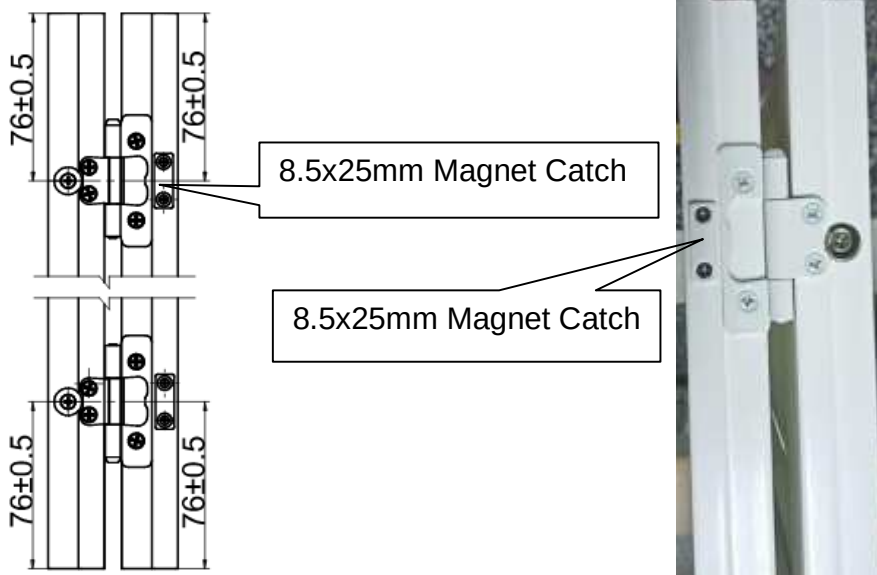
Track Dimension:

Top Track

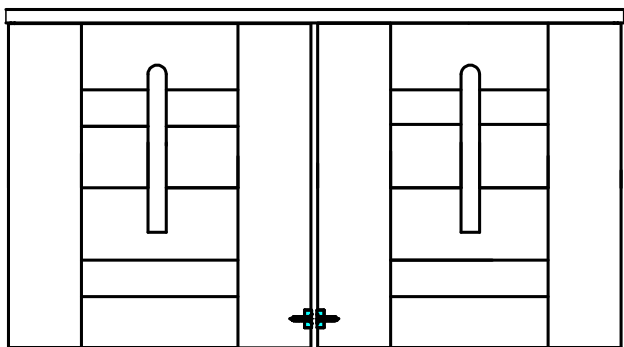


M Shape Bottom Track

Extra 8.5x25mm Magnet Catches and magnets will be supplied for Bifold hinge: (only for the most top and bottom hinges)



Side Magnet: applied to the meeting stiles of LR panels



Extra magnet and Side Magnet

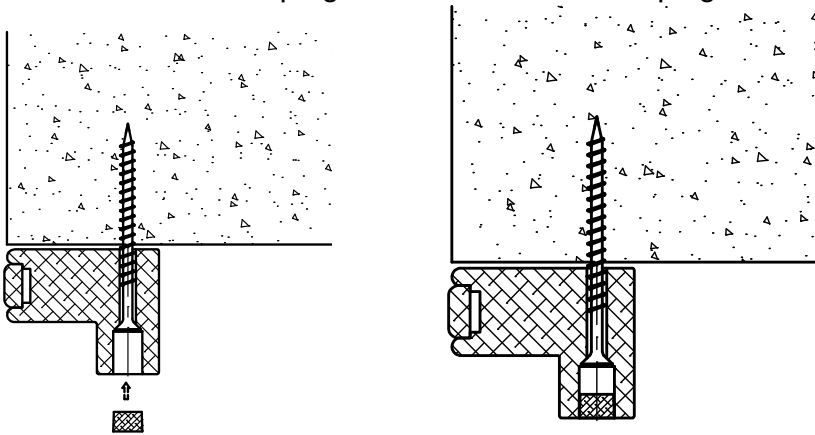


90° Bi Fold Track Shutters Hardware Box

Parts
Wrench
Hinge Pin
Triangle Bracket (based on the length of header)
2" screws for Triangle Bracket
¾" screws for Triangle Bracket
Bottom Pivot Bracket
38.1mm screws for bottom pivot bracket
63.5mm screws for Frame with pre-drilled screw holes

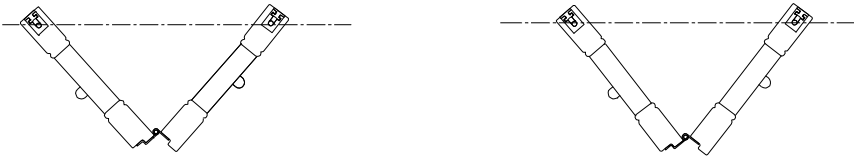
Notes:

1. for all bifold track shutter Outside Mount, the header will be pre-drilled with screw holes in the insert and screws will be supplied by default;
2. for all bifold track shutter Inside/Semi-Inside Mount, the header will be pre-drilled with holes for screws and wood plug and screws and wood plug will be supplied by default

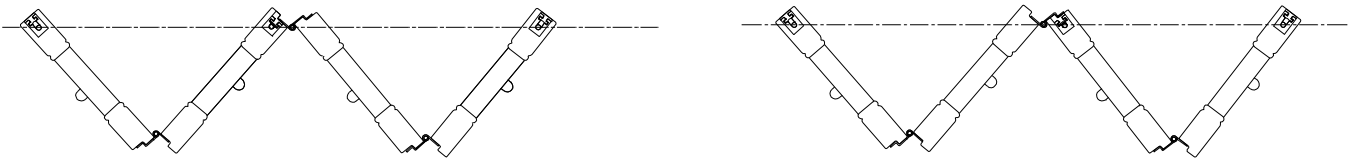


90 Degree Bifold Track & Floating Panels Hanger Location:

LL/RR/LLRR

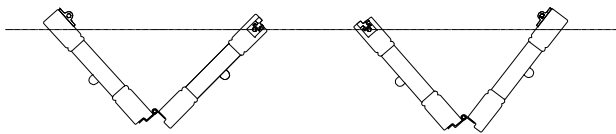


LLLL/RRRR/LLLLRRRR

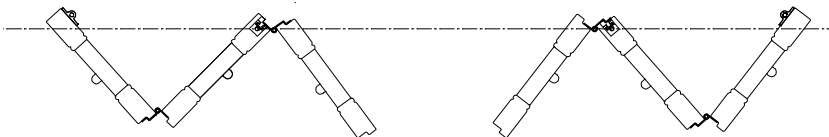


180 Degree Bifold Track Panel Hanger Location:

LL/RR/LLRR

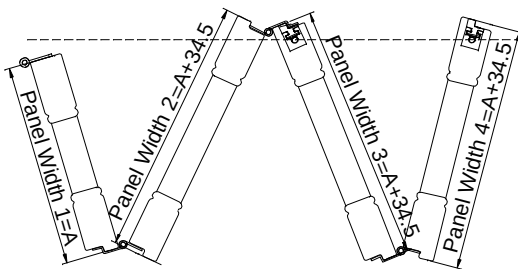
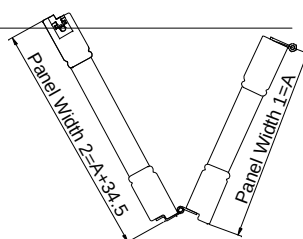
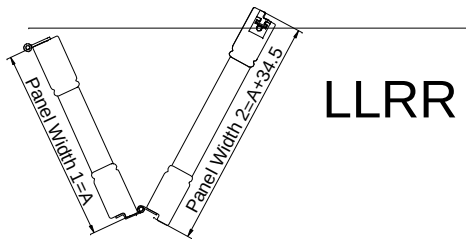
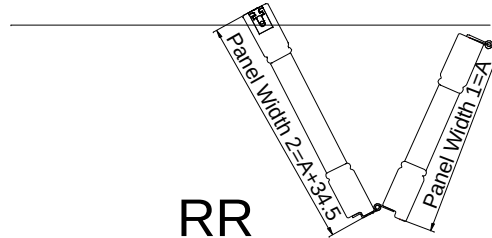
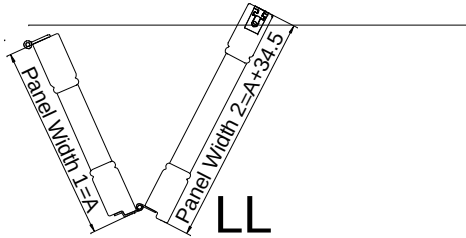


LLL/RRR/LLLRRR

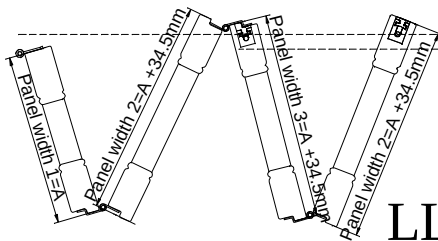
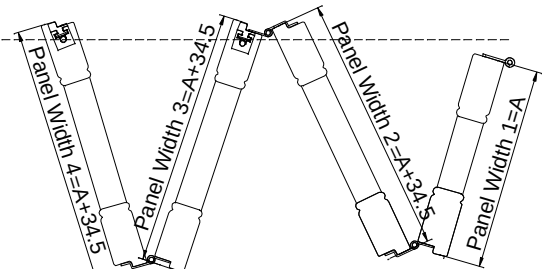


Frame Hinged Bifold Track Panel Hanger Location:

Take the drawings of the hinge type between panel and frame is 2 3/8" Self Mortise Hinge as examples:



LLLLRRRR

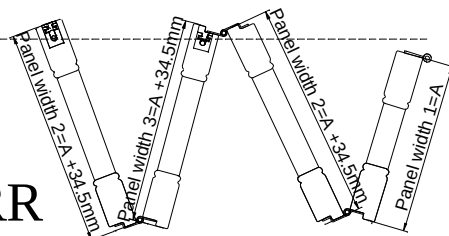


LLLL

Vintage Hang Strip at
Right (frame height
matched with that of the
left frame)

Vintage Hang Strip at
Left (frame height
matched with that of the
right frame)

RRRR

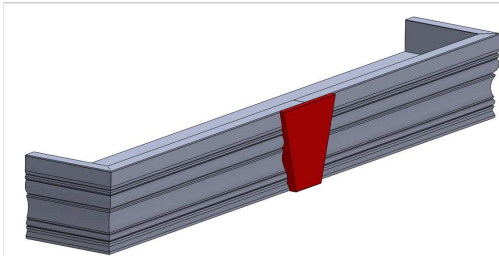


Keystone for Fascia of the Track Shutter

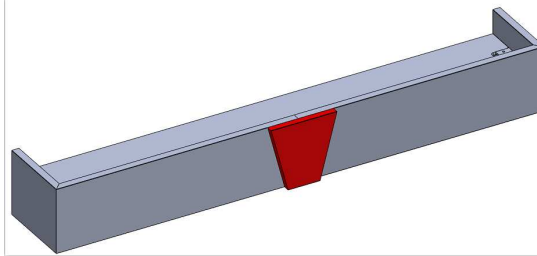
When fascia length exceeds the limitation and needs to split, keystone design fascia will be supplied without surcharge for track shutters.

For fascia length within the split limitation, keystone can also be supplied as an option with surcharge.

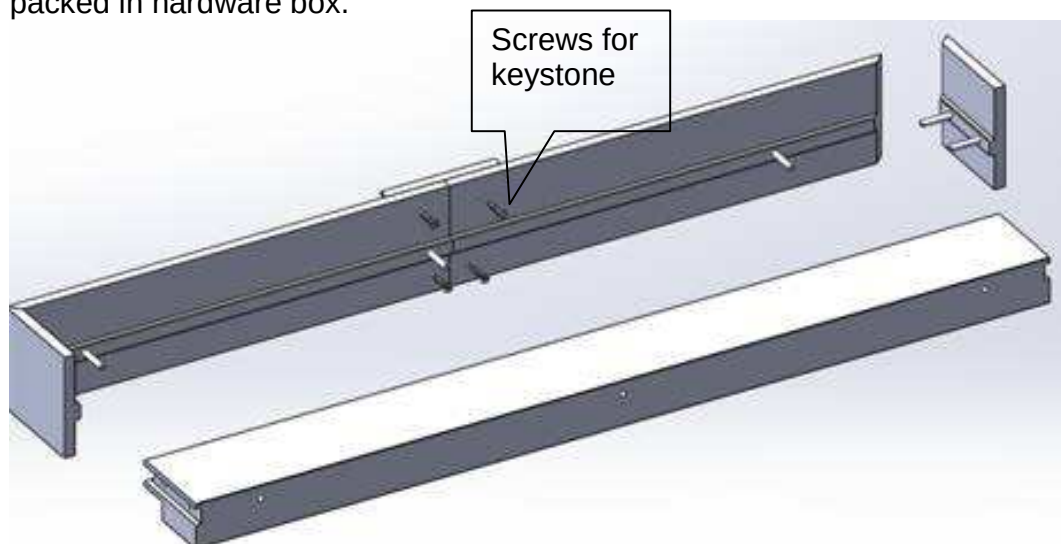
For Deco Fascia:



For Flat Fascia:



The keystone will be installed by screws. We will pre-drill the screw holes and the screws will be packed in hardware box.



Track Shutters—2020 Bypass

Definitions

Closed Bypass - louvers must be in the closed position for panels to bypass each other when sliding.

Open Bypass - louvers can remain open position when panels bypass each other allowing for a clearer view when sliding.

Combi-joined panels - two or more panels combined or hinged together to form a single, wider sliding section, please see the drawings of where hinges will be installed.

Note: the interlocking bottom guide is standard on the closed bypass and optional on the open bypass.

Guidelines and Specs:

<u>Description</u>	<u>Minimum</u>	<u>Maximum</u>		
		1 $\frac{7}{8}$ "	2 $\frac{1}{2}$ " ; 3"	3 $\frac{1}{2}$ " ; 4 $\frac{1}{2}$ "
Single-Panel Width (Wood)	6"	30"	36"	42"
Single-Panel Height	10"	132"	132"	132"

Outside Mount

Max frame-to-frame width: window width + 3"

Max frame-to-frame height: window height + 4 $\frac{1}{2}$ "

Semi-Inside Mount

Max frame-to-frame width: window width - $\frac{1}{8}$ "

Max frame-to-frame height: window height + 1 $\frac{1}{8}$ "

Inside Mount

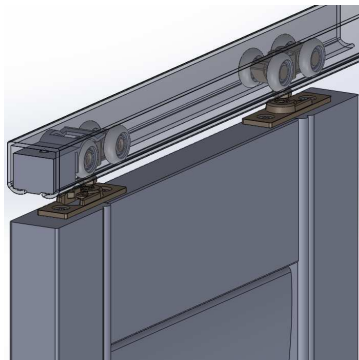
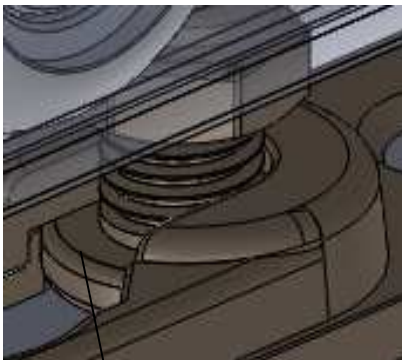
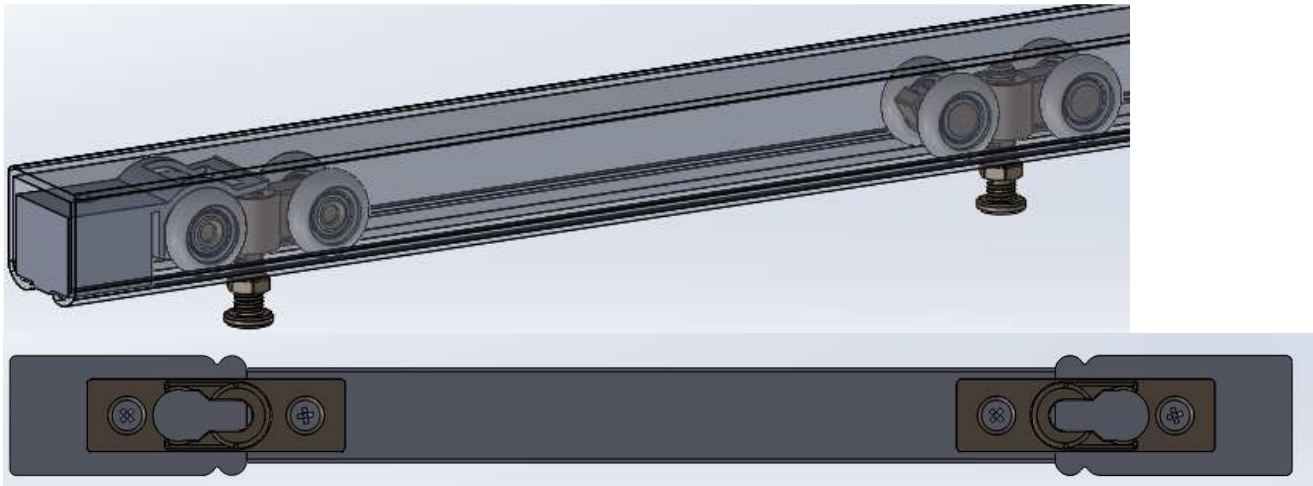
Max frame-to-frame width: window width - $\frac{1}{8}$ "

Max frame-to-frame height: window height - $\frac{1}{8}$ "

Track Shutters—2020 Bypass

Wheel Carriers are in the Track / Wheel Carrier Brackets are on the top of Panel:

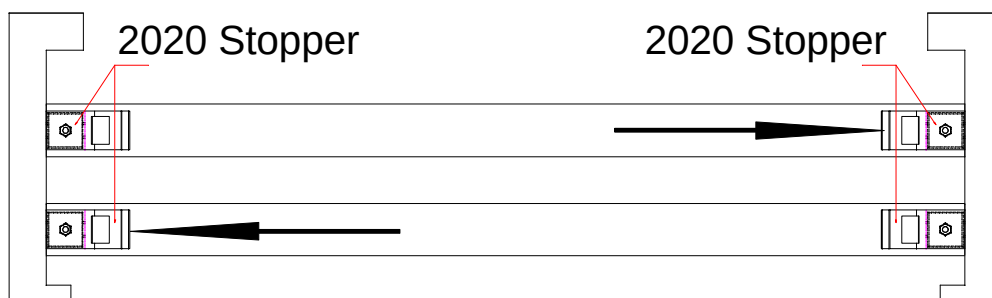
Adjustments to panel alignment are made by turning the wheel carrier bolt with the wrench included in the Hardware Box



Push wheel carrier into the wheel carrier bracket to seat

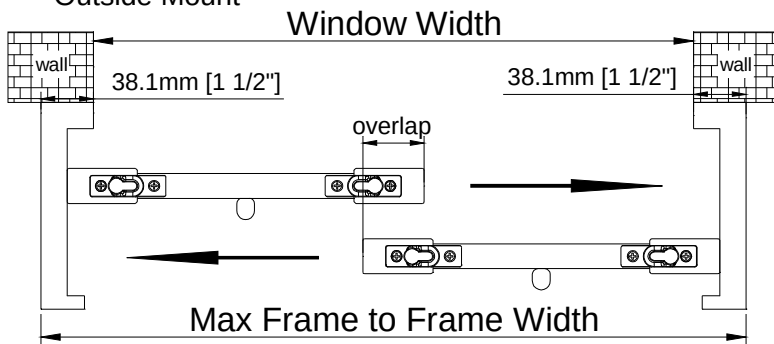
Note: 2020 Stopper will be supplied at both ends of top track by default.

Top View

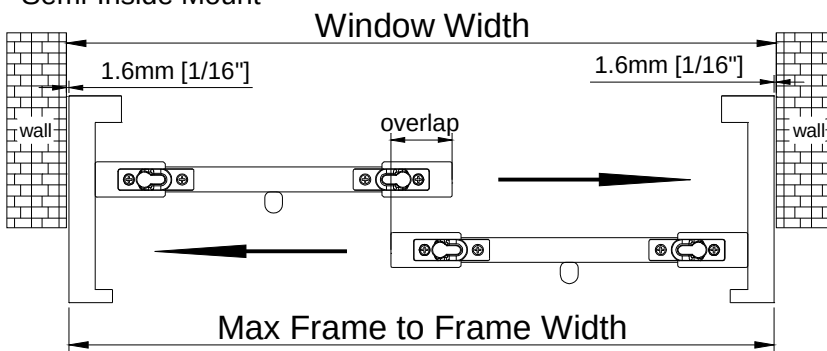


Track Shutters—2020 Bypass

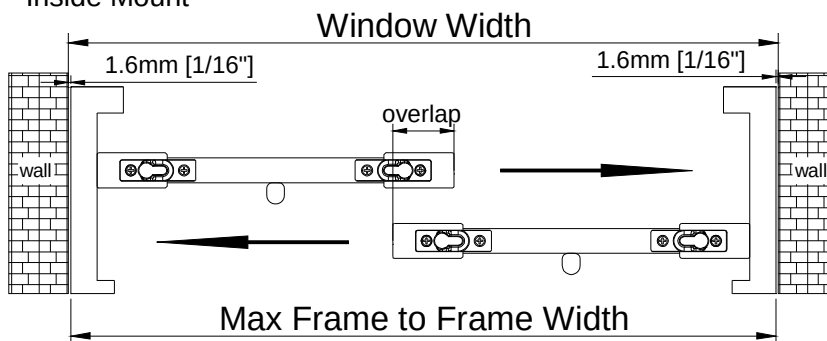
Outside Mount



Semi-Inside Mount

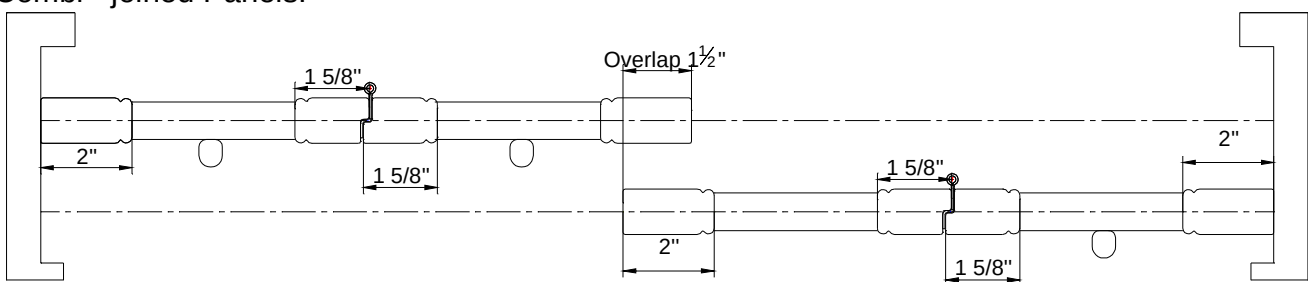


Inside Mount



Top View

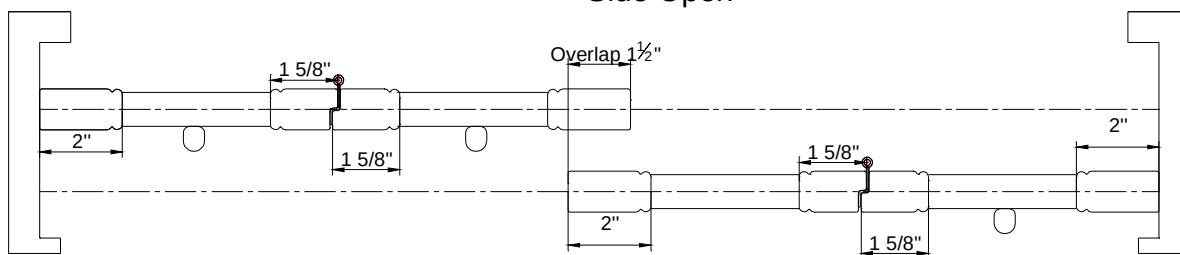
Combi - joined Panels:



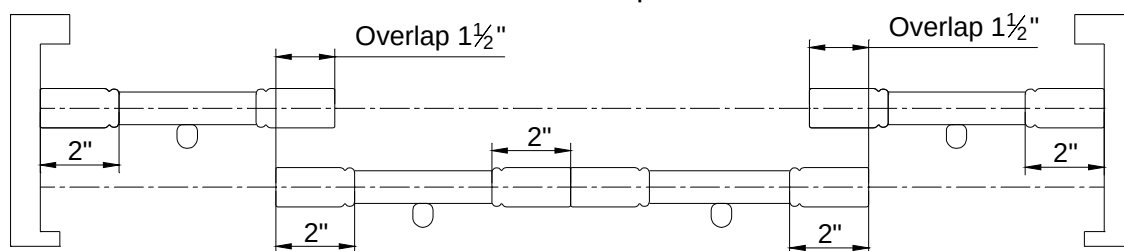
Track Shutters—2020 Bypass

For shutters with 4 or more panels, please select “side open” or “center open”

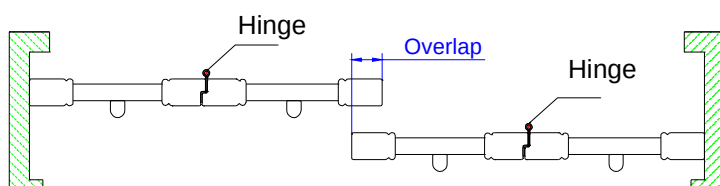
Side Open



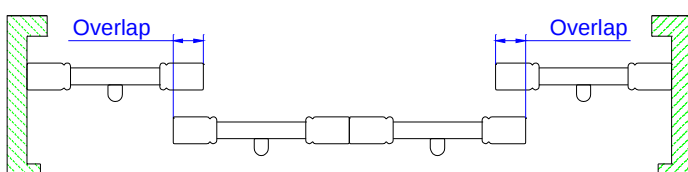
Center Open



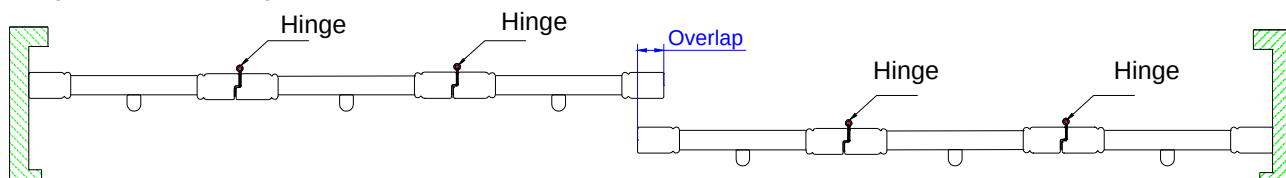
4 panels side open w/Rabbet stile



4 panels center open w/Butt stile



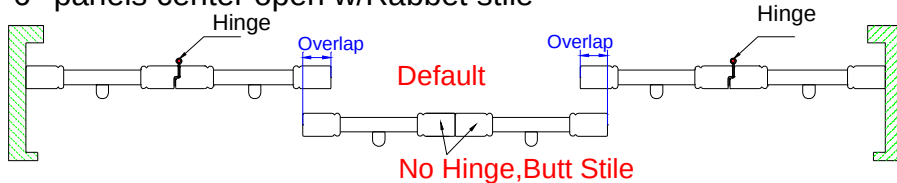
6 panels side open w/Rabbet stile



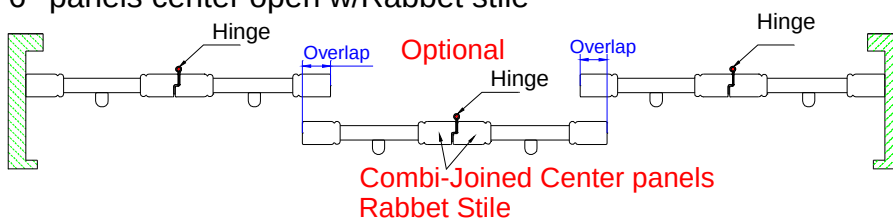
Track Shutters—2020 Bypass

For 6 panels center open configurations, the center panels should split (1/1) by default with the option for them to be combi-joined into a single 2 panel section.

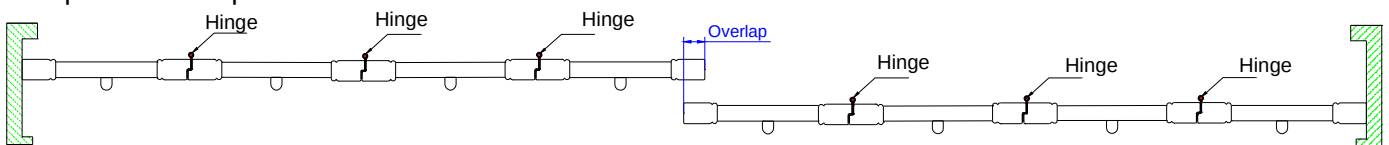
6 panels center open w/Rabbit stile



6 panels center open w/Rabbit stile

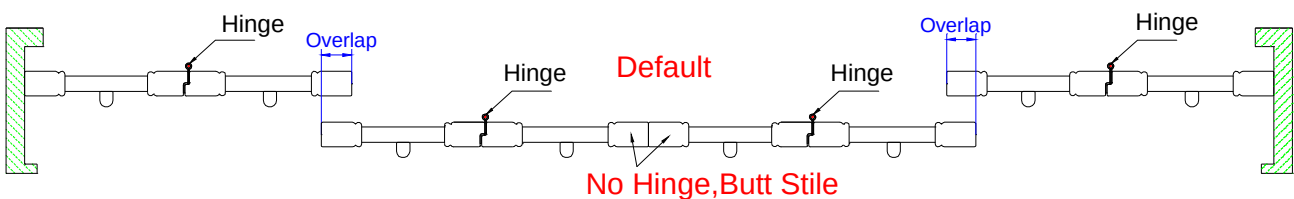


8 panels side open w/Rabbit stile

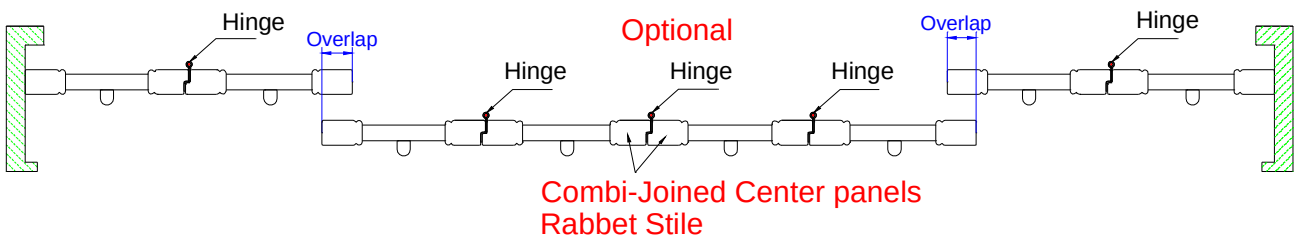


For 8 panels center open configurations, the center panels should split (2/2) by default with the option for them to be combi-joined into a single 4 panel section.

8 panels center open w/Rabbit stile



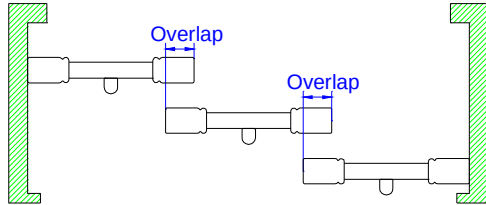
8 panels center open w/Rabbit stile



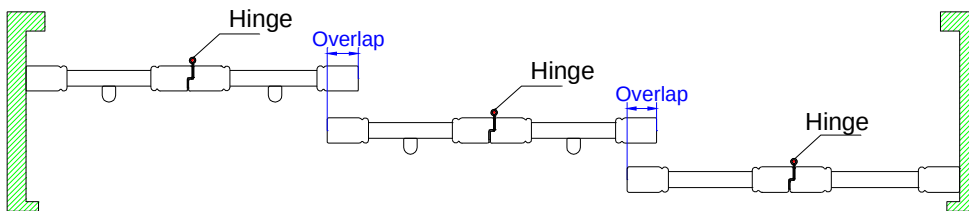
Track Shutters—2020 Bypass

Triple Track

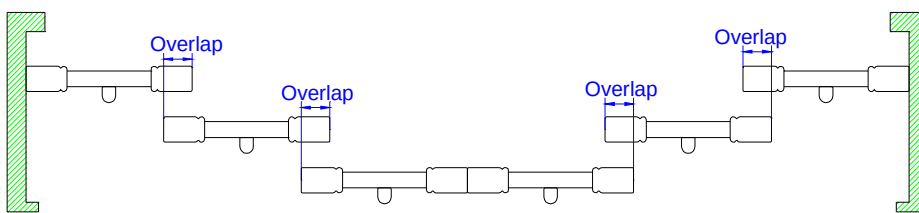
3 panels side open w/Butt stile



6 panels side open w/Rabbit stile



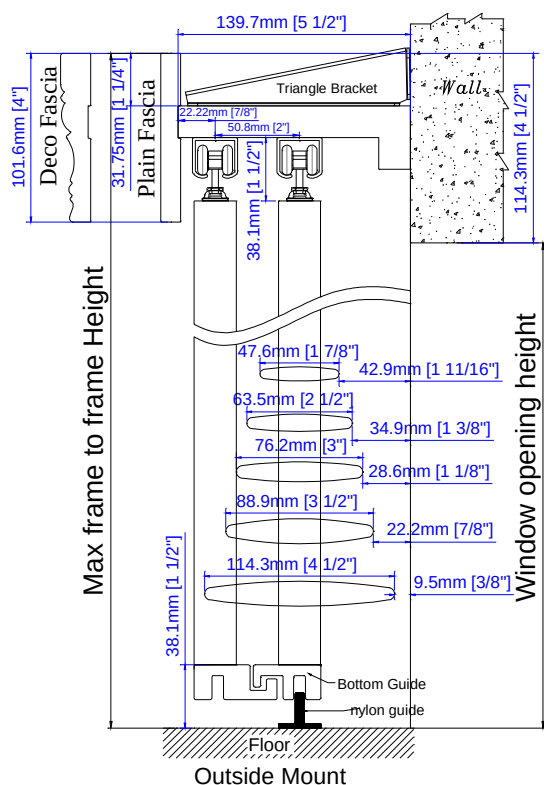
6 panels center open w/Butt stile



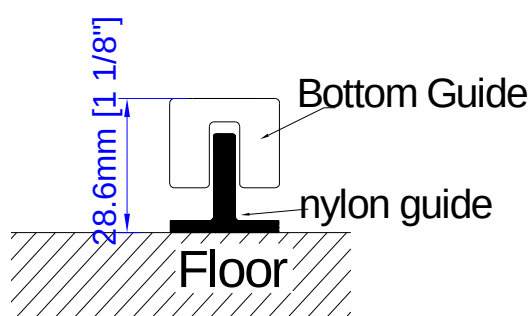
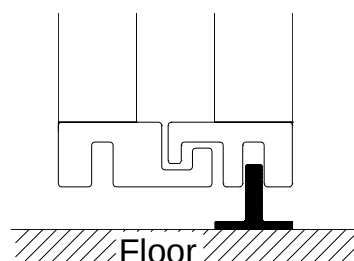
Stile: Only 2", 2 1/4" butt and rabbit stiles are available,
1 5/8" rabbit stile only used for combi-joined panels.
Center stile is not available.

Track Shutters—2020 Bypass Mounting Options (Closed Bypass)

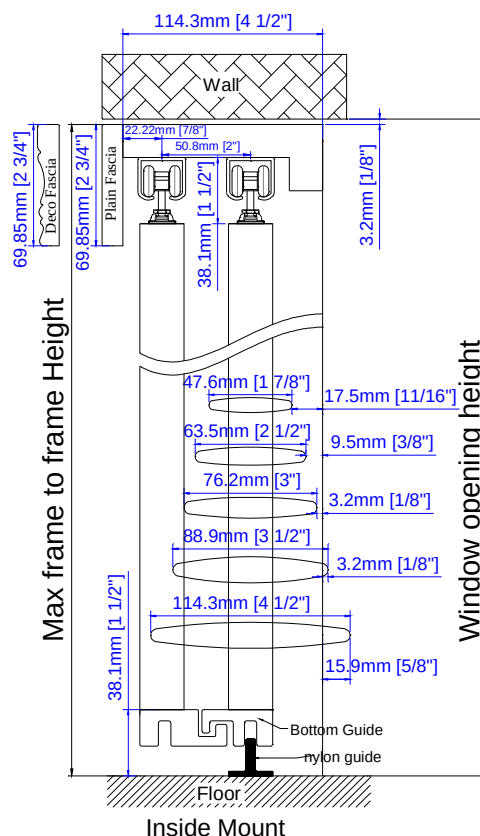
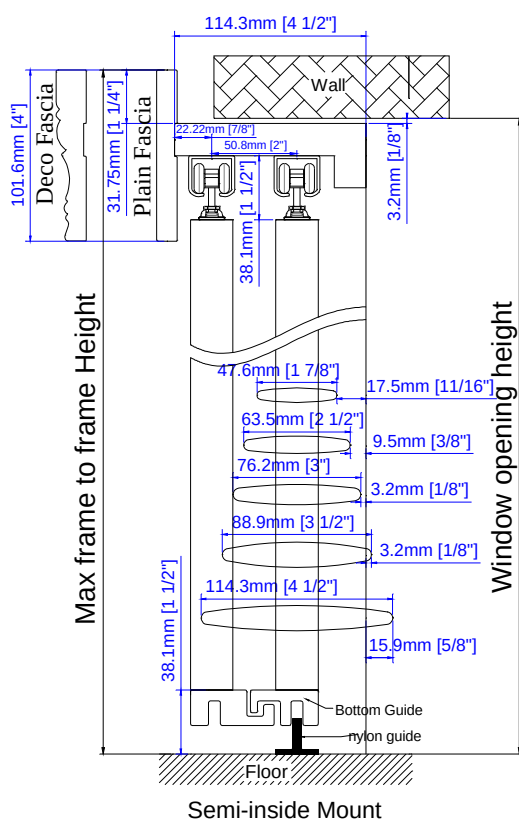
Side View



Bottom Guide

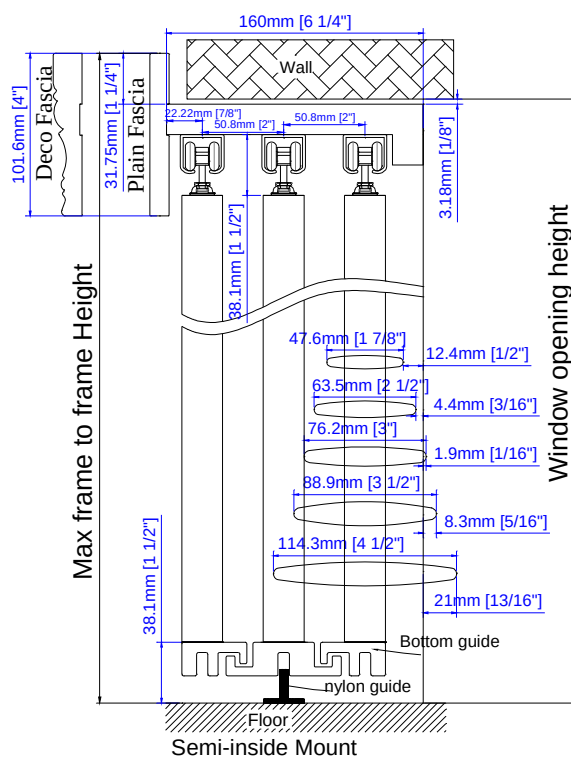
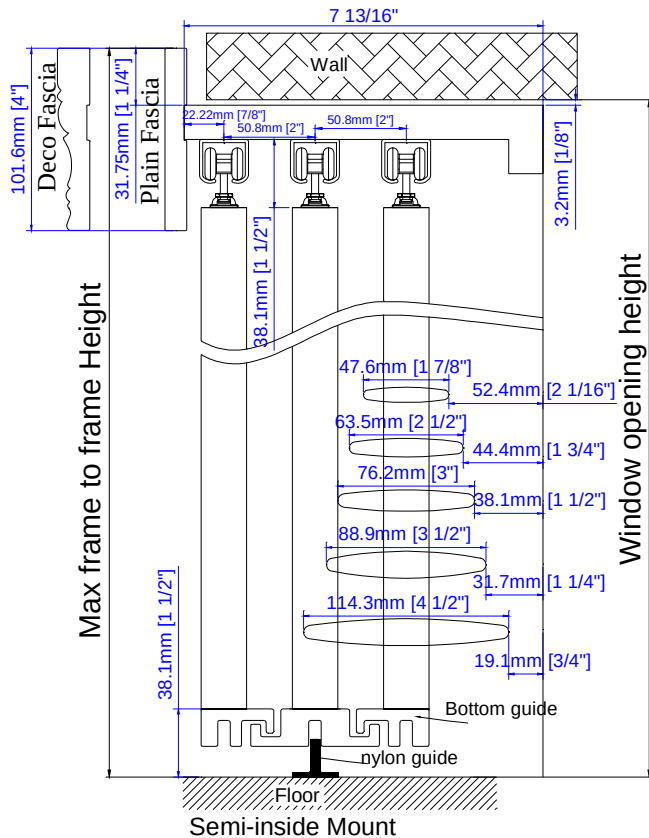
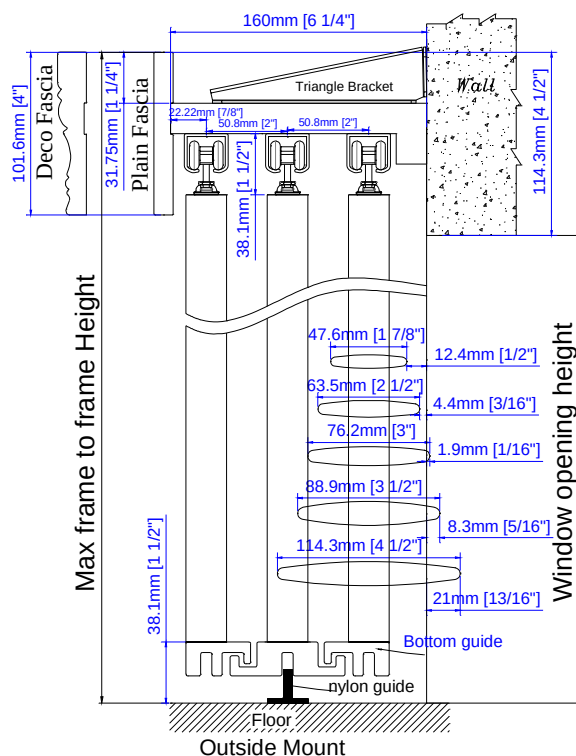
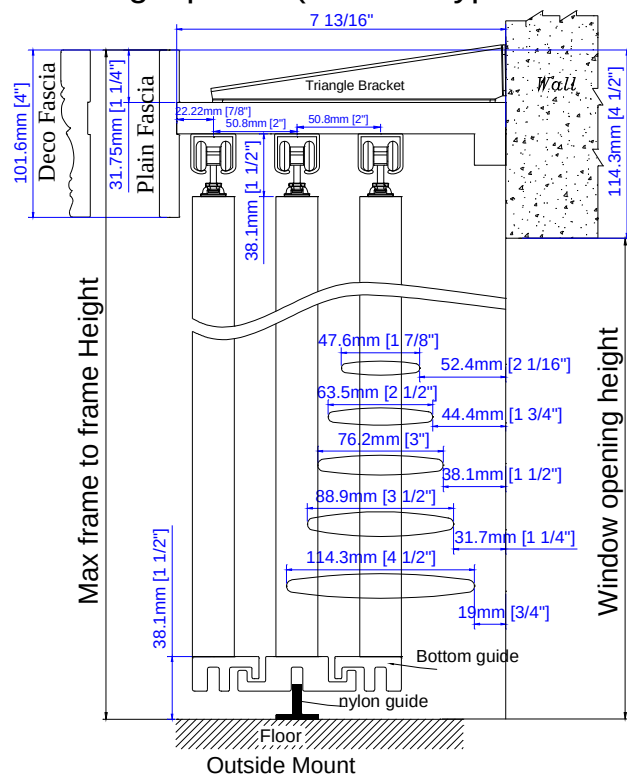


If the closed bypass only uses one track, the U-shaped bottom guide for the open bypass will be used at the bottom of the panel.

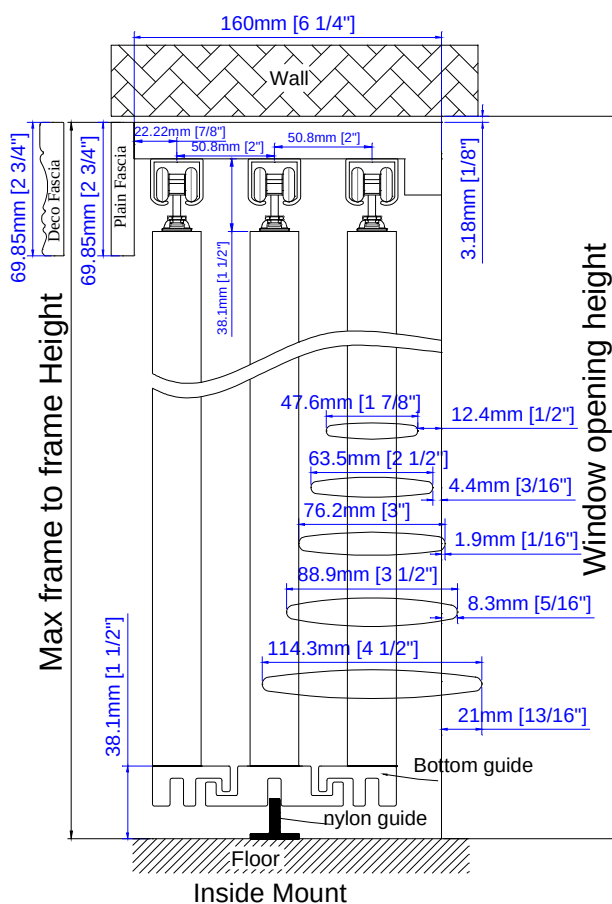
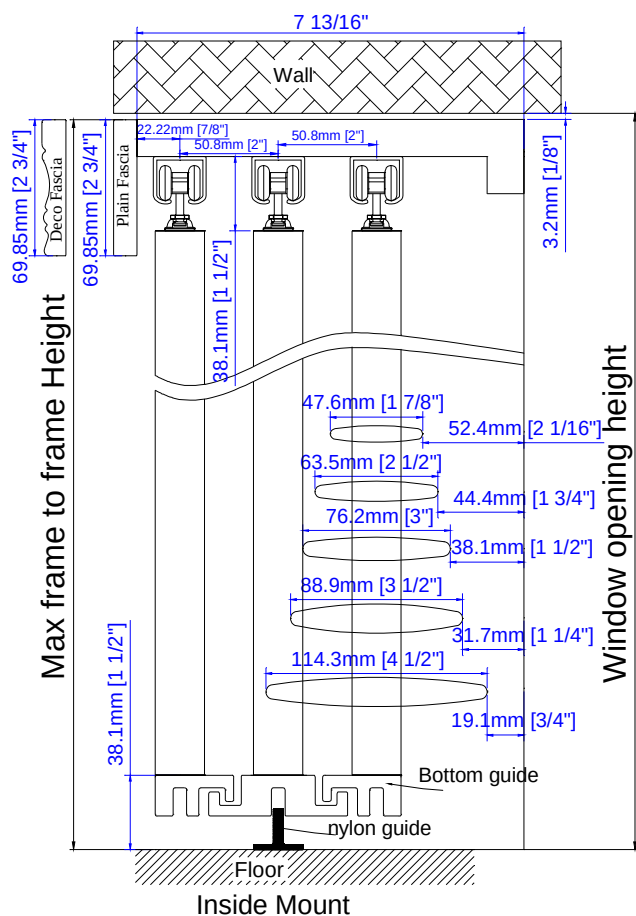


Track Shutters—2020 Bypass

Mounting Options (Closed Bypass with 3 Tracks, Triple tracks):



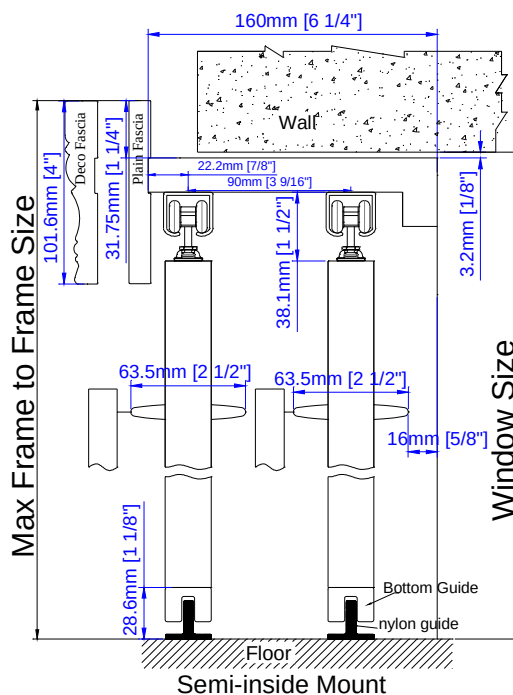
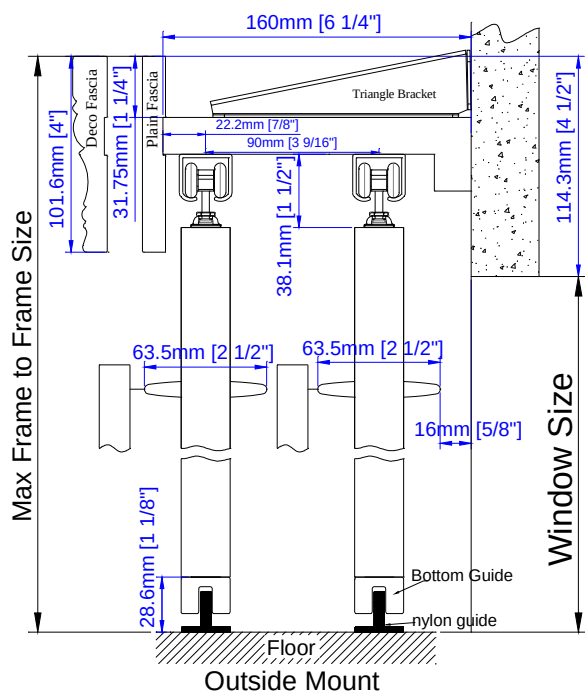
Track Shutters—2020 Bypass



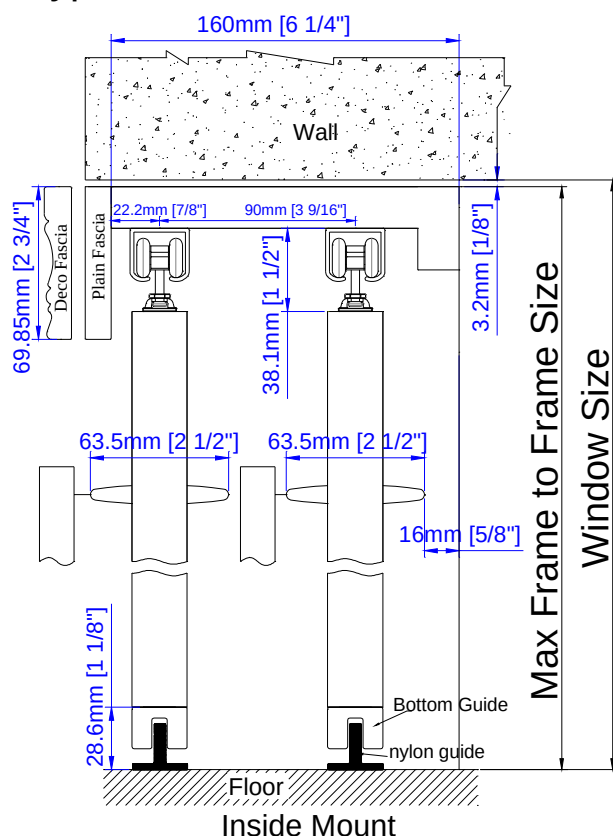
Mounting Options (Opened Bypass)

Only available in 2 1/2", 3", 3 1/2" and 4 1/2" louver sizes

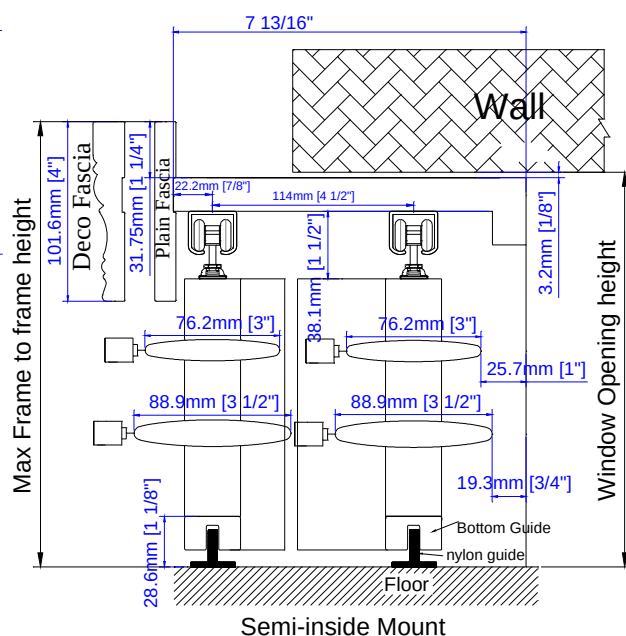
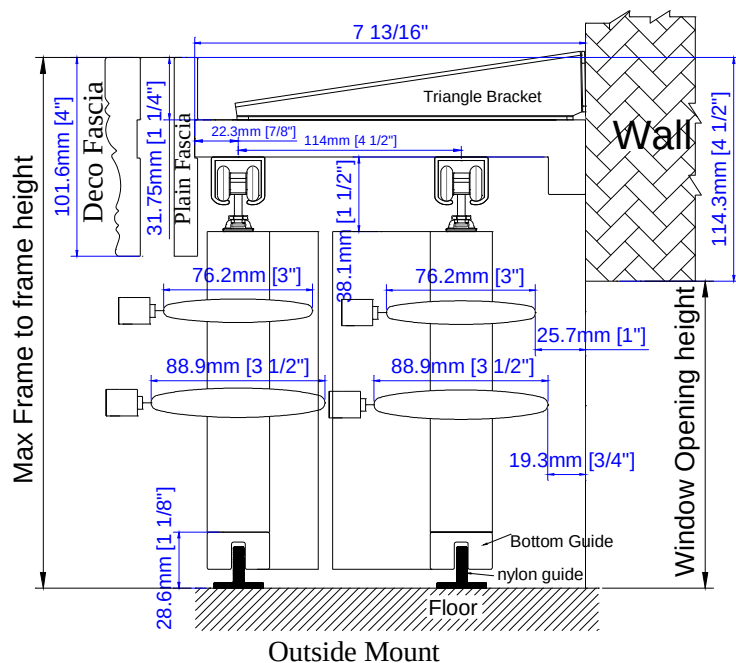
2 1/2" Louver



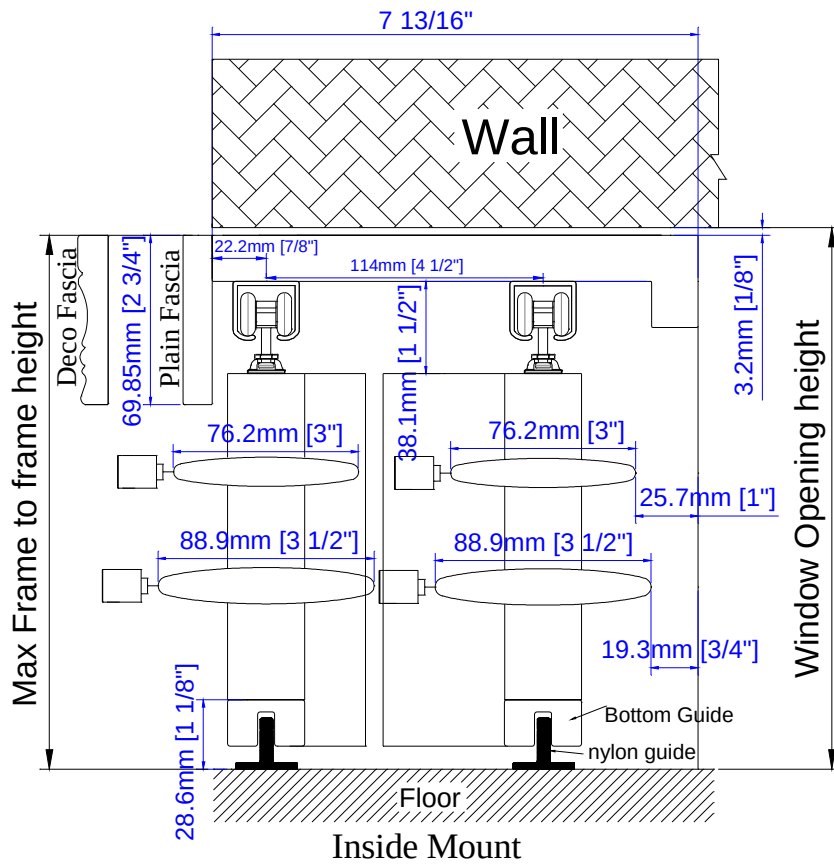
Track Shutters—2020 Bypass



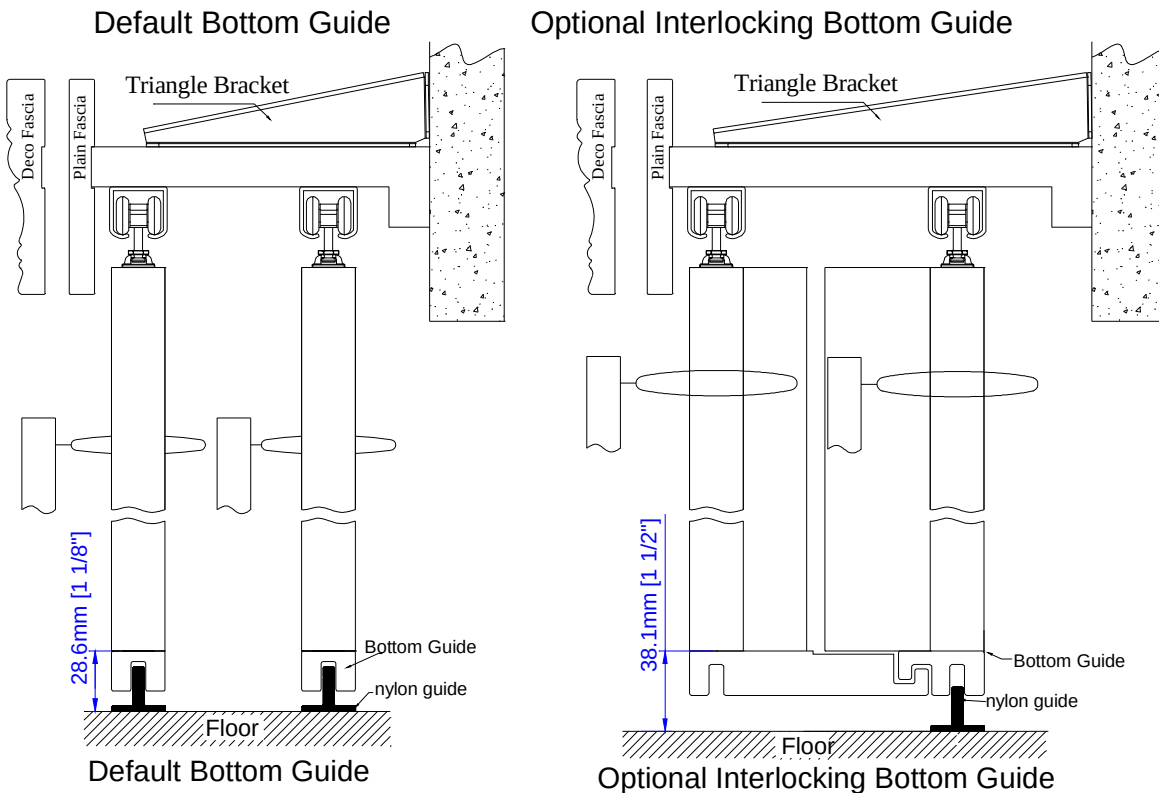
3" / 3 1/2" Louver



Track Shutters—2020 Bypass

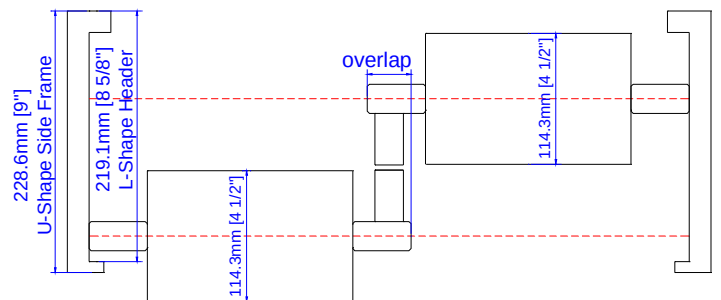
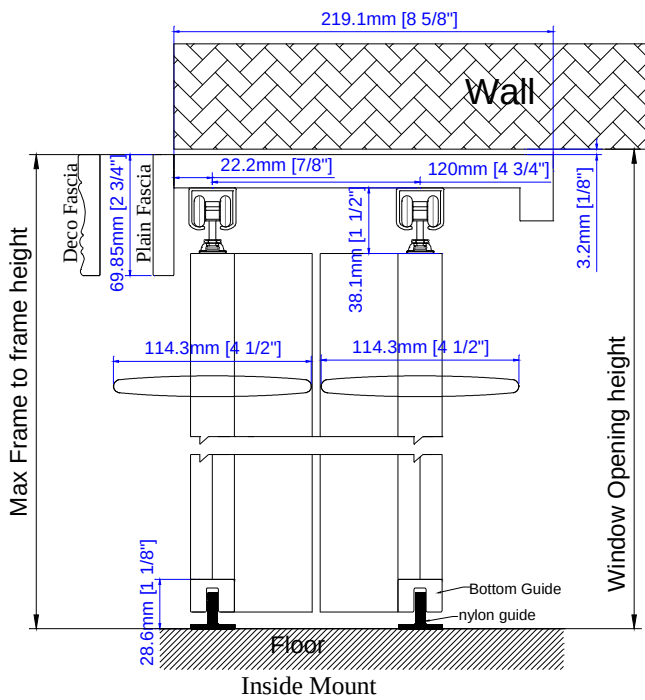
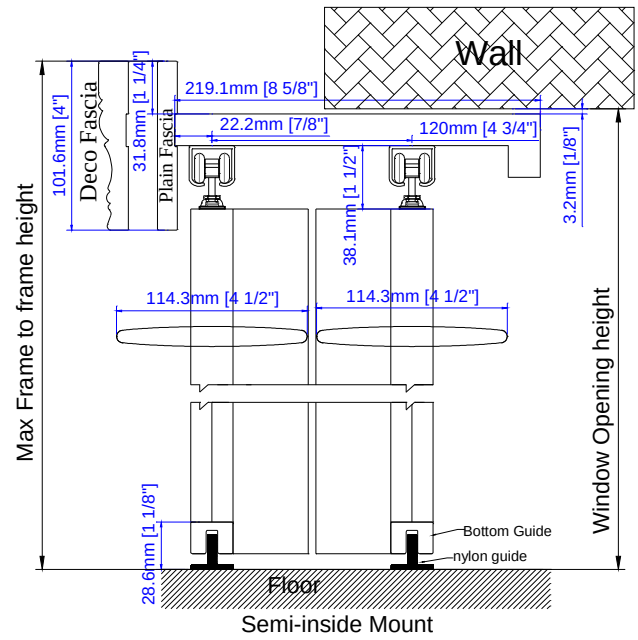
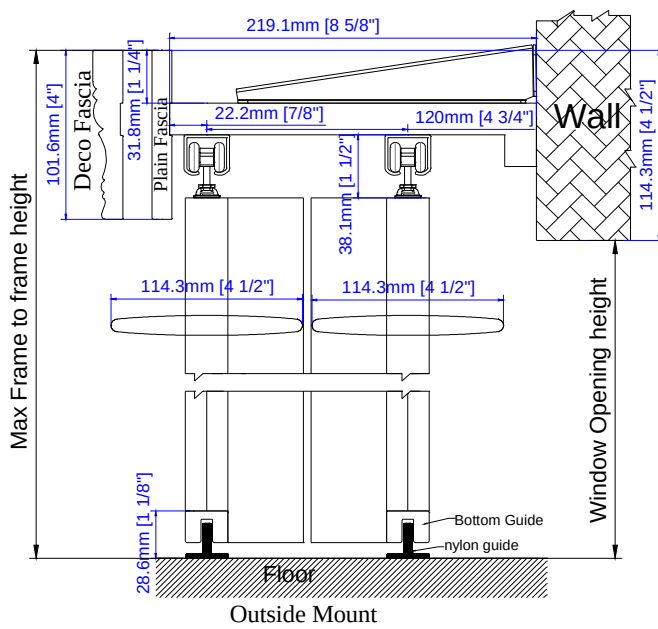


Bottom guide for Open Bypass:



Track Shutters—2020 Bypass

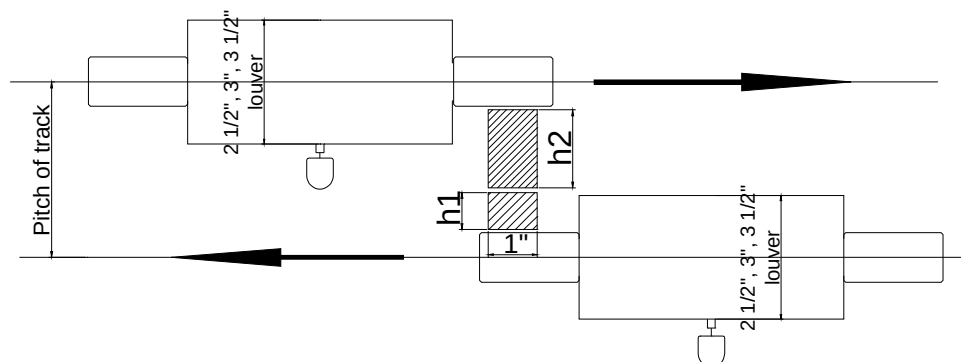
4 ½" Louver (Only available for the order with U-shape Bottom Guide and Invisible Tilt)



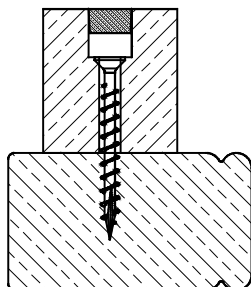
Track Shutters—2020 Bypass

Opened Bypass

Top View



Note: The light block for open bypass will be changed to use screws with wood plugs instead of current dowels.



Separation of track

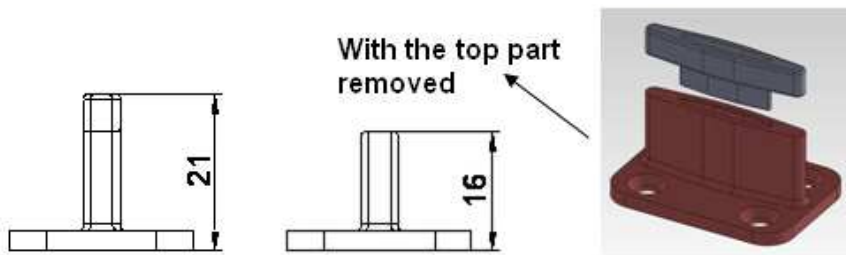
Louver	U shape bottom guide		Interlocking bottom guide	
	Regular Tilt Rod	Invisible tilt	Regular Tilt Rod	Invisible tilt
2.5"	3 9/16"	2 3/4"	3 9/16"	3 9/16"
3" / 3.5"	4 1/2"	3 5/8"	4 1/2"	4 1/2"
4.5"	NA	4 3/4"	NA	NA

Light block thickness for panel

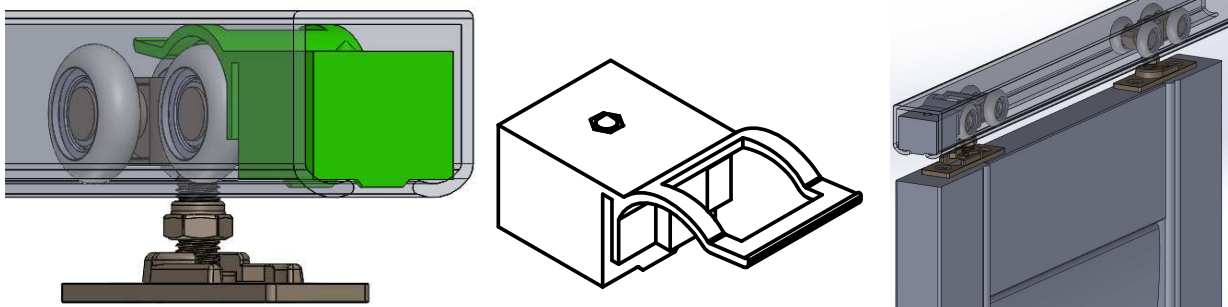
Louver		U shape bottom guide		Interlocking bottom guide	
		Regular Tilt Rod	Invisible tilt	Regular Tilt Rod	Invisible tilt
2.5"	h1	3/4"	3/4"	3/4"	3/4"
	h2	1 9/16"	3/4"	1 9/16"	1 9/16"
3" / 3.5"	h1	1 3/16"	1 3/16"	1 3/16"	1 3/16"
	h2	1 15/16"	1 3/16"	1 15/16"	1 15/16"
4.5"	h1	NA	1 3/4"	NA	NA
	h2	NA	1 3/4"	NA	NA

Track Shutters—2020 Bypass

Nylon Guide: The nylon guide is composed by 2 parts and will be supplied as a whole set by default (bottom part 16mm and top part 5mm), please remove the top part if necessary.

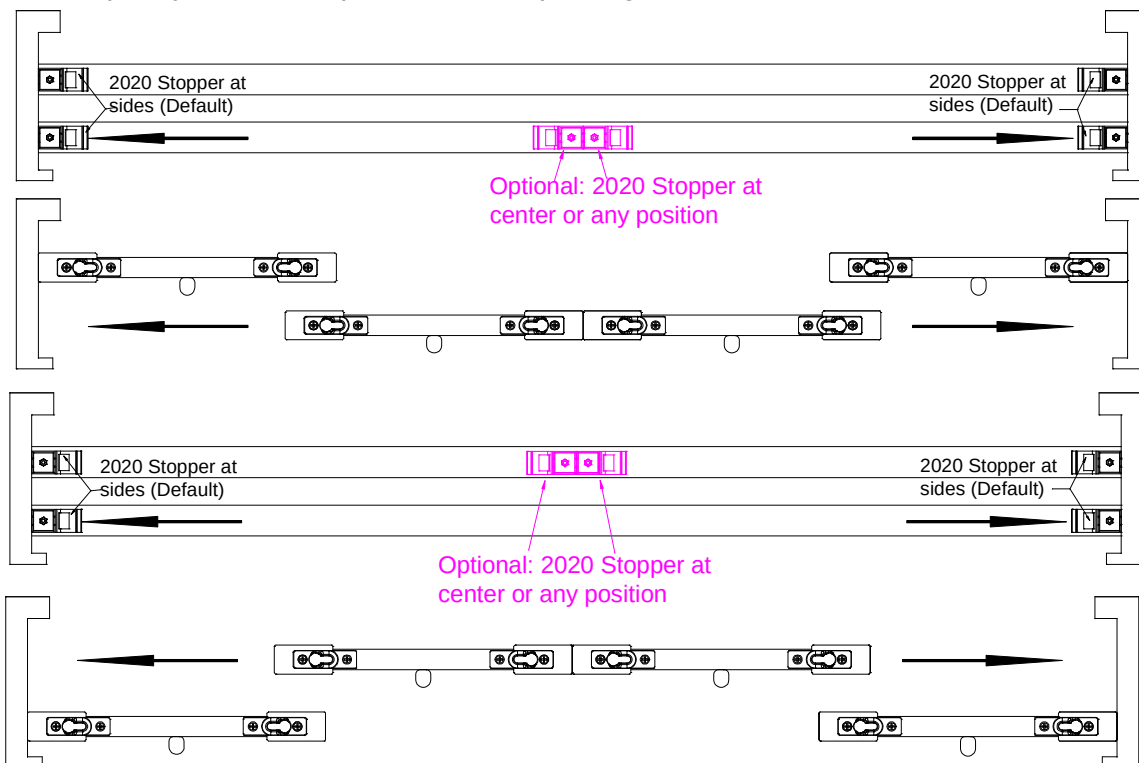


The 2020 Stopper



Note: 2020 Stopper will be supplied at both ends of top track by default.

The 2020 stopper can also be used in the center or any position of the top track as an option. Please specify the actual position when placing an order.



Track Shutters—2020 Bypass

Track Shutters Framing Materials

Framing	Size				Pc(s)
L Shape Header	Header (Closed Bypass)	Header (Opened Bypass)	Supporting Strip		1 pc
(2 in 1)	$\frac{3}{4}" \times 5\frac{1}{2}"^*$ (Default for Outside Mount; Optional for Semi-Inside and Inside Mount)	$\frac{3}{4}" \times 6\frac{1}{4}"^*$ (2½" Louver)	$\frac{3}{4}" \times \frac{3}{4}"$		
Header with Supporting Strip	$\frac{3}{4}" \times 4\frac{1}{2}"^*$ (Default for Semi-Inside and Inside Mount; Optional for Outside Mount)	$\frac{3}{4}" \times 7\frac{13}{16}"^*$ (3", 3½" Louver)			
		$\frac{3}{4}" \times 8\frac{5}{8}"^*$ (4½" Louver)			
	Custom * $\leq 8"$ *	Custom * $\leq 8"$ *			
U Shape Side Frame	Side Frame (Closed Bypass)	Side Frame (Opened Bypass)	Supporting Strip	Side Fascia	2 pcs
(3 in 1)	$\frac{3}{4}" \times 5\frac{7}{8}"^*$ (Default for Outside Mount; Optional for Semi-Inside and Inside Mount)	$\frac{3}{4}" \times 6\frac{5}{8}"^*$ (2½" Louver)	$\frac{3}{4}" \times \frac{3}{4}"$	$\frac{3}{8}" \times 1\frac{1}{4}"$	
Side Frame with Supporting Strip & Side Fascia	$\frac{3}{4}" \times 4\frac{7}{8}"^*$ (Default for Semi-Inside and Inside Mount; Optional for Outside Mount)	$\frac{3}{4}" \times 8\frac{1}{4}"^*$ (3", 3½" Louver)			
		$\frac{3}{4}" \times 9"$ (4½" Louver)			
	Custom * $\leq 8"$ *	Custom * $\leq 8"$ *			
Top Fascia (Plain/Deco)	4" (Semi-inside and Outside Mount)				1 pc
	$\frac{1}{2}" \times 2\frac{3}{4}"$ (Inside Mount)				
Top Fascia Return	4" (Semi-inside and Outside Mount)				2 pcs
	N/A (Inside Mount)				

Note:

- When the header size is $4\frac{1}{2}"^*$, $5\frac{1}{2}"^*$, $6\frac{1}{4}"^*$, $7\frac{13}{16}"^*$ or $8\frac{5}{8}"^*$, the L Shape header and the U Shape Side Frame will be sent integrally. That is:
 1. The header and the supporting strip will be sent integrally as L Shape Header;
 2. The side frame, supporting strip and the side fascia will be sent integrally as U Shape Side Frame.
- 3. $8\frac{5}{8}"^*$ L Shape header and $9"$ U shape side frame are only available for $4\frac{1}{2}"$ louver Open Bypass with U-shape Bottom Guide and Invisible Tilt.
- For any custom header sizes other than $4\frac{1}{2}"^*$, $5\frac{1}{2}"^*$, $6\frac{1}{4}"^*$, $7\frac{13}{16}"^*$ or $8\frac{5}{8}"^*$ the L Shape Header and the U Shape Side Frame will be sent separately. That is:
 1. The header and the supporting strip will be sent separately for L Shape Header;
 2. The side frame, supporting strip and the side fascia will be sent separately for U Shape Side Frame.

Track Shutters—2020 Bypass

- Custom Boxout Width is available for both closed and opened bypass shutters.
- For Outside Mount Bypass, the custom header size can reach to 8", and the total size of custom header size with extension thickness can't be exceeded 8 5/8".
- For Inside Mount and Semi-Inside Mount Bypass, the custom header size can reach to 8", and the extension thickness of header can reach to 2".

Hardware Box

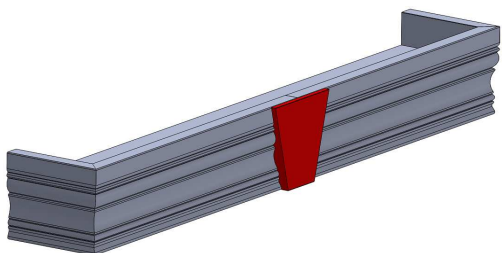
Hardware Box
Part
Wrench
Nylon Guide
3/4" screws for Nylon Guide
Triangle Bracket (based on the length of header)
2" screws for Triangle bracket
3/4" screws for Triangle bracket

Keystone for Fascia of the Track Shutter

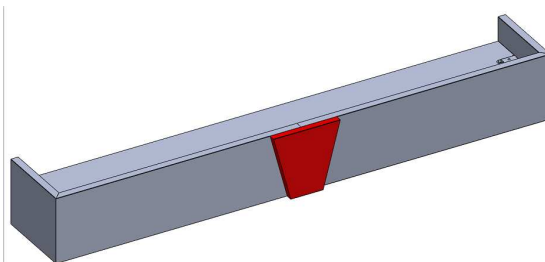
When fascia length exceeds the limitation and needs to split, keystone design fascia will be supplied without surcharge for track shutters.

For fascia length within the split limitation, keystone can also be supplied as an option with surcharge.

For Deco Fascia:

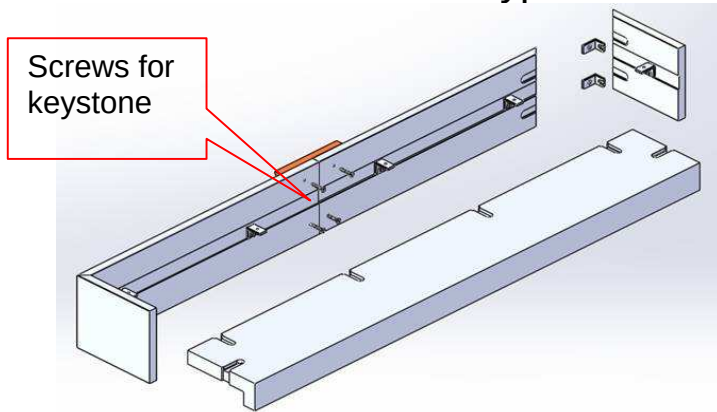


For Plain Fascia:



The keystone will be installed by screws. We will pre-drill the screw holes and the screws will be packed in hardware box.

Track Shutters—2020 Bypass

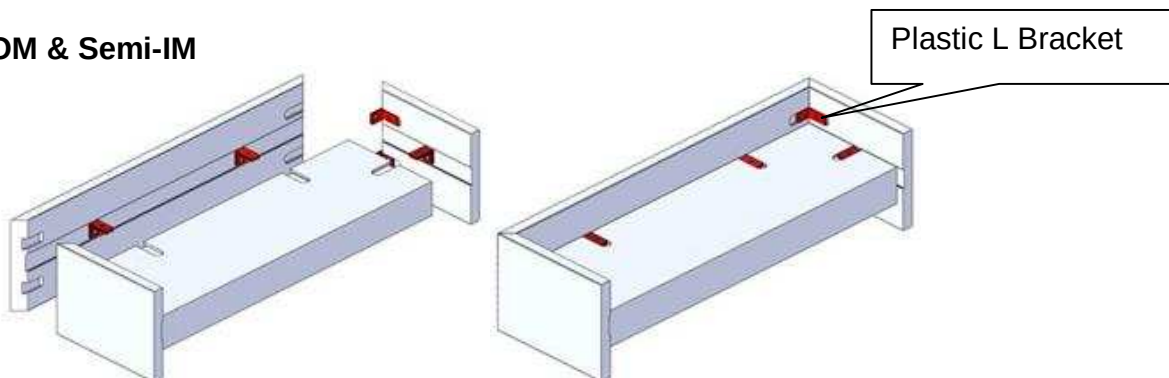


Plastic L Bracket to Connect the Header, Fascia & Fascia Returns

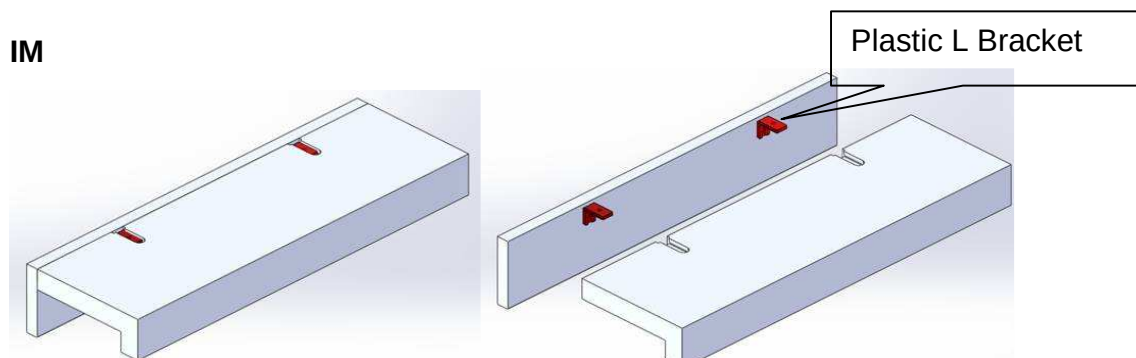


L Bracket will be used to connect the header, fascia and fascia returns for Bypass shutter. This applies for Inside Mount, Semi-Inside Mount & Outside Mount.

OM & Semi-IM



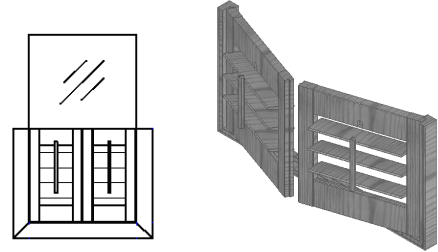
IM



Special Options

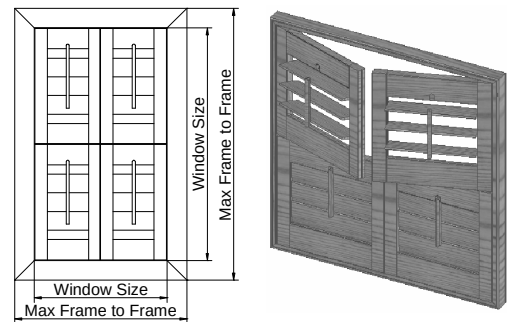
Café Shutters*

- The top of the panel requires a special finish. Please indicate “café shutter” when placing orders.
- The shutter does not come with recessed magnet on top.
- When the height of the framing material (including t-posts) exceeds the panel height, by more than 5", a surcharge will apply.

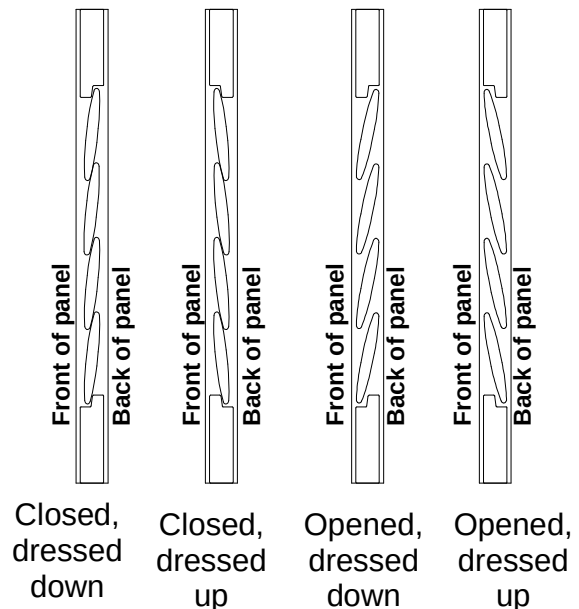
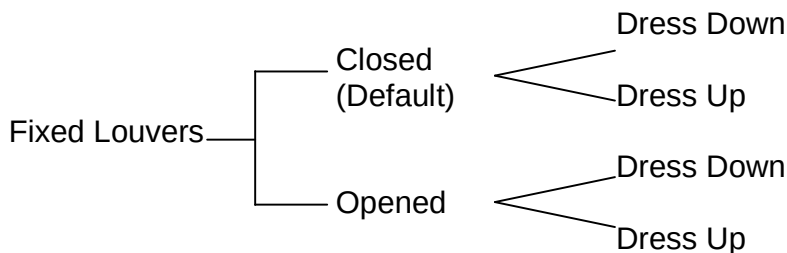


Double-Hung Shutters*

- For total number of panels, please include panels in both rows.
- For panel configuration, please provide panel fold for one row.
- The default location for the division point is the center unless specified otherwise.
- Horizontal T Post* can be added for increased structural integrity.
- The length of Horizontal T post in each section must be less than or equal to 48".
- For Double Hung shutter, whenever an upper or lower panel is larger than 78", we will add one more extra hinge for it, we will also attempt to create larger top/bottom rails for additional strength by reducing the number of louvers by one as long as the rail sizes remain within spec.



Fixed Louvers



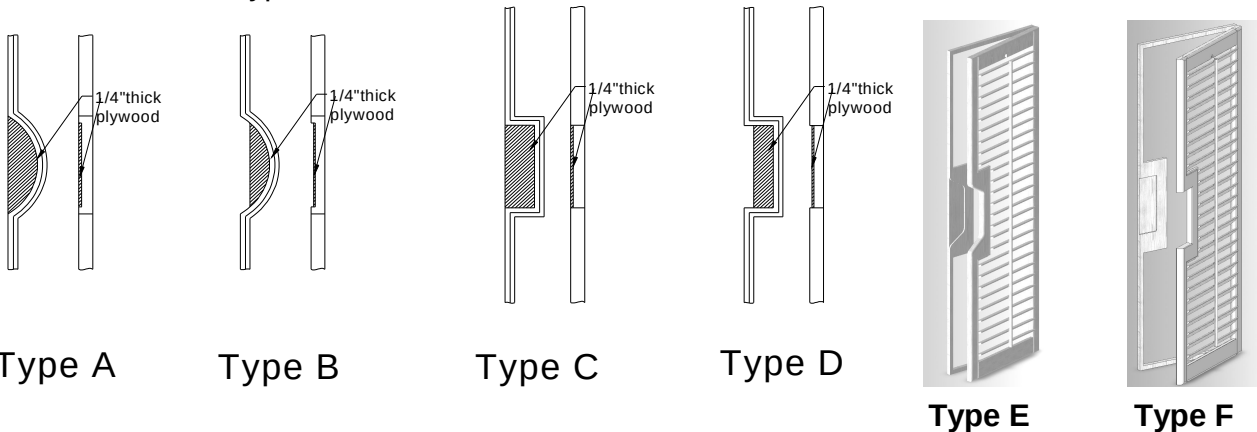
Fixed louvers Closed Dress Down and Fixed louvers Opened Dress Down are not available for all specialty shutters except all French Door Type E,F and Type A,B,C&D without fixed louver(s) which did not choose “Arch type” option.

French Door Cut-out Collection*

- Flush and offset cutout options are available in both curved and square French door cutout. Offset cutout is good option for French Door with molding around the glass.
- All French door cutouts are only available in L frame except Type E and F. The batten is placed behind the Type E/F panel to block light.

Louver size	Batten Width	Batten thickness
1 7/8" & 2 1/2"	3/4"	1"
3" & 3 1/2"	3/4"	1 3/8"
4 1/2"	3/4"	1 7/8"

- For measurement details please refer to Form section.
- French Door cutouts with Arch/Quarter Arch at the top are also available except Type E and F.
- Vintage L frame with 1/4" light block and Beaded L frame with 1/4" light block are not available for French door Type A, B, C & D.



Custom Colors*

Available in Normandy® shutters.

Option (Color Code)	Sample Size	<u>Confirmation Process</u>
Custom Solid (C01)	A 4"x4" coated foam core or poster board.	All orders require customers to confirm an actual swatch which will be returned in the mail before production will commence. An additional shipping and handling fee applies.
Custom Stain (C02)	Raw shutter louvers (Available on the sample site for purchase: SKU# CSKN-01 for Normandy)	

*Samples sent to the factory will not be returned.

*Some surcharges apply

ND k- 2 / 12

Note:

- 1) All custom colors will be finished with the standard gloss used on all Norman products.
- 2) Please be aware that we will only be able to match the tone and not the texture or finish of the sample you submit.
- 3) Only one color revision will be allowed during the color confirmation process.
- 4) Custom color does not apply to hardware or accessories.
- 5) For custom color orders, please provide a color sample in the form of: a 4" x 4" coated foam core, a poster board if you are requesting a custom paint color, or a raw shutter louver if you are requesting a custom stain color. You are able to send in your custom color sample in a format that does not fit the three listed forms; however, please anticipate a longer lead time as the custom color sample may be required to be processed through customs. In the event that you are sending in a custom color sample that is not in the three listed formats, providing any supplemental details about the sample will assist us in getting the right color quickly, e.g. material, the Latin scientific name of the material, detail Size, a photo of the color sample, the package method, net weight, whether a premium is included in the value, certificate of origin, plant inspection certificate etc.

Due to varying materials corresponding to different colors, if a custom color sample is sent in, we will match the color on a louver and send the physical louver sample to you for confirmation. In order to streamline the process, in the event that the color variation between the physical louver swatch and the custom color sample falls outside of our acceptable range, we will verify the color digitally first with you before mailing the physical color swatch for final confirmation.

Disclaimer:

**Sherwin Williams ColorSnap (MWCS),
Sherwin Williams Designer Expressions (MWDE),
Benjamin Moore Classic (MBMC) (M2460009TL) &
Benjamin Moore Preview (MBMP) (M2440564TL)
Benjamin Moore, Off White Collection M2451888TL**

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*For better tracking and minimal customs delays we recommend using Fedex/DHL. Please do not use the USPS - United States Postal Service.

Shutters to be installed in Angled or Horizontal Skylight Window Openings

If shutters are going to be installed in angled or horizontal skylight window openings, please specify on the order form or in special instructions.

Available Material	Normandy Shutters
Available Frame Type	Frame with light block thickness not less than 3/4"; Frames with frame insert on the face side (such as all vintage L frames / all Insert T-posts) are not available.
Max Panel Size	26" W x 78"H
Tilt Rod	Invisible Tilt
Panel Configuration	single panel or single panels with T-Posts
Hardware for unhinged side	Panel Lock on top and bottom + 2 Button Catches

Auto Tilt™ (Patented) *

Available in Bifold or Multi-fold panel configuration

Available with Invisible Tilt Only

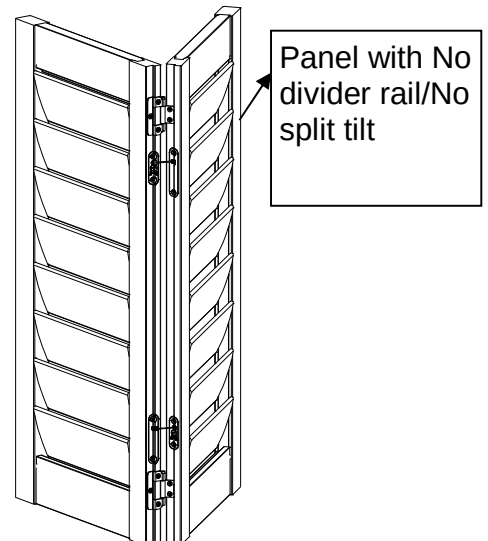
Available in 2" Butt, Astragal and Rabbet Stile

Minimum Louver Qty (Please see below for details)

Ring pulls included. For ring pull location, please refer to the ring pull section.

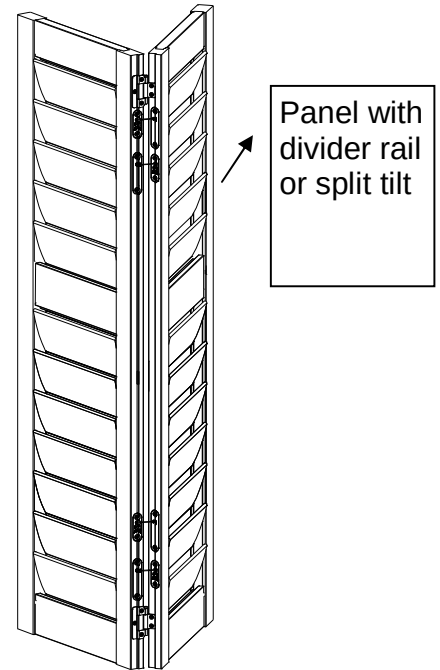
When there is only one louver section (no divider rail/no split tilt)

Louver Size	Rail Size	Minimum Louver qty
2.5"	<4.25"	6
	>=4.25"	4
3"	<3.75"	5
	>=3.75"	4
3.5"	2.75"~6.5"	4
4.5"	2.75"~6.5"	3



When there is more than one louver sections (with divider rail/ split tilt)

Louver Size	Rail Size	Minimum Louver qty
2.5"	<4.25"	5
	>=4.25"	4
3"	2.75" – 6"	4
3.5"	2.75"~6.5"	3
4.5"	2.75"~6.5"	3



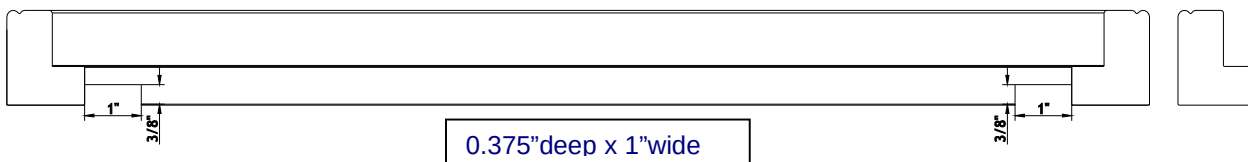
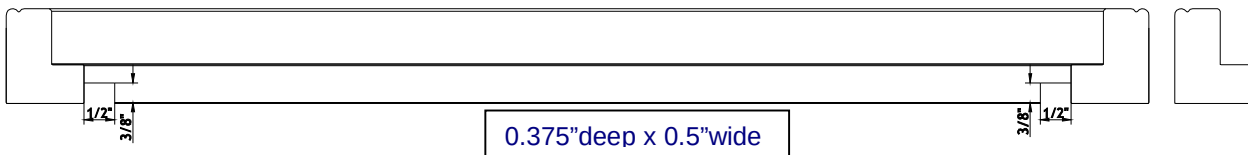
Standard frame cutouts: R cutout, S cutout, L cutout, C cutout. The sizes can vary.

- a. **R cutout**---Ridge cut out: 0.375"deep(0.25" deep for Deco Frames and L Frame with 1/4" Light block) x 0.5"wide and 0.375"deep (0.25" deep for Deco Frames and L Frame with 1/4" Light block) x 1.0" wide (see below).

The cutout can be on the top and the bottom of the frame

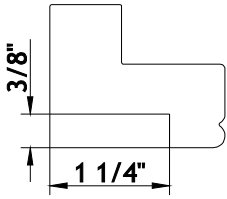
Available Frame Types:

- 1) All types of the L/Deco/Z frames;
- 2) All standard sill plates for Deco/Z Frame.



- b. **S cutout**---Sill cut out: 0.375"deep x 1.25"wide. The cutout is always on bottom of bottom frame.
Available frame type:

- 1) All L-frames except for L frame with 1/4" Light block
- 2) All Z Frame and its standard sill plates.



c. **L cutout**---Lock cutout. L cutout will be on the left or right frame or T Posts.

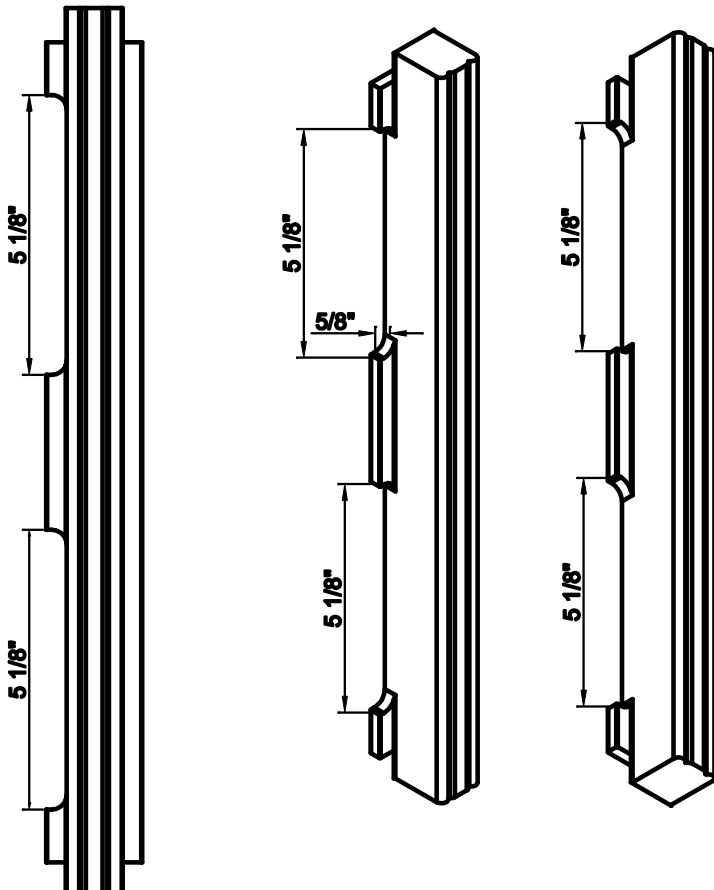
There will be one or two cutouts on the T-posts or frames depending on the height of the window;

- 1) For window around 42-48" height, 1 cutout is available;
- 2) For window around 60" or above, 2 cutouts are available.

Measuring for L cut out: The default size is 5.125". Measurements are from the bottom of frame if size type is max frame to frame size and from bottom of window if it is window size to the center of the cut out (2.5625" down and 2.5625" up).

Available frame type and T Post:

- 1) All types of L/Deco/Z frames except Tilt Out Z;
- 2) All standard sill plates for Deco/Z Frame;
- 3) All T Posts except for T Post with 3/8" wide light block.



d. **C cutout**---Crank cut outs (C1 cutout, C2 cutout, C3 cutout): Always on bottom of bottom frame.

Available frame type for C1&C2:

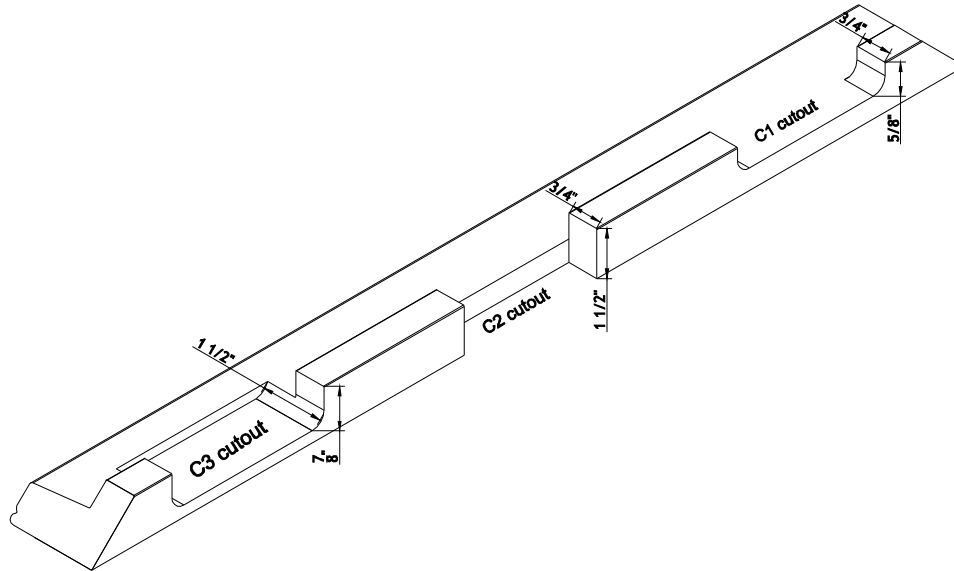
- 1) All types of L/Z Frames except Tilt Out Z

*Some surcharges apply

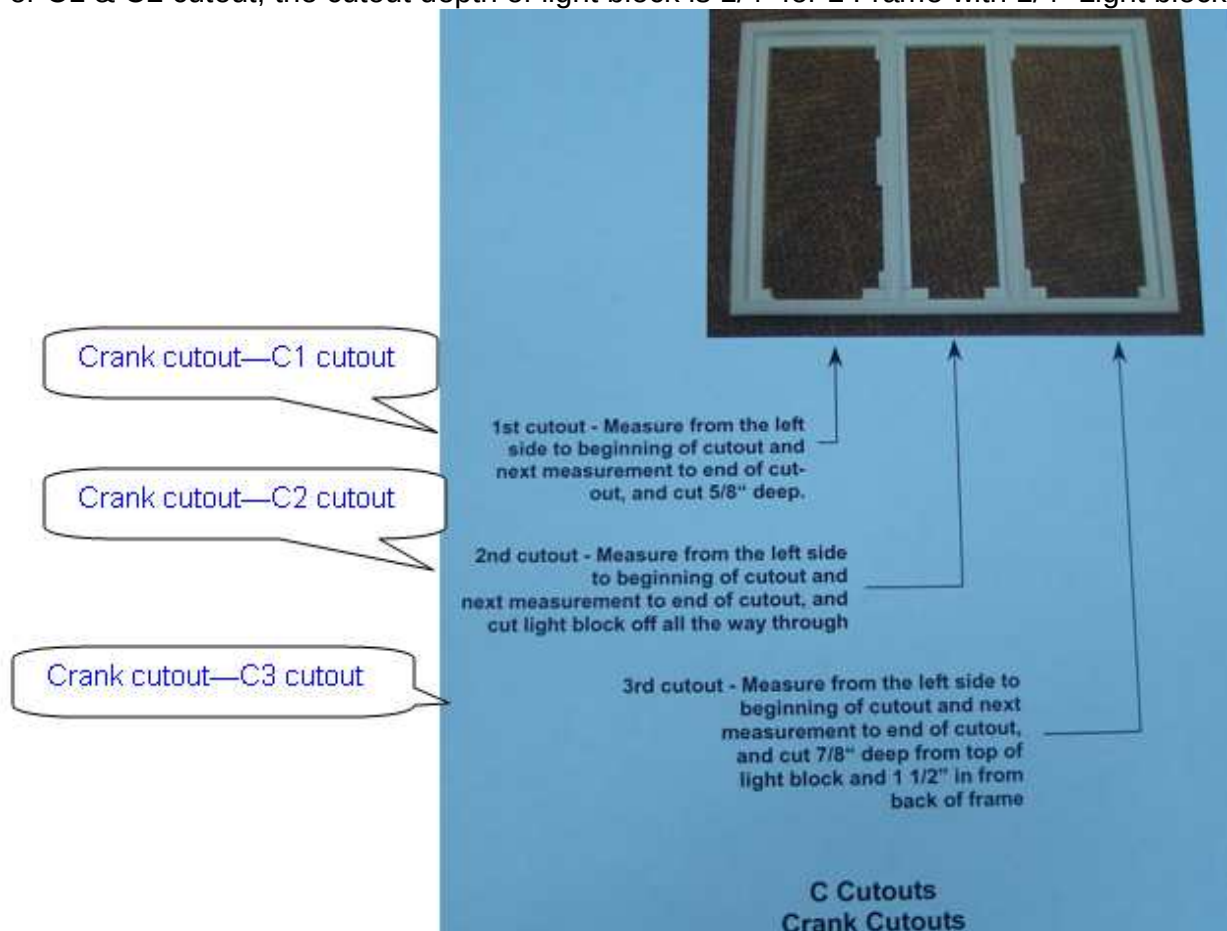
2) All standard sill plates for Z Frame

Available frame type for C3:

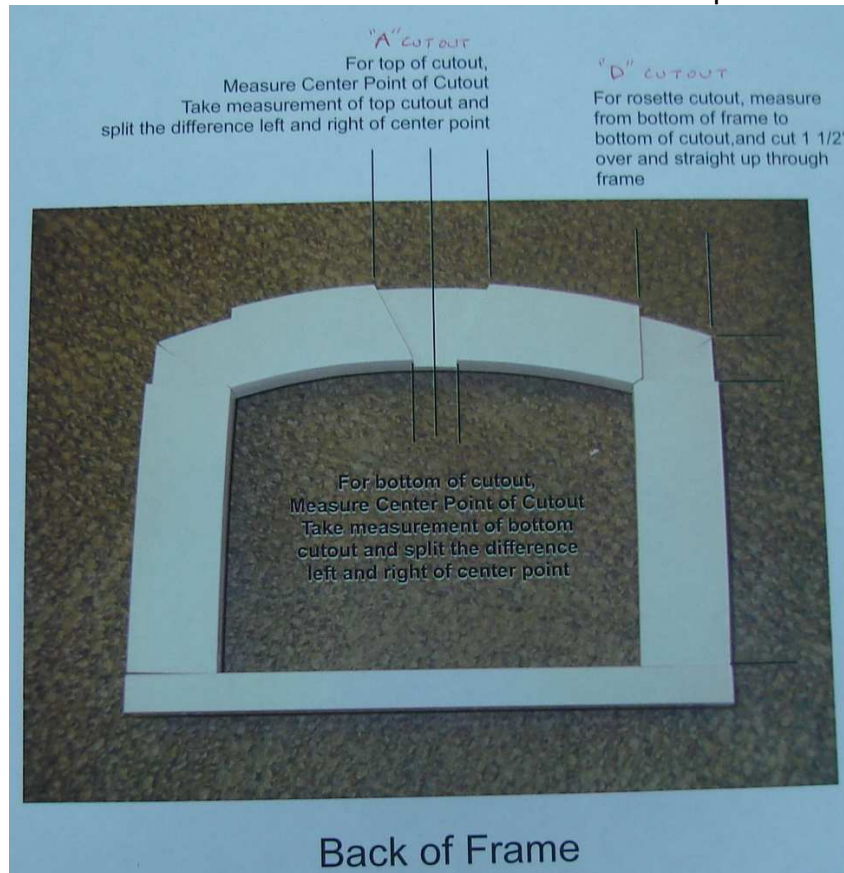
- 1) All types Z-frames except Tilt Out Z
- 2) All standard sill plates for Z Frame
- 3) All type of L Frame except for all Vintage L Frames with different height or Beaded L Frame with 1/4" light block.



For C1 & C2 cutout, the cutout depth of light block is 1/4" for L Frame with 1/4" Light block.

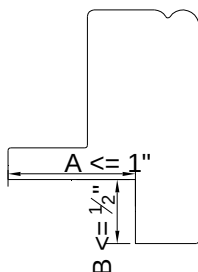


- e. **A & D cutouts:** This should be for the outside mount frames, deco frames, L-frames. Normally a build out or an extension is used to build the frame out so the frame can set on top of the "keystone":

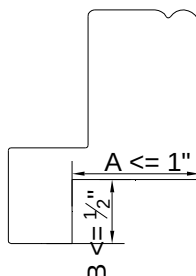


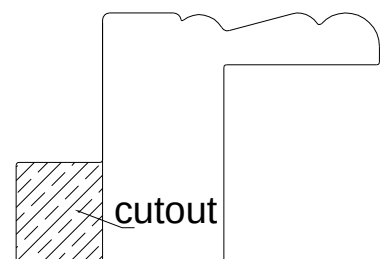
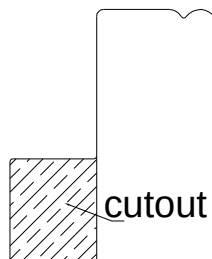
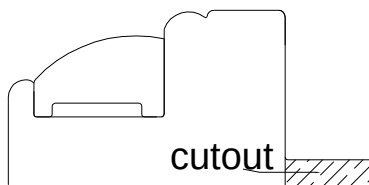
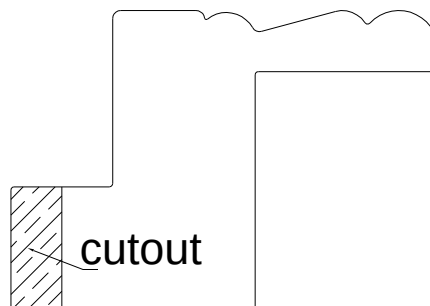
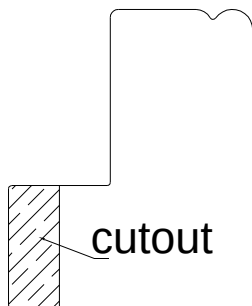
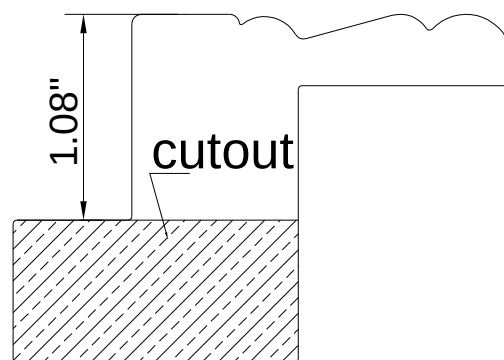
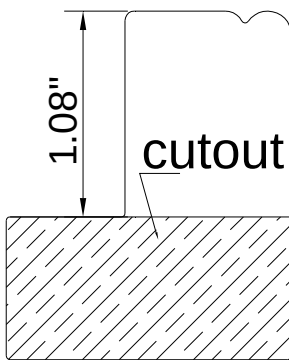
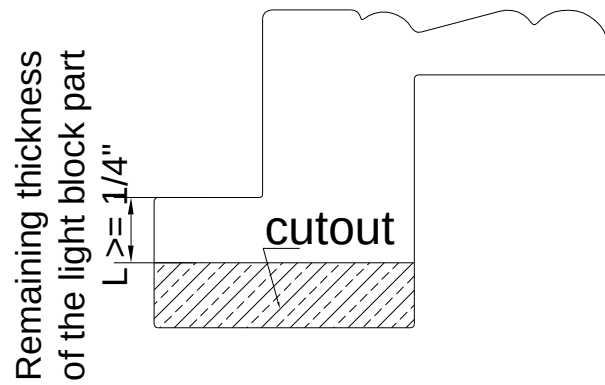
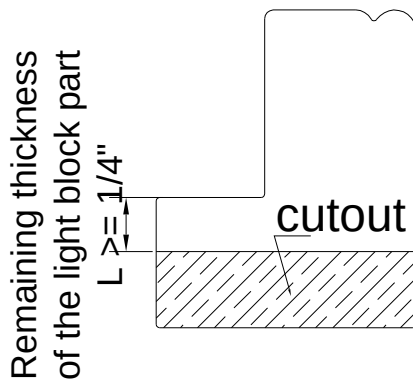
- For cut out frames, if there is not enough room to install the round magnet catch, the L-shaped magnet catch will replace it.
- For cut out frame sizes, take following drawings as examples; please refer to the available limit size in detail:

Notchout L frame
Type A

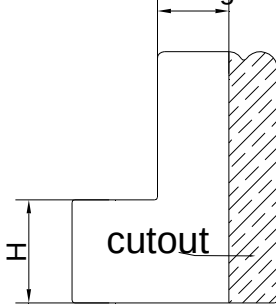


Notchout L frame
Type B

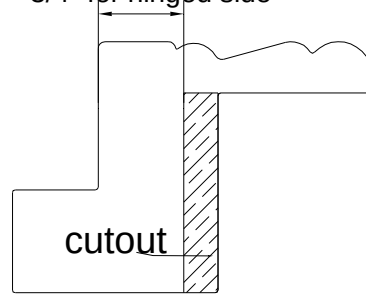




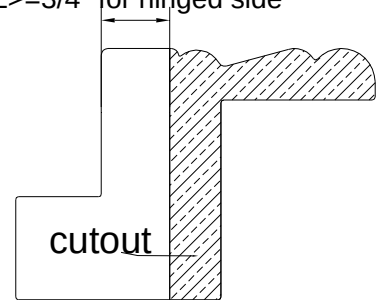
$L \geq 1/2"$ for Non-hinged side
 $L \geq 3/4"$ for hinged side



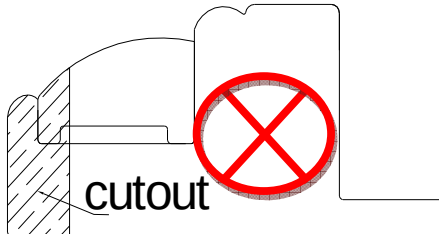
$L \geq 1/2"$ for Non-hinged side
 $L \geq 3/4"$ for hinged side



$L \geq 1/2"$ for Non-hinged side
 $L \geq 3/4"$ for hinged side

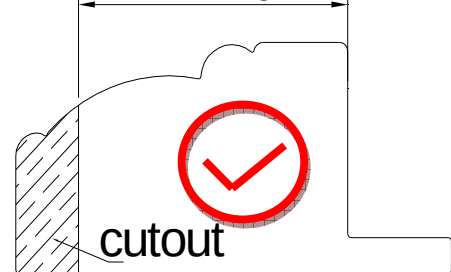


Note: It is not available for Vintage L frames with insert; It is available for Vintage L frames without insert; It can't be installed Invisible Hinges.



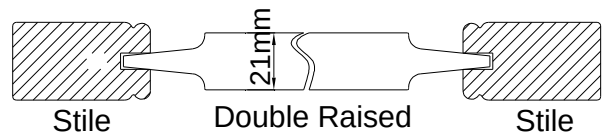
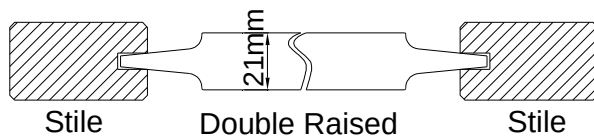
Note: It is not available for Deco frames with insert; It can't be installed Invisible Hinges.

$L \geq 1/2"$ for Non-hinged side
 $L \geq 3/4"$ for hinged side



It is available for Deco frames without insert;

Raised Panel *



Available in

- Raised panel with louver combination.
 Raised panel only.

Raised Panel Location(s) Note: the divider rail option must be selected in order entry.

Top.
 Middle.
 Bottom.
 All

Specifications

$12" \leq$ raised panel section height (including rail size) $\leq 48"$;
 For $1 \frac{5}{8}"$ Stile, the single panel width can't be less than $6 \frac{7}{8}"$;
 For $2"$ Stile, the single panel width can't be less than $7 \frac{5}{8}"$;
 For $2 \frac{1}{4}"$ Stile, the single panel width can't be less than $8 \frac{1}{8}"$;

Raised panel only (without louvers):

The maximum panel width will be 36" for single panel (such as L / R / LR / LTR etc.).

The maximum panel width will be same as the normal shutter spec for Bifold & Multiple fold panel configurations.

Raised panel with louver combination:

The maximum panel width will be 36" for single panel with 3.5" & 4.5" louvers;

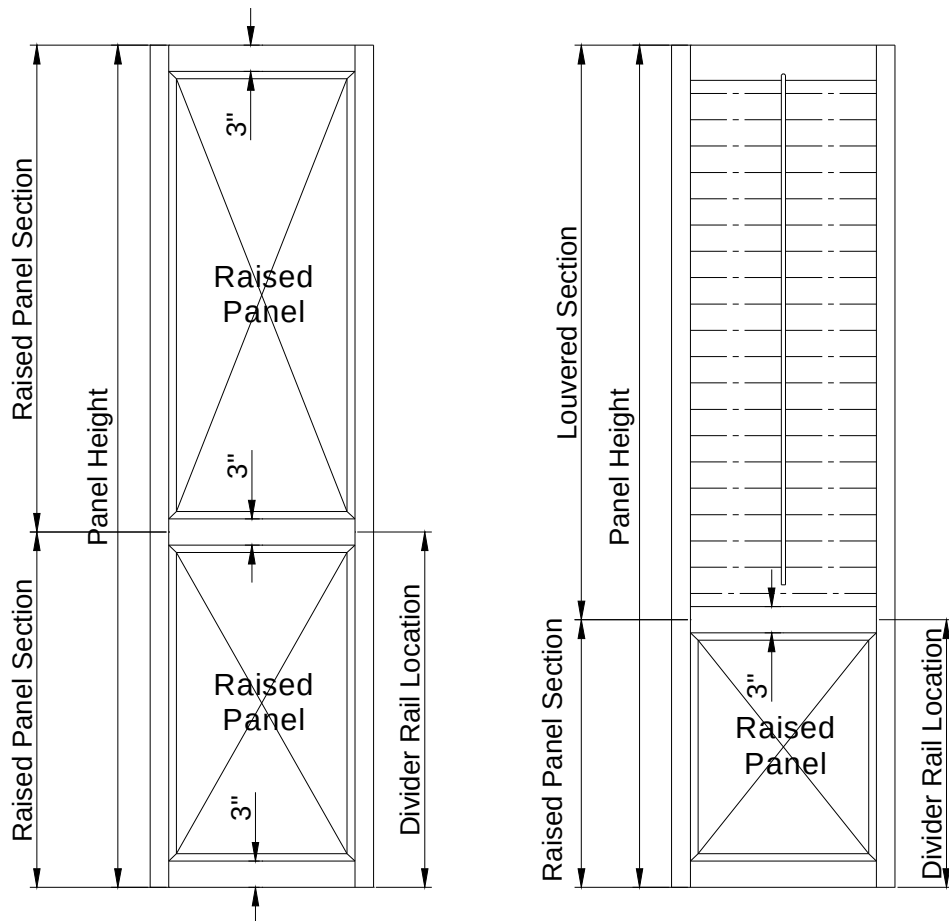
The maximum panel width will be same as the normal shutter spec for single panel with other louver sizes except 3.5" & 4.5" louvers.

The maximum panel width will be same as the normal shutter spec for Bifold & Multiple fold panel configurations.

Rails

For a panel with raised panel section only, 3" is the standard size for top, bottom and divider rails. The minimum size for top and bottom rails is 3".

If you are matching raised panels to standard shutter panels, the top rail of the standard shutter panel will match the top rail of the raised panel. Please specify clearly when placing order.



Hand-Carved Rail *

Available for top and/or bottom rails.

Please note that the proportion of the design is based on the height of the rail, not the width of the panel.



Vine



Wheat



Fleur de Lis



Rosette



Star



Fish



Swan

Liberty Arch *

The arch starts and ends 1" from the rail sides when panel width is more than 8";

While the arch starts and ends 1/2" from the rail sides when panel width is less than or equal to 8";

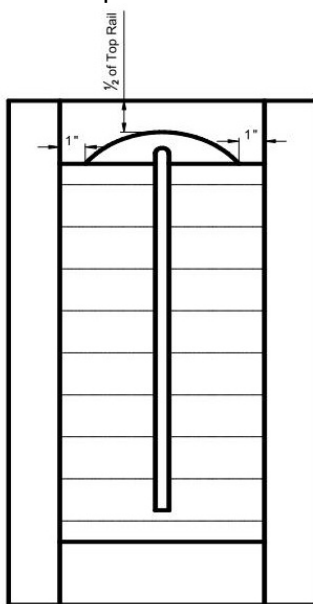
The arch height is half of the top rail size (rounded up to 1/8" increment).

The shape of the Arch is based on 3 points taken from the rail.

1st point: 1" (or 1/2") from the left stile

2nd point: 1" (or 1/2") from the right stile

3rd point: Midpoint of the rail front



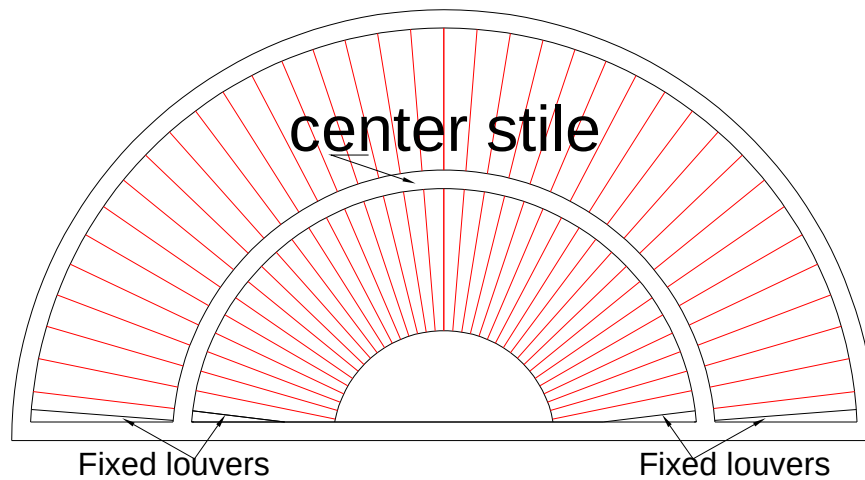
Specialties

Specialty Shutters^{*}

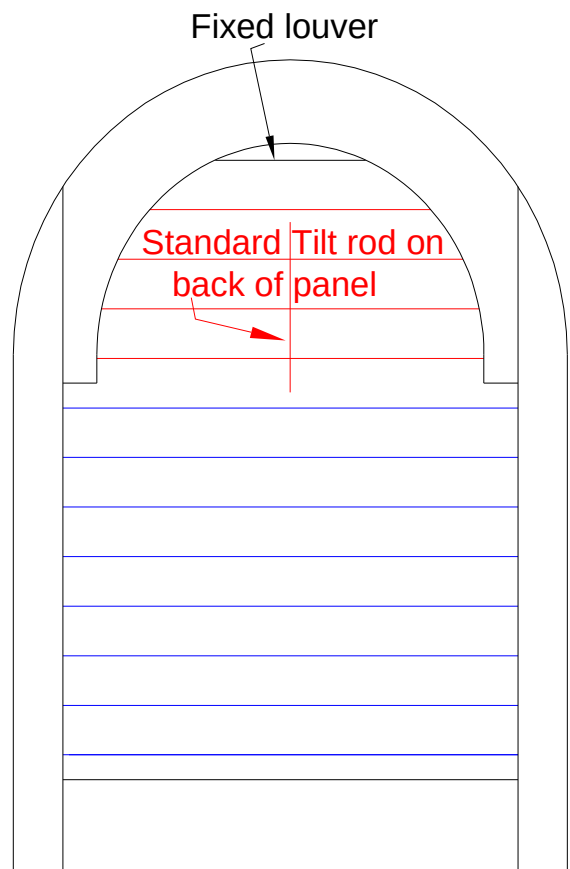
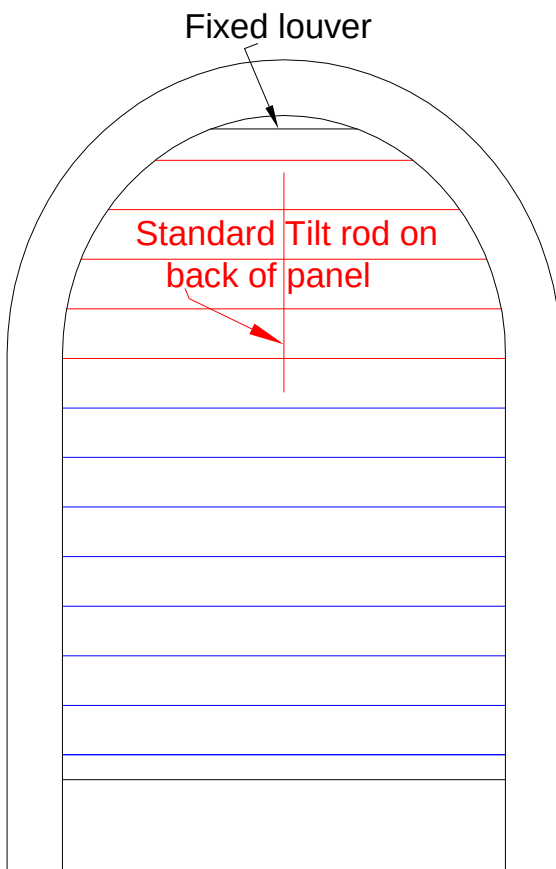
Specialty shutters are made of premium solid wood, finished to match the beauty of all Norman shutters. Templates are required if it is an imperfect specialty. Delivery time is 8 weeks from the order confirmation date for all specialty shutters. Please refer to Specialties and Template Requirement sections for details.

Regarding frame with an insert or solid frame issue:

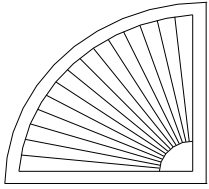
- 1). For Half Round Sunburst, Quarter Round Sunburst, Eyebrow Sunburst & Elongated Eyebrow Sunburst:
 - All deco frames will be made as solid frames;
 - All Vintage L frames & Vintage hang strip: all top frames (curve shape frame) will be made as a solid frame but the bottom & both sided frames (straight frames) will be made as regular frames with inserts
- 2). For Triangle, Hexagon, Octagon, Peak & Angle top: all frames will be made as regular frames with inserts.
- 3). For Circle, Oval: all frames will be made as solid frames (not with an insert).
- 4). For Sunburst on top with Divider strip: the top frame (curve shape frame) will be made as a solid frame but the bottom & both side frames (straight frames) will be made as regular frames with inserts.
- 5). For Louver Arch with Eyebrow curve shape:
 - All deco frames excluding 2.5" Mission deco frame: all frames will be made as solid frames (not with an insert).
 - All Vintage L frames & Vintage hang strip & 2.5" Mission deco frame: the top frame (curve shape frame) will be made as a solid frame but the bottom & both sided frames (straight frames) will be made as regular frames with inserts;
- 6). For other Specialty Shutters when curve shape is similar as perfect arch (such as Louver Arch except Louver Arch with Eyebrow curve shape in point 5), Horizontal louvers center arch with quarter round side panels(-Standard louvered arch & Continuous arch), Sunburst louvers center arch with quarter round side panels, Arch Top Picture Window with horizontal louvers, Arch Top Picture Window with Sunburst, Standard unit with horizontal arch center etc.):
 - All deco frames will be made as solid frames;
 - All Vintage L frames & Vintage hang strip: all top frames (curve shape frame) will be made as a solid frame but the bottom & both sided frames (straight frames) will be made as regular frames with inserts.
- 7). Hang strip behind the panel is not available for Hexagon and octagon with horizontal louvers.
- 8). All sunburst hubs must be less than or equal to 12".
- 9). If a requested half round sunburst arch will result in louvers that are out of spec, a center stile may be added in order to produce the unit.



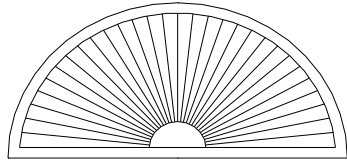
The louvers in the curved section can not be operated by the Invisible Tilt system. They can be independently operated by a separate standard tilt rod installed on back of the panel.



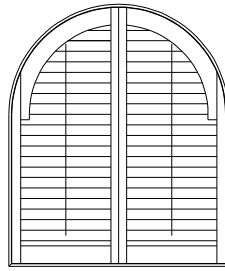
Normandy® Shutters



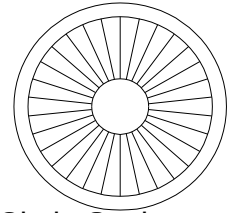
Quarter Round
Sunburst



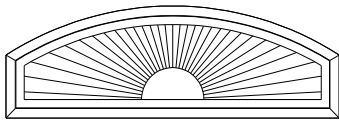
Half Round
Sunburst



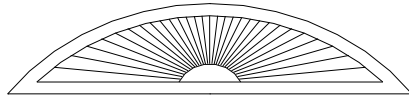
Louvered Arch
w/ Center Tilt Rod



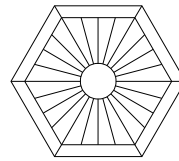
Circle Sunburst



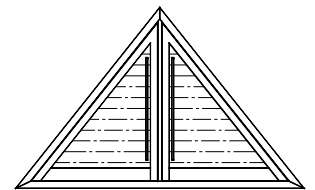
Elongated Eyebrow
Sunburst



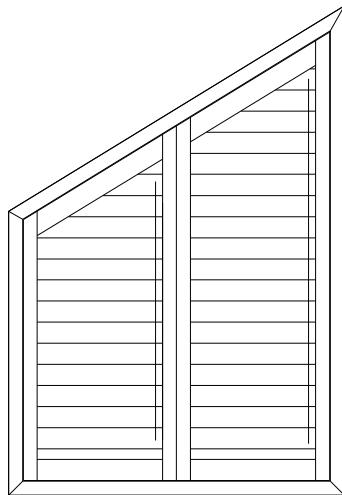
Eyebrow Sunburst



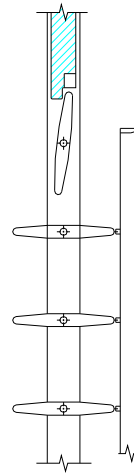
Hexagon Sunburst



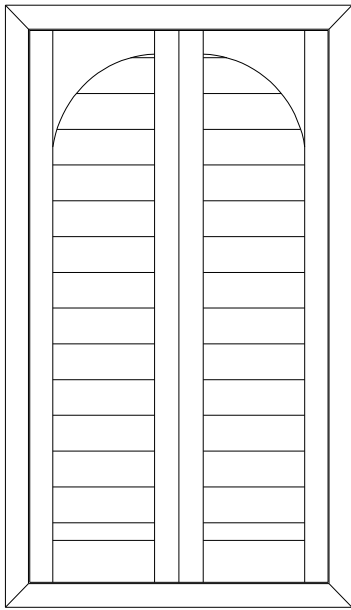
Triangle



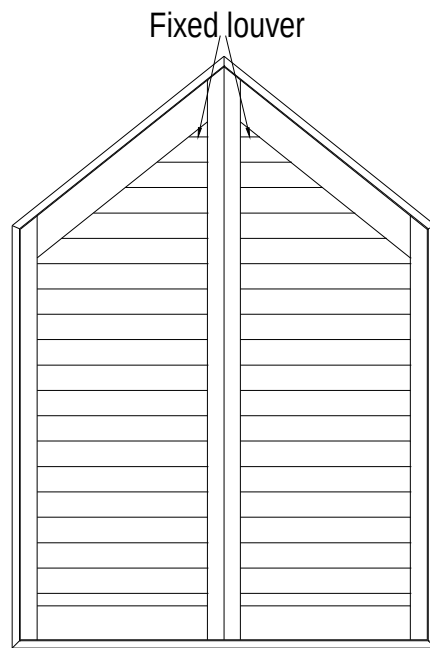
Angle Top



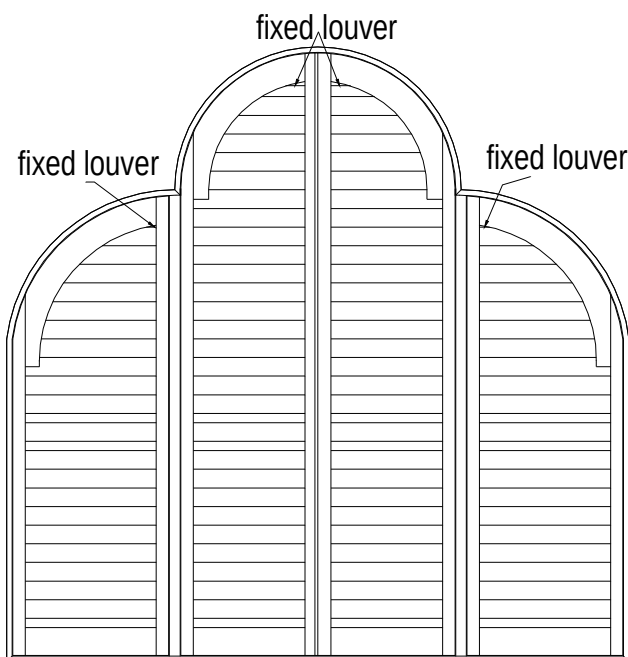
*Top louver of angle top
might be fixed as shown
above



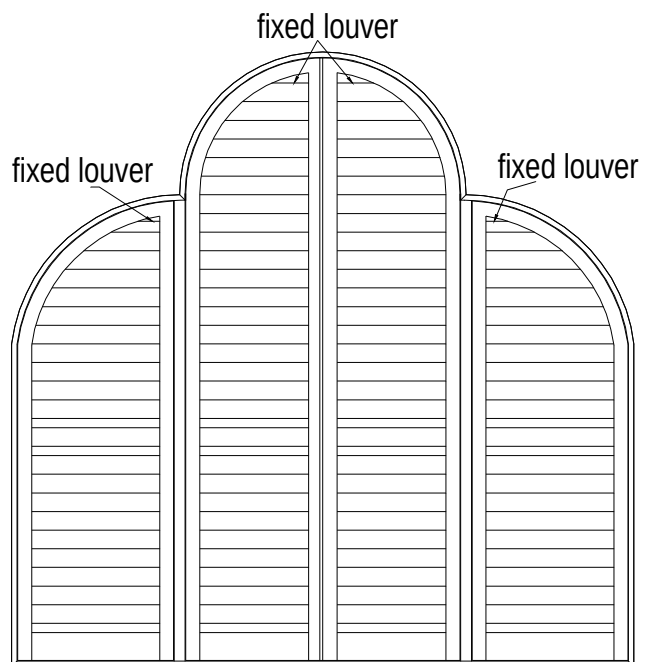
Louver Arch in a Rectangle



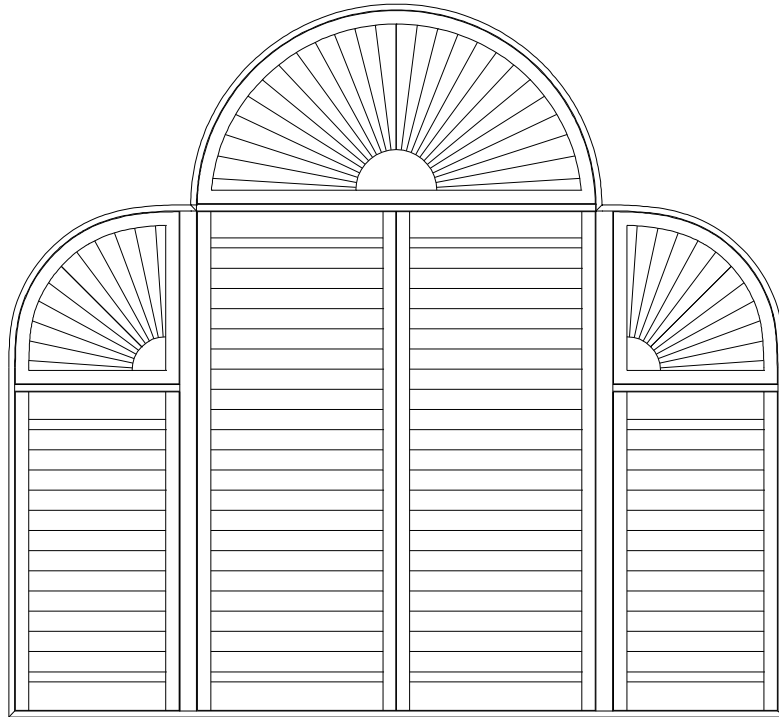
Peak



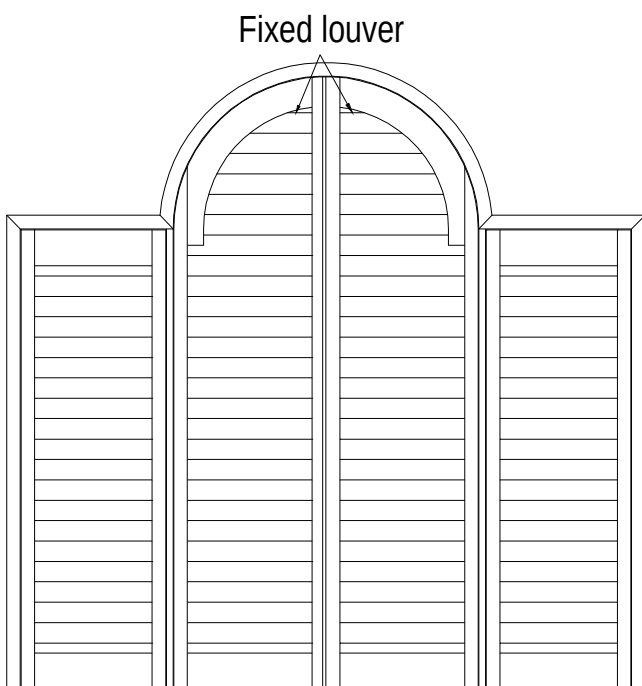
Horizontal louvers center arch
with quarter round side panels
-Standard louvered arch



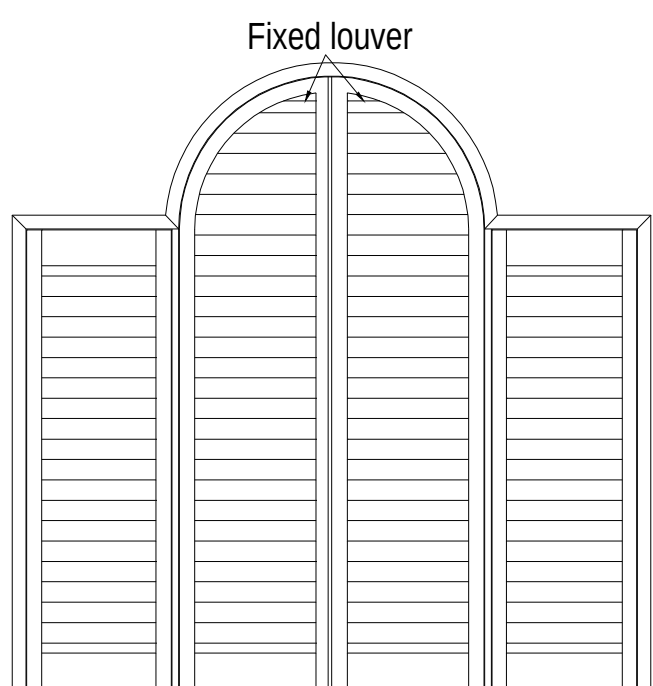
Horizontal louvers center arch
with quarter round side panels
-Continuous arch



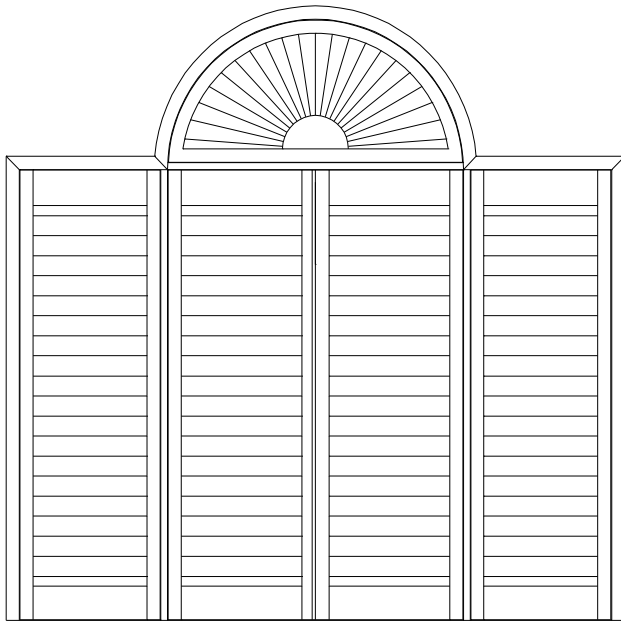
Sunburst louvers center arch with quarter round side panels



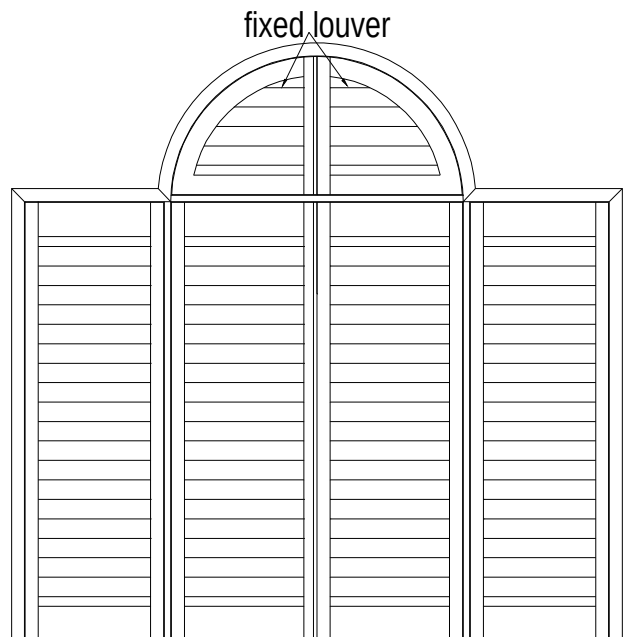
Arch Top Picture Window with
horizontal louvers
-Standard louvered arch



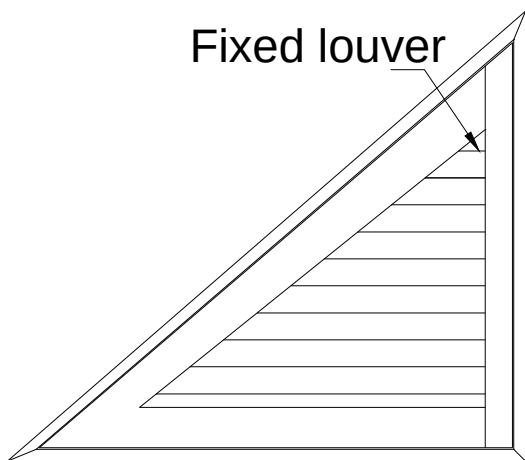
Arch Top Picture Window with
horizontal louvers
-Continuous arch



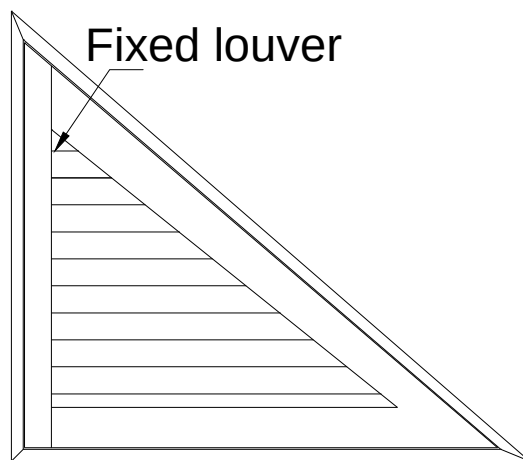
Arch Top Picture Window with Sunburst



Standard unit with horizontal arch center



Left Triangle



Right Triangle

Pricing

1) By Max Frame-to-Frame Measurement:							
Regular Shutter							
Inside and Outside Mount	Width	x	Height	/ 144	x	\$ /sq ft	
Track Shutter**							
Inside	Width	x	Height	/ 144	x	\$ /sq ft	
Mini-Inside and Outside Mount	Width	x	(Height-1½")	/ 144	x	\$ /sq ft	
2) By Net Panel Measurement:	Width	x	Height	/ 144	x	\$ /sq ft	
3) By Window Size:							
2" Camber Deco Frame	4-sided	(Width+4")	x	(Height+4")	/ 144	x	\$ /sq ft
	3-sided with Sill Plate	(Width+4")	x	(Height+4")	/ 144	x	\$ /sq ft
	3-sided	(Width+4")	x	(Height+2")	/ 144	x	\$ /sq ft
	2 sided with Sill Plate	(Width+4")	x	(Height+2")	/ 144	x	\$ /sq ft
	2-sided	(Width+4")	x	(Height)	/ 144	x	\$ /sq ft
3" Ridge Deco Frame	4-sided	(Width+6")	x	(Height+6")	/ 144	x	\$ /sq ft
	3-sided with Sill Plate	(Width+6")	x	(Height+6")	/ 144	x	\$ /sq ft
	3-sided	(Width+6")	x	(Height+3")	/ 144	x	\$ /sq ft
	2-sided with Sill Plate	(Width+6")	x	(Height+3")	/ 144	x	\$ /sq ft
	2-sided	(Width+6")	x	(Height)	/ 144	x	\$ /sq ft
2.5" Mission	4-sided	(Width+5")	x	(Height+5")	/ 144	x	\$ /sq ft
deco frame	3-sided	(Width+5")	x	(Height+2.5")	/ 144	x	\$ /sq ft

	2-sided	(Width+5")	x	(Height)	/ 144	x	\$/sq ft
3" Crown Z Frame	4-sided	(Width+4 ½"	x	Height+4½")	/ 144	x	\$/sq ft
	3-sided with Sill Plate	(Width+4 ½")	x	Height+4½")	/ 144	x	\$/sq ft
	3-sided	(Width+4 ½)	x	Height+2¼")	/ 144	x	\$/sq ft
	2-sided with Sill Plate	(Width+4 ½)	x	Height+2¼")	/ 144	x	\$/sq ft
	2-sided	(Width+4 ½")	x	(Height)	/ 144	x	\$/sq ft
2" Bel Air Z Frame 2" Bullnose Z Frame	4-sided	(Width+2 ½")	x	Height+2½")	/ 144	x	\$/sq ft
	3-sided with Sill Plate	(Width+2 ½")	x	Height+2½")	/ 144	x	\$/sq ft
	3-sided	(Width+2 ½")	x	Height+1¼")	/ 144	x	\$/sq ft
	2-sided with Sill Plate	(Width+2 ½")	x	Height+1¼")	/ 144	x	\$/sq ft
	2-sided	(Width+2 ½")	x	(Height)	/ 144	x	\$/sq ft
1 ½ " Bullnose Z Frame	4-sided	(Width+2")	x	(Height+2")	/ 144	x	\$/sq ft
	3-sided with Sill Plate	(Width+2")	x	(Height+2")	/ 144	x	\$/sq ft
	3-sided	(Width+2")	x	(Height+1")	/ 144	x	\$/sq ft
	2-sided with Sill Plate	(Width+2")	x	(Height+1")	/ 144	x	\$/sq ft
	2-sided	(Width+2")	x	(Height)	/ 144	x	\$/sq ft
Vintage Hang Strip	Inside	Width	x	Height	/ 144	x	\$/sq ft
	Outside 4-sided	(Width+3")	x	(Height+3")	/ 144	x	\$/sq ft
	Outside 3-sided	(Width+3")	x	(Height+1½")	/ 144	x	\$/sq ft
	Outside 2-sided	(Width+3")	x	(Height)	/ 144	x	\$/sq ft
Beaded	Outside 4-sided	(Width+3")	x	(Height+3")	/ 144	x	\$/sq ft

L frame /	Outside 3-sided	(Width+3")	x	Height+1½")	/ 144	x	\$/sq ft
Vintage L Frame	Outside 2-sided	(Width+3")	x	(Height)	/ 144	x	\$/sq ft
Track Shutter**							
Inside and Semi-Inside Mount		Width	x	Height	/144	x	\$/sq ft
Outside Mount		(Width+3")	x	(Height+3")	/144	x	\$/sq ft

1side frame price:

2" Camber Deco Frame	1-sided	(Width+2")	x	(Height)	/ 144	x	\$/sq ft
2" Classic Deco Frame	1-sided	(Width)	x	(Height+2")	/ 144	x	\$/sq ft
3" Ridge Deco Frame	1-sided	(Width+3")	x	(Height)	/ 144	x	\$/sq ft
	1-sided	(Width)	x	(Height+3")	/ 144	x	\$/sq ft
2 1/2" Mission deco frame	1-sided	(Width+2 ½")	x	(Height)	/ 144	x	\$/sq ft
	1-sided	(Width)	x	(Height+2 ½")	/ 144	x	\$/sq ft
3" Crown Z Frame	1-sided	(Width+2 ¼")	x	(Height)	/ 144	x	\$/sq ft
	1-sided	(Width)	x	(Height+2 ¼")	/ 144	x	\$/sq ft
2" Bel Air Z Frame	1-sided	(Width+1 ¼")	x	(Height)	/ 144	x	\$/sq ft
2" Bullnose Z Frame	1-sided	(Width)	x	(Height+1 ¼")	/ 144	x	\$/sq ft
1 ½" Bullnose Z Frame	1-sided	(Width+1")	x	(Height)	/ 144	x	\$/sq ft
1 ¼ " Beaded Z Frame	1-sided	(Width)	x	(Height+1")	/ 144	x	\$/sq ft
Bullnose Tilt Out Z Frame							

Vintage Hang Strip	Outside 1-sided	(Width+1 ½")	x	(Height)	/ 144	x	\$/sq ft
	Outside 1-sided	(Width)	x	(Height+1 ½")	/ 144	x	\$/sq ft
Beaded L frame	Outside 1-sided	(Width+1 ½")	x	(Height)	/ 144	x	\$/sq ft
Vintage L Frame							
Beaded L with 1/4" Light Block							
Beaded L with 1/2" Buildout	Outside 1-sided	(Width)	x	(Height+1 ½")	/ 144	x	\$/sq ft
Beaded L with 1" Buildout							
Vintage L with 1/4" Light Block							
Vintage L with 1/2" Buildout							
Vintage L with 1" Buildout							